Engineering Mathematics

Detailed Solutions

GATE ENGINEERING MATHEMATICS

PREPFUSION

Previous Year Questions Solutions Series

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SOLUTIONS — UNIT

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Differential Equations

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Differential Equations

Engineering Mathematics

Q6.26.1 MCQ | 2026 | 1M | Ans: A | ● ○ ○

Given solution is

$$\vec{w}(t) = e^t \vec{u}_x + e^{-2t} \vec{u}_y$$

Since $\vec{u}_x$ and $\vec{u}_y$ are unit vectors along the positive x and y axes respectively, $\vec{u}_x = \begin{bmatrix}1\\0\end{bmatrix}$ and $\vec{u}_y = \begin{bmatrix}0\\1\end{bmatrix}$,

$$\vec{w}(t) = e^t \begin{bmatrix}1\\0\end{bmatrix} + e^{-2t} \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}e^t\\e^{-2t}\end{bmatrix}$$

The differential equation is

$$\dot{\vec{w}} = A\vec{w}$$

Solving LHS by differentiating $\vec{w}(t)$ with respect to t,

$$\dot{\vec{w}}(t) = \begin{bmatrix}e^t\\-2e^{-2t}\end{bmatrix}$$

Let

$$A = \begin{bmatrix}a & b\\c & d\end{bmatrix}$$

Then solving RHS,

$$A\vec{w}(t) = \begin{bmatrix}a & b\\c & d\end{bmatrix}\begin{bmatrix}e^t\\e^{-2t}\end{bmatrix} = \begin{bmatrix}ae^t + be^{-2t}\\ce^t + de^{-2t}\end{bmatrix}$$

On comparing, we get

$$ae^t + be^{-2t} = e^t$$

$$ce^t + de^{-2t} = -2e^{-2t}$$

Equating coefficients,

$$a = 1, \qquad b = 0$$

$$c = 0, \qquad d = -2$$

Hence,

$$A = \begin{bmatrix}1 & 0\\0 & -2\end{bmatrix}$$

$$\boxed{\text{(A) } A = \begin{bmatrix}1 & 0\\0 & -2\end{bmatrix}}$$

Q6.25.1 NAT | 2025 | 1M | Ans: 0.25 | ● ○ ○

Given differential equation is

$$t^2 y''(t) - 2ty'(t) + 2y(t) = 0$$

Assume a trial solution of the form

$$y(t) = t^m$$

$$\Rightarrow \; y'(t) = mt^{m-1}$$

$$\Rightarrow \; y''(t) = m(m-1)t^{m-2}$$

Substituting into the differential equation,

$$t^2\left(m(m-1)t^{m-2}\right) - 2t\left(mt^{m-1}\right) + 2t^m = 0$$

$$\Rightarrow \left(m(m-1) - 2m + 2\right)t^m = 0$$

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2)=0$$

Hence, the general solution is

$$y(t)=c_1t+c_2t^2$$

Using the boundary conditions:

From $y'(0)=1$,	From $y'(1)=-1$,
$y'(t)=c_1+2c_2t$	$1+2c_2=-1$
$y'(0)=c_1=1$	$c_2=-1$

Therefore,

$$y(t)=t-t^2$$

To find the maximum value over $[0,1]$, Setting $y'(t)=0$,

$$y'(t)=1-2t=0$$

$$t=\frac{1}{2}$$

Now,

$$y\left(\frac{1}{2}\right)=\frac{1}{2}-\left(\frac{1}{2}\right)^2=\frac{1}{2}-\frac{1}{4}=\frac{1}{4}=0.25$$

$$\boxed{0.25}$$

Key Points

- This is a Cauchy-Euler differential equation.
- For Cauchy-Euler equations, assume a solution of the form $y=t^m$.
- Maximum or minimum values are obtained by setting the first derivative equal to zero.

Q6.24.1 MCQ 2024 1M Ans: A ● ○ ○

Given complementary function is

$$y(t)=(At+B)e^{-2t}$$

A repeated root form of the complementary function is $y(t)=(C_1t+C_2)e^{mt}$.
Hence, the auxiliary equation has repeated roots at $m=-2,-2$
Therefore, the auxiliary equation is

$$(m+2)^2=0$$

$$m^2+4m+4=0$$

Replacing m by the differential operator D,

$$D^2+4D+4=0$$

Thus, the corresponding differential equation is

$$\frac{d^2y}{dt^2}+4\frac{dy}{dt}+4y=f(t)$$

$$\boxed{\text{(A) } \frac{d^2y}{dt^2}+4\frac{dy}{dt}+4y=f(t)}$$

Key Points

- A non-homogeneous term $f(t)$ does not affect the complementary function.

Q6.22.1 MSQ 2022 1M Ans: A,B ● ○ ○

Given,

$$f(x,y) = e^{(\xi x + \eta y)}$$

Differentiating two times with respect to x,

$$\frac{\partial^2 f(x,y)}{\partial x^2} = \xi^2 e^{(\xi x+\eta y)} = \xi^2 f(x,y)$$

Differentiating two times with respect to y,

$$\frac{\partial^2 f(x,y)}{\partial y^2} = \eta^2 e^{(\xi x+\eta y)} = \eta^2 f(x,y)$$

Substituting into

$$a\frac{\partial^2 f(x,y)}{\partial x^2} + b\frac{\partial^2 f(x,y)}{\partial y^2} = f(x,y)$$

$$a\xi^2 f(x,y) + b\eta^2 f(x,y) = f(x,y)$$

$$a\xi^2 + b\eta^2 = 1$$

Checking option (A):

$$\xi = \frac{1}{\sqrt{2a}}, \qquad \eta = \frac{1}{\sqrt{2b}}$$

Then,

$$a\left(\frac{1}{\sqrt{2a}}\right)^2 + b\left(\frac{1}{\sqrt{2b}}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1$$

Hence, option (A) satisfies the equation.

Checking option (B):

$$\xi = \frac{1}{\sqrt{a}}, \qquad \eta = 0$$

Then,

$$a\left(\frac{1}{\sqrt{a}}\right)^2 + b(0)^2 = 1$$

Hence, option (B) also satisfies the equation.

Similarly on checking, options (C) and (D) do not satisfy $a\xi^2 + b\eta^2 = 1$

$$\boxed{\text{(A) } \xi = \frac{1}{\sqrt{2a}}, \eta = \frac{1}{\sqrt{2b}};\ \text{(B) } \xi = \frac{1}{\sqrt{a}}, \eta = 0}$$

Q6.21.1 MCQ 2021 1M Ans: B ● ● ○

Given differential equation is

$$\frac{dy}{dx} + \frac{x}{1-x^2}y = x\sqrt{y}$$

Divide throughout by $\sqrt{y}$,

$$\frac{1}{\sqrt{y}}\frac{dy}{dx} + \frac{x}{1-x^2}\sqrt{y} = x$$

Let $t = \sqrt{y} \Rightarrow \ y = t^2$

Differentiating,

$$\frac{dy}{dx} = 2t\frac{dt}{dx}$$

Substituting into the equation,

$$2\frac{dt}{dx} + \frac{x}{1-x^2}t = x$$

$$\frac{dt}{dx} + \frac{x}{2(1-x^2)}t = \frac{x}{2}$$

This is in the standard linear form

$$\frac{dt}{dx} + Pt = Q \qquad \text{where } P = \frac{x}{2(1-x^2)}$$

Hence, the integrating factor is

$$\text{I.F.} = e^{\int P\,dx} = e^{\int \frac{x}{2(1-x^2)}dx}$$

Let $u = 1 - x^2 \Rightarrow du = -2x\,dx$
Therefore,

$$\int \frac{x}{2(1-x^2)}dx = -\frac{1}{4}\int \frac{du}{u} = -\frac{1}{4}\ln(1-x^2)$$

Thus,

$$\text{I.F.} = e^{-\frac{1}{4}\ln(1-x^2)} = (1-x^2)^{-1/4}$$

$$\boxed{\text{(B) } (1-x^2)^{-1/4}}$$

Key Points

- Use substitution $t = \sqrt{y}$ to convert the equation into a linear differential equation.
- The integrating factor for $\frac{dt}{dx} + Pt = Q$ is $e^{\int P\,dx}$.
- Logarithmic integration commonly appears in rational functions of the form $\frac{f'(x)}{f(x)}$.

Q6.20.1 MCQ 2020 1M Ans: B ● ○ ○

Given function is

$$f(x,y,z) = e^{1-x\cos y} + xze^{-1/(1+y^2)}$$

Partially differentiating with respect to x,

$$\frac{\partial f}{\partial x} = e^{1-x\cos y}(-\cos y) + ze^{-1/(1+y^2)}$$

At the point $(1,0,e)$, $\cos 0 = 1$
Hence,

$$\left.\frac{\partial f}{\partial x}\right|_{(1,0,e)} = \left[e^{1-1}(-1)\right] + \left[e \cdot e^{-1}\right] = -1 + 1 = 0$$

$$\boxed{\text{(B) } 0}$$

Mistakes to be avoided

- Do not substitute the point $(1,0,e)$ before computing the partial derivative

Q6.20.2 MCQ 2020 1M Ans: C ● ○ ○

Given differential equation is

$$\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 9y = 0$$

Replacing $\frac{d}{dt}$ by D,

$$(D^2 - 6D + 9)y = 0$$

The auxiliary equation is

$$m^2 - 6m + 9 = 0$$
$$(m-3)^2 = 0$$

Hence, the roots are repeated:$m = 3, 3$
For equal repeated roots $m = a$, the complementary function is

$$y = (C_1 + C_2x)e^{ax}$$

Substituting $a = 3$,

$$y = (C_1 + C_2x)e^{3x}$$

$$\boxed{\text{(C) } y = (C_1 + C_2x)e^{3x}}$$

Key Points

- A homogeneous differential equation has only the complementary function as its general solution.

Q6.20.3 MCQ 2020 2M Ans: A ● ○ ○

Given differential equation is

$$\frac{dy}{dx} = (y-1)x$$

Case-1: For $y \neq 1$, separate the variables:

$$\frac{dy}{y-1} = x\,dx$$

Integrating both sides,

$$\int \frac{dy}{y-1} = \int x\,dx$$

$$\ln|y-1| = \frac{x^2}{2} + C$$

$$\ln|y-1| = 0.5x^2 + C$$

Case-2: Check for constant solutions. If $y = 1$, $\frac{dy}{dx} = 0$
Substituting into the differential equation,

$$0 = (1-1)x = 0$$

$$y = 1$$

Therefore, the two solutions are

$$\ln|y-1| = 0.5x^2 + C \quad \text{and} \quad y = 1$$

$$\boxed{\text{(A) } \ln|y-1| = 0.5x^2 + C \text{ and } y = 1}$$

Mistakes to be avoided

- Constant solutions may be lost during division and must be checked separately.

Q6.19.1 MCQ 2019 1M Ans: C ● ○ ○

Given differential equation is

$$\frac{dy}{dx} = -\left(\frac{x}{y}\right)^n$$

For $n = -1$,

$$\frac{dy}{dx} = -\left(\frac{x}{y}\right)^{-1} = -\frac{y}{x}$$

$$\frac{1}{y}dy = -\frac{1}{x}dx$$

Integrating,

$$\ln y = -\ln x + C$$

$$\ln(xy) = C$$

$$\Rightarrow\ xy = c \rightarrow \text{ represents a family of hyperbolas.}$$

For $n = 1$,

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$y\,dy = -x\,dx$$

Integrating,

$$\frac{y^2}{2} = -\frac{x^2}{2} + C$$

$$\Rightarrow\ x^2 + y^2 = c \rightarrow \text{ represents a family of circles.}$$

Hence, for $n = -1$ and $n = 1$, the families of curves are hyperbolas and circles, respectively.

$$\boxed{\text{(B) Hyperbolas and Circles}}$$

Key Points

- The equation $xy = c$ represents a rectangular hyperbola.
- The equation $x^2 + y^2 = c$ represents a family of circles centered at the origin.

Mistakes to be avoided

- After integration, simplify the equation into a standard geometric form before identifying the curve.

Q6.19.2 NAT 2019 2M Ans: 5.25 ● ● ○

Given differential equation is

$$x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 3y = 0$$

Assume a solution of the form

$$y = x^m$$

$$\Rightarrow\ \frac{dy}{dx} = mx^{m-1}$$

$$\frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Substituting into the differential equation,

$$x^2\left(m(m-1)x^{m-2}\right) - 3x\left(mx^{m-1}\right) + 3x^m = 0$$

$$\Rightarrow \quad \left(m(m-1) - 3m + 3\right)x^m = 0$$

$$m^2 - 4m + 3 = 0$$

$$(m-1)(m-3) = 0$$

Hence, the general solution is $y = c_1x + c_2x^3$
Using the given conditions:

$$y(1) = 1 \Rightarrow \; c_1 + c_2 = 1$$

$$y(2) = 14 \Rightarrow \; 2c_1 + 8c_2 = 14$$

Solving, $c_1 = -1, \; c_2 = 2$
Therefore,

$$y = -x + 2x^3$$

At $x = 1.5$,

$$y(1.5) = -(1.5) + 2(1.5)^3$$

$$= -1.5 + 6.75 = 5.25$$

$$\boxed{5.25}$$

Q6.18.1 MCQ 2018 2M Ans: A ●●○

Given differential equation is

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2y} + \frac{y}{x}$$

$$\Rightarrow \; \frac{dy}{dx} = \frac{x^2 + y^2}{2y} + \frac{xy}{x^2}$$

Let $y = tx$, where $t = \frac{y}{x}$. Then,

$$\frac{dy}{dx} = t + x\frac{dt}{dx}$$

Substituting into the differential equation,

$$t + x\frac{dt}{dx} = \frac{x^2 + t^2x^2}{2tx} + t$$

$$x\frac{dt}{dx} = \frac{x(1+t^2)}{2t}$$

$$\frac{2t}{1+t^2}\,dt = dx$$

Integrating both sides,

$$\int \frac{2t}{1+t^2}\,dt = \int dx$$

$$\ln(1+t^2) = x + C$$

Substituting $t = \frac{y}{x}$,

$$\ln\left(1 + \frac{y^2}{x^2}\right) = x + C$$

Using the condition that the curve passes through $(1, 0)$,

$$\ln\left(1 + \frac{0}{1}\right) = 1 + C$$

$$0 = 1 + C$$

$$C = -1$$

Hence,

$$\ln\left(1 + \frac{y^2}{x^2}\right) = x - 1$$

$$\boxed{\text{(A) } \ln\left(1 + \frac{y^2}{x^2}\right) = x - 1}$$

Mistakes to be avoided

- Do not forget to differentiate $y = tx$ using the product rule.

Q6.18.2 NAT 2018 2M Ans: -0.21 ● ● ○

Given differential equation is

$$\frac{d^2y}{dt^2} = -\frac{dy}{dt} - \frac{5}{4}y$$

$$\Rightarrow \frac{d^2y}{dt^2} + \frac{dy}{dt} + \frac{5}{4}y = 0$$

The auxiliary equation is

$$m^2 + m + \frac{5}{4} = 0$$

$$4m^2 + 4m + 5 = 0$$

$$m = \frac{-4 \pm \sqrt{16 - 80}}{8}$$

$$m = -\frac{1}{2} \pm i$$

Hence, the general solution is

$$y(t) = e^{-t/2}\left(c_1 \cos t + c_2 \sin t\right)$$

Using the initial condition $y(0) = 1$,

$$1 = e^0(c_1 \cos 0 + c_2 \sin 0)$$

$$c_1 = 1$$

Differentiating,

$$\frac{dy}{dt} = e^{-t/2}\left(-c_1 \sin t + c_2 \cos t\right) - \frac{1}{2}e^{-t/2}\left(c_1 \cos t + c_2 \sin t\right)$$

Using $\left.\frac{dy}{dt}\right|_{t=0} = 0$,

$$0 = c_2 - \frac{1}{2}c_1$$

$$c_2 = \frac{1}{2}$$

Therefore,

$$y(t) = e^{-t/2}\left(\cos t + \frac{1}{2}\sin t\right)$$

At $t = \pi$,

$$y(\pi) = e^{-\pi/2}\left(\cos\pi + \frac{1}{2}\sin\pi\right)$$

$$= e^{-\pi/2}(-1+0) = -e^{-\pi/2} \approx -0.2079$$

Rounded to two decimal places,

$$\boxed{-0.21}$$

Q6.17.1 MCQ 2017 2M Ans: D ● ○ ○

Given differential equation is

$$\frac{dy}{dx} = (x+y-1)^2$$

Let $p = x + y \Rightarrow \frac{dp}{dx} = 1 + \frac{dy}{dx}$
Substituting into the differential equation,

$$\frac{dp}{dx} - 1 = (p-1)^2$$

$$\frac{dp}{dx} = (p-1)^2 + 1$$

$$\frac{dp}{(p-1)^2+1} = dx$$

Integrating both sides,

$$\int \frac{dp}{(p-1)^2+1} = \int dx$$

$$\tan^{-1}(p-1) = x + C$$

$$p - 1 = \tan(x+C)$$

Substituting $p = x + y$,

$$x + y - 1 = \tan(x+C)$$

$$y = 1 - x + \tan(x+C)$$

$$\boxed{\text{(D) } y = 1 - x + \tan(x+C)\text{, where } c \text{ is a constant}}$$

Key Points

- The standard integral

$$\int \frac{du}{u^2+1} = \tan^{-1} u$$

is used here.

Q6.17.2 MCQ 2017 1M Ans: A ● ○ ○

Given differential equation is

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 5y = 0$$

Replacing $\frac{d}{dx}$ by D,

$$(D^2 + 2D - 5)y = 0$$

The auxiliary equation is

$$m^2 + 2m - 5 = 0$$

$$m = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-5)}}{2} = \frac{-2 \pm \sqrt{24}}{2} = \frac{-2 \pm 2\sqrt{6}}{2}$$

$$m_1 = -1 + \sqrt{6}, \qquad m_2 = -1 - \sqrt{6}$$

Since the roots are real and distinct, the general solution is

$$y = K_1 e^{m_1 x} + K_2 e^{m_2 x}$$

Substituting the roots,

$$y = K_1 e^{(-1+\sqrt{6})x} + K_2 e^{(-1-\sqrt{6})x}$$

$$\boxed{\text{(A) } K_1 e^{(-1+\sqrt{6})x} + K_2 e^{(-1-\sqrt{6})x}}$$

Q6.16.1 MCQ | 2016 | 2M | Ans: A | ● ○ ○

Given differential equation is

$$\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 36y = 0 \qquad \text{with } y(0) = 3, \ \left.\frac{dy}{dx}\right|_{x=0} = -36$$

Replacing $\frac{d}{dx}$ by D,

$$(D^2 + 12D + 36)y = 0$$

The auxiliary equation is

$$m^2 + 12m + 36 = 0$$

$$(m+6)^2 = 0$$

Hence, the repeated root is $m = -6$
Therefore, the complementary function is

$$y = (c_1 + c_2 x)e^{-6x}$$

Differentiating,

$$\frac{dy}{dx} = c_2 e^{-6x} + (c_1 + c_2 x)(-6)e^{-6x}$$

$$= [c_2 - 6(c_1 + c_2 x)]\, e^{-6x}$$

Using the initial conditions:

$y(0) = 3$	$\left.\frac{dy}{dx}\right\|_{x=0} = -36$
$(c_1 + 0)\, e^0 = 3$	$(c_2 - 6c_1)\, e^0 = -36$
$c_1 = 3$	$c_2 - 18 = -36$
	$c_2 = -18$

Substituting $c_1 = 3$ and $c_2 = -18$,

$$y = (3 - 18x)e^{-6x}$$

$$\boxed{\text{(A) } (3 - 18x)e^{-6x}}$$

Q6.15.1 MCQ 2015 2M Ans: B ● ● ○

Given differential equation is

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0 \text{ with initial conditions } y(0) = 1,\ y'(0) = 1$$

Method 1

Replacing $\frac{d}{dt}$ by D,

$$(D^2 + 2D + 1)y = 0$$

The auxiliary equation is

$$m^2 + 2m + 1 = 0$$
$$(m + 1)^2 = 0$$

Hence, the repeated root is $m = -1$
Therefore, the general solution is

$$y = (C_1 + C_2 t)e^{-t}$$

Differentiating,

$$y' = C_2 e^{-t} - (C_1 + C_2 t)e^{-t}$$
$$= (C_2 - C_1 - C_2 t)\, e^{-t}$$

Using the initial conditions:

$y(0) = 1$	$y'(0) = 1$
$(C_1 + 0)\, e^0 = 1$	$(C_2 - C_1)\, e^0 = 1$
$C_1 = 1$	$C_2 - 1 = 1$
	$C_2 = 2$

Substituting $C_1 = 1$ and $C_2 = 2$,

$$y = (1 + 2t)e^{-t}$$

Method 2

Using Laplace transform:

$$\mathcal{L}\left\{\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y\right\} = 0$$
$$(s^2Y - s - 1) + 2(sY - 1) + Y = 0$$
$$(s^2 + 2s + 1)Y = s + 3$$
$$Y = \frac{s + 3}{(s + 1)^2}$$

Taking inverse Laplace transform,

$$y(t) = (1 + 2t)e^{-t}$$

$$\boxed{\text{(B) } (1 + 2t)\, e^{-t}}$$

Q6.15.2 MCQ | 2015 | 1M | Ans: C | ● ○ ○

Given differential equation is

$$\frac{dy}{dx} = \frac{1+\cos 2y}{1-\cos 2x}$$
$$\frac{dy}{1+\cos 2y} = \frac{dx}{1-\cos 2x}$$

Using the identities $1+\cos 2y = 2\cos^2 y$ and $1-\cos 2x = 2\sin^2 x$, we get

$$\frac{dy}{2\cos^2 y} = \frac{dx}{2\sin^2 x}$$

$$\sec^2 y\, dy = \csc^2 x\, dx$$

Integrating both sides,

$$\int \sec^2 y\, dy = \int \csc^2 x\, dx$$

$$\tan y = -\cot x + C$$

$$\tan y + \cot x = C$$

$$\boxed{\text{(C)}\ \tan y + \cot x = c\ (c \text{ is a constant})}$$

Key Points

- Standard integrals:

$$\int \sec^2\theta\, d\theta = \tan\theta, \qquad \int \csc^2\theta\, d\theta = -\cot\theta.$$

Q6.15.3 MCQ | 2015 | 2M | Ans: C | ● ● ○

Given differential equation is

$$\frac{dx}{dt} = 10 - 0.2x \quad \text{with initial condition } x(0) = 1$$

Method 1

This is a first-order linear differential equation of the form

$$\frac{dx}{dt} + Px = Q \ \text{ where } P = 0.2, Q = 10$$

The integrating factor is

$$\text{I.F.} = e^{\int 0.2\,dt} = e^{0.2t}$$

Multiplying throughout by the integrating factor,

$$e^{0.2t}\frac{dx}{dt} + 0.2\,e^{0.2t}x = 10\,e^{0.2t}$$

$$\frac{d}{dt}\left(xe^{0.2t}\right) = 10e^{0.2t}$$

Integrating,

$$xe^{0.2t} = \int 10\,e^{0.2t}\,dt$$

$$x\,e^{0.2t} = 50\,e^{0.2t} + C$$

$$x = 50 + Ce^{-0.2t}$$

Using $x(0) = 1$,

$$1 = 50 + C \Rightarrow C = -49$$

Hence,

$$x(t) = 50 - 49e^{-0.2t}$$

Method 2

Using steady-state and transient response:
The steady-state value is obtained from

$$10 - 0.2x = 0 \Rightarrow x = 50$$

Hence, the solution has the form

$$x(t) = 50 + Ce^{-0.2t}$$

Using $x(0) = 1$,

$$1 = 50 + C \Rightarrow C = -49$$

Therefore,

$$x(t) = 50 - 49e^{-0.2t}$$

$$\boxed{\text{(C) } 50 - 49e^{-0.2t}}$$

Key Points

- The transient response decays exponentially when the coefficient of x is positive.

Q6.15.4 NAT 2015 2M Ans: 0.86 ● ● ○

Given differential equation is

$$\frac{d^2x(t)}{dt^2} + 3\frac{dx(t)}{dt} + 2x(t) = 0 \text{ with conditions } x(0) = 20,\ x(1) = \frac{10}{e}$$

Method 1

Replacing $\frac{d}{dt}$ by D,

$$(D^2 + 3D + 2)x = 0$$

The auxiliary equation is

$$m^2 + 3m + 2 = 0$$

$$(m + 1)(m + 2) = 0$$

Hence, the roots are $m_1 = -1, m_2 = -2$ Therefore, the general solution is

$$x(t) = C_1e^{-t} + C_2e^{-2t}$$

Using the boundary conditions:

$x(0) = 20$	$x(1) = \frac{10}{e}$
$C_1 + C_2 = 20$	$C_1e^{-1} + C_2e^{-2} = \frac{10}{e}$
	$C_1 + C_2e^{-1} = 10$

Solving,

$$C_2 = \frac{10e - 20}{e - 1} \approx 15.82$$

$$C_1 = 20 - C_2 \approx 4.18$$

Hence,

$$x(t) = 4.18e^{-t} + 15.82e^{-2t}$$

At $t = 2$,

$$x(2) = 4.18e^{-2} + 15.82e^{-4} \approx 0.86$$

Method 2

Using the exact constants directly:
From

$$C_1 + C_2 = 20 \Rightarrow C_1 = 20 - C_2$$

Substituting into

$$C_1 + C_2e^{-1} = 10,$$
$$20 - C_2 + \frac{C_2}{e} = 10$$
$$C_2\left(\frac{1}{e} - 1\right) = -10$$
$$C_2 = \frac{10e}{e-1} \Rightarrow C_1 = 20 - \frac{10e}{e-1}$$

Thus,

$$x(2) = \left(20 - \frac{10e}{e-1}\right)e^{-2} + \frac{10e}{e-1}e^{-4} x(2) \approx 0.86$$

$$\boxed{0.86}$$

Q6.14.1 MCQ | 2014 | 1M | Ans: A | ● ○ ○

Given differential equation is

$$\frac{d^2y}{dx^2} + 2\alpha\frac{dy}{dx} + y = 0$$

Replacing $\frac{d}{dx}$ by D,

$$(D^2 + 2\alpha D + 1)y = 0$$

The auxiliary equation is

$$m^2 + 2\alpha m + 1 = 0$$
$$m = \frac{-2\alpha \pm \sqrt{(2\alpha)^2 - 4}}{2} = \frac{-2\alpha \pm \sqrt{4\alpha^2 - 4}}{2}$$

According to the question, both roots are equal.
Hence, the discriminant must be zero:

$$4\alpha^2 - 4 = 0 \Rightarrow \alpha^2 = 1$$

$$\alpha = \pm 1$$

$$\boxed{\text{(A)} \ \pm 1}$$

Key Points

- Equal roots occur when the discriminant of the auxiliary equation is zero.

Q6.14.2 MCQ 2014 1M Ans: A ● ○ ○

A first-order linear non-homogeneous differential equation has the standard form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where $P(x)$ and $Q(x)$ are functions of the independent variable only.

Checking option (A):

$$\frac{dy}{dx} + xy = e^{-x}$$

All derivatives are first order and the equation is linear in y.
The right-hand side is non-zero.
Hence, it is a first order linear non-homogeneous differential equation.

Checking option (B):

$$\frac{dy}{dx} + xy = 0$$

All derivatives are first order and the equation is linear in y.
Also, the right-hand side is zero.
Hence, it is a first-order linear homogeneous differential equation.

Checking option (C):

$$\frac{dy}{dx} + xy = e^{-y}$$

The term e^{-y} is a nonlinear function of the dependent variable y.
Hence, the equation is nonlinear.

Checking option (D):

$$\frac{dy}{dx} + e^{-y} = 0$$

Since e^{-y} is nonlinear in y, the equation is not linear.
Therefore, the correct option is

$$\boxed{\text{(A) } \frac{dy}{dx} + xy = e^{-x}}$$

Key Points

- A linear differential equation must contain the dependent variable and its derivatives only to the first power.
- Terms like e^{-y}, y^2, or $\sin y$ make the equation nonlinear.

Q6.14.3 MCQ 2014 1M Ans: C ● ○ ○

Given,

$$z = xy\ln(xy)$$

Partially differentiating with respect to x,

$$\frac{\partial z}{\partial x} = y\ln(xy) + xy\left(\frac{1}{xy}\right)y$$

$$\frac{\partial z}{\partial x} = y\ln(xy) + y$$

Multiplying by x,

$$x\frac{\partial z}{\partial x} = xy\ln(xy) + xy \qquad \cdots (1)$$

Similarly, partially differentiating with respect to y,

$$\frac{\partial z}{\partial y} = x\ln(xy) + xy\left(\frac{1}{xy}\right)x$$

$$\frac{\partial z}{\partial y} = x \ln(xy) + x$$

Multiplying by y,

$$y\frac{\partial z}{\partial y} = xy \ln(xy) + xy \qquad \cdots (2)$$

From equations (1) and (2),

$$x\frac{\partial z}{\partial x} = y\frac{\partial z}{\partial y}$$

$$\boxed{\text{(C) } x\frac{\partial z}{\partial x} = y\frac{\partial z}{\partial y}}$$

Key Points

- The logarithmic derivative

$$\frac{d}{du}(\ln u) = \frac{1}{u}$$

  is used with $u = xy$.

Q6.14.4 MCQ 2014 1M Ans: B ● ○ ○

Given differential equation is

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x = 0$$

Replacing $\frac{d}{dt}$ by D,

$$(D^2 + 2D + 1)x = 0$$

The auxiliary equation is

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0$$

Hence, the repeated root is $m = -1$
For repeated roots $m = a$, the general solution is

$$x(t) = (C_1 + C_2 t)e^{at}$$

Substituting $a = -1$,

$$x(t) = (C_1 + C_2 t)e^{-t}$$

Since a and b are arbitrary constants, let

$$C_1 = a, \qquad C_2 = b$$

Therefore,

$$x(t) = ae^{-t} + bte^{-t}$$

$$\boxed{\text{(B) } ae^{-t} + bte^{-t}}$$

Mistakes to be avoided

- Do not write the solution as $C_1 e^{-t} + C_2 e^{-t}$; both terms are linearly dependent.

Q6.14.5 NAT 2014 2M Ans: 0.541 ● ○ ○

Given differential equation is

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0 \quad \text{with initial conditions } y(0) = 1,\ y'(0) = 1$$

Replacing $\frac{d}{dx}$ by D,

$$(D^2 + 4D + 4)y = 0$$

The auxiliary equation is

$$m^2 + 4m + 4 = 0$$
$$(m + 2)^2 = 0$$

Hence, the repeated root is $m = -2$
Therefore, the general solution is

$$y = (C_1 + C_2x)e^{-2x}$$

Using the initial condition $y(0) = 1$,

$$(C_1 + 0)e^0 = 1 \Rightarrow\ C_1 = 1$$

Differentiating,

$$y' = C_2e^{-2x} - 2(C_1 + C_2x)e^{-2x}$$
$$= \Big[C_2 - 2C_1 - 2C_2x\Big]e^{-2x}$$

Using $y'(0) = 1$,

$$(C_2 - 2C_1)e^0 = 1$$
$$C_2 - 2 = 1 \Rightarrow\ C_2 = 3$$

Hence,

$$y = (1 + 3x)e^{-2x}$$

At $x = 1$,

$$y(1) = 4e^{-2} \approx 0.541$$
$$\boxed{0.541}$$

Q6.13.1 MCQ 2013 2M Ans: D ● ● ○

Consider the first-order linear system

$$\frac{dy(t)}{dt} + ay(t) = bx(t) \quad \text{with initial condition } y(0).$$

Method 1

Taking Laplace transform,

$$sY(s) - y(0) + aY(s) = bX(s)$$
$$Y(s) = \frac{bX(s) + y(0)}{s + a}$$

Suppose the desired output is $y_1(t) = -2y(t)$.
Taking Laplace transform,

$$Y_1(s) = -2Y(s) = -2\left[\frac{bX(s) + y(0)}{s + a}\right].$$

Let the modified input and initial condition be $x_1(t)$ and $y_1(0)$.

Then,

$$Y_1(s) = \frac{bX_1(s) + y_1(0)}{s + a}.$$

Equating the two expressions for $Y_1(s)$,

$$\frac{bX_1(s) + y_1(0)}{s + a} = -2\frac{bX(s) + y(0)}{s + a}$$

Hence,

$$bX_1(s) + y_1(0) = -2bX(s) - 2y(0).$$

Comparing corresponding terms,

$$X_1(s) = -2X(s)$$

$$y_1(0) = -2y(0)$$

Taking inverse Laplace transform,

$$x_1(t) = -2x(t).$$

Method 2

Using the linearity property of linear differential equations:
If $y(t)$ satisfies

$$\frac{dy}{dt} + ay = bx(t)$$

with input $x(t)$ and initial condition $y(0)$, then multiplying the entire solution by a constant k gives

$$k\frac{dy}{dt} + aky = kb\,x(t).$$

This is equivalent to the same system driven by

$$x_{\text{new}}(t) = k\,x(t) \quad \text{with initial condition} \quad y_{\text{new}}(0) = k\,y(0).$$

For $k = -2$,

$$x_{\text{new}}(t) = -2x(t), \qquad y_{\text{new}}(0) = -2y(0).$$

Therefore, to obtain $y_{\text{new}}(t) = -2y(t)$,
we must choose

$$x_{\text{new}}(t) = -2x(t)$$

$$y_{\text{new}}(0) = -2y(0).$$

$$\boxed{\text{(D) change the initial condition to } -2y(0) \text{ and the forcing function to } -2x(t)}$$

Key Points

- Linear differential systems satisfy the scaling property, scaling the input and initial condition by the same factor scales the output by that factor.

Mistakes to be avoided

- Remember that the desired output is $-2y(t)$, not $2y(t)$.

Q6.12.1 MCQ 2012 1M Ans: D ● ○ ○

Given differential equation is

$$t\frac{dx}{dt} + x = t \ \text{ with initial condition } \ x(1) = 0.5$$

$$\frac{dx}{dt} + \frac{x}{t} = 1$$

This is a first-order linear differential equation of the form

$$\frac{dx}{dt} + Px = Q \ \text{ where } P = \frac{1}{t}, \ Q = 1.$$

The integrating factor is

$$\text{I.F.} = e^{\int \frac{1}{t}\,dt} = e^{\ln t} = t$$

The complete solution is,

$$x \times \text{I.F.} = \int Q \times \text{I.F.}\ dt + C$$

$$xt = \int t\,dt + C = \frac{t^2}{2} + C$$

Using the initial condition $x(1) = 0.5$,

$$0.5(1) = \frac{1}{2} + C \Rightarrow \ C = 0$$

Hence,

$$xt = \frac{t^2}{2} \Rightarrow \ x = \frac{t}{2}$$

$$\boxed{\text{(D) } x = \frac{t}{2}}$$

Key Points

- Convert the equation into the standard linear form before finding the integrating factor.
- After multiplying by the integrating factor, the left-hand side becomes an exact derivative.

Q6.12.2 MCQ 2012 2M Ans: D ● ● ○

Given differential equation is

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = \delta(t) \ \text{ with initial conditions } \ y(0^-) = -2, \ \left.\frac{dy}{dt}\right|_{t=0^-} = 0.$$

Method 1

Taking Laplace transform,

$$s^2Y(s) - sy(0^-) - y'(0^-) + 2\left[sY(s) - y(0^-)\right] + Y(s) = 1$$

Substituting the initial conditions,

$$Y(s)(s^2 + 2s + 1) + 2s + 4 = 1$$

$$Y(s)(s+1)^2 = -(2s+3)$$

$$Y(s) = -\frac{2s+3}{(s+1)^2}$$

Using partial fractions,

$$\frac{2s+3}{(s+1)^2} = \frac{2}{s+1} + \frac{1}{(s+1)^2}$$

Hence,

$$Y(s) = -\frac{2}{s+1} - \frac{1}{(s+1)^2}$$

Taking inverse Laplace transform,

$$y(t) = -2e^{-t} - te^{-t}$$

Differentiating,

$$\frac{dy}{dt} = 2e^{-t} - \left(e^{-t} - te^{-t}\right)$$

$$\frac{dy}{dt} = e^{-t} + te^{-t}$$

Therefore,

$$\left.\frac{dy}{dt}\right|_{t=0^+} = e^0 + 0 \cdot e^0 = 1$$

Method 2

Integrate the differential equation across an infinitesimal interval around $t = 0$:

$$\int_{0^-}^{0^+} \left(\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y\right) dt = \int_{0^-}^{0^+} \delta(t)\, dt$$

As the interval shrinks to zero,

$$\left.\frac{dy}{dt}\right|_{0^+} - \left.\frac{dy}{dt}\right|_{0^-} = 1$$

since the integrals involving y and y' vanish.
Given

$$\left.\frac{dy}{dt}\right|_{0^-} = 0,$$

we obtain

$$\left.\frac{dy}{dt}\right|_{0^+} = 1.$$

This method uses the jump property of the impulse input directly.

$\boxed{\text{(D) 1}}$

Key Points

- An impulse input causes a jump in the first derivative of the response.
- Integrating the differential equation across the impulse location provides a quick way to determine the jump condition.

Q6.11.1 MCQ 2011 1M Ans: C ● ○ ○

Given differential equation is

$$\frac{dy}{dx} = ky \quad \text{with initial condition} \quad y(0) = c$$

Separating variables,

$$\frac{dy}{y} = k\, dx$$

Integrating both sides,

$$\int \frac{dy}{y} = \int k\,dx$$

$$\ln y = kx + C$$

$$y = e^{kx+C} \Rightarrow y = C_1 e^{kx}$$

Using $y(0) = c$,

$$c = C_1 e^0 \Rightarrow C_1 = c$$

Hence,

$$y = ce^{kx}.$$

$$\boxed{\text{(C) } y = ce^{kx}}$$

Q6.10.1 MCQ 2010 1M Ans: D ● ○ ○

Given differential equation is

$$\frac{d^2 n(x)}{dx^2} - \frac{n(x)}{L^2} = 0 \text{ with boundary conditions } n(0) = K,\ n(\infty) = 0$$

Replacing $\frac{d}{dx}$ by D,

$$\left(D^2 - \frac{1}{L^2}\right) n(x) = 0$$

The auxiliary equation is

$$m^2 - \frac{1}{L^2} = 0$$

$$\left(m - \frac{1}{L}\right)\left(m + \frac{1}{L}\right) = 0$$

Hence, the roots are

$$m_1 = \frac{1}{L}, \qquad m_2 = -\frac{1}{L}$$

Therefore, the general solution is

$$n(x) = C_1 e^{x/L} + C_2 e^{-x/L}$$

Using the boundary condition $n(\infty) = 0$,

$$\lim_{x\to\infty} e^{x/L} = \infty$$

Hence, the term $C_1 e^{x/L}$ cannot be present.
Therefore,

$$C_1 = 0 \text{ and } n(x) = C_2{}^{-x/L}.$$

Using $n(0) = K$,

$$K = C_2 e^0 \Rightarrow C_2 = K$$

Thus,

$$n(x) = Ke^{-x/L}$$

$$\boxed{\text{(D) } n(x) = Ke^{-x/L}}$$

Key Points

- Boundary conditions at infinity are often used to eliminate exponentially growing terms.

Mistakes to be avoided

- Do not discard the wrong exponential term; always check its behavior as $x \to \infty$.

Q6.09.1 MCQ 2009 1M Ans: B ● ○ ○

Given differential equation is

$$\frac{d^2y}{dt^2} + \left(\frac{dy}{dt}\right)^3 + y^4 = e^{-t}$$

The **order** of a differential equation is defined as the order of the highest derivative present in the equation. Let us identify the derivatives appearing in the given equation:

- Although the first derivative appears with power 3, $\left(\frac{dy}{dt}\right)^3$, its order remains 1, because order depends on the *derivative itself*, not on the power to which it is raised.
- Similarly, y^4 contains no derivative and therefore does not affect the order.

The highest-order derivative present is

$$\frac{d^2y}{dt^2}$$

Hence,

$$\text{Order of the differential equation} = 2.$$

$$\boxed{\text{(B) } 2}$$

Key Points

- **Order** of a differential equation = order of the highest derivative present.
- The power of a derivative does *not* affect the order.
- Terms without derivatives do not contribute to the order.

Q6.09.2 MCQ 2009 2M Ans: A ● ● ○

To identify the family of curves corresponding to each differential equation, solve them one by one and compare the resulting curve with the given families.

For P:

$$\frac{dy}{dx} = \frac{y}{x}$$

Separating variables and Integrating,

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\ln y = \ln x + C$$

$$y = Cx$$

This represents a family of straight lines passing through the origin.

$$P \to 2$$

For Q:

$$\frac{dy}{dx} = -\frac{y}{x}$$

Separating variables and Integrating,

$$\int \frac{dy}{y} = \int -\frac{dx}{x}$$

$$\ln y = -\ln x + C$$
$$\ln(xy) = C$$
$$xy = C_1$$

This is a rectangular hyperbola.

$$Q \rightarrow 3$$

For R:

$$\frac{dy}{dx} = \frac{x}{y}$$

Separating variables and Integrating,

$$\int y\,dy = \int x\,dx$$
$$\frac{y^2}{2} = \frac{x^2}{2} + C$$
$$y^2 - x^2 = C_1$$

This is a hyperbola.

$$R \rightarrow 3$$

For S:

$$\frac{dy}{dx} = -\frac{x}{y}$$

Separating variables and Integrating,

$$\int y\,dy = \int -x\,dx$$
$$\frac{y^2}{2} = -\frac{x^2}{2} + C$$
$$x^2 + y^2 = C_1$$

This represents a family of circles centered at the origin.

$$S \rightarrow 1$$

Hence,

$$P \rightarrow 2, \qquad Q \rightarrow 3, \qquad R \rightarrow 3, \qquad S \rightarrow 1$$

$$\boxed{\text{(A) P-2, Q-3, R-3, S-1}}$$

Q6.08.1 MCQ | 2008 | 1M | Ans: B | ● ○ ○

Given differential equation is

$$\frac{dx(t)}{dt} + 3x(t) = 0 \text{ with initial condition } x(0) = 2$$

Method 1

Separating variables & Integrating both sides,

$$\int \frac{dx}{x} = -3\int dt$$
$$\ln x = -3t + C$$
$$x = Ce^{-3t}$$

Using the initial condition $x(0) = 2$,

$$2 = Ce^0 \Rightarrow C = 2$$

Hence,

$$x(t) = 2e^{-3t}$$

Method 2

Replacing $\frac{d}{dt}$ by D,

$$(D + 3)x = 0$$

The auxiliary equation is

$$m + 3 = 0 \Rightarrow m = -3$$

Therefore, the complementary function is

$$x(t) = Ce^{-3t}$$

Using $x(0) = 2, C = 2$
Hence,

$$x(t) = 2e^{-3t}$$

Therefore, the correct option is

$$\boxed{\text{(B) } x(t) = 2e^{-3t}}$$

Q6.07.1 MCQ 2007 2M Ans: D ● ● ○

Given differential equation is

$$k^2 \frac{d^2y}{dx^2} - y = -y_2 \text{ with boundary conditions } y(0) = y_1, \; y(\infty) = y_2,$$

where y_1 and y_2 are constants.

Method 1

The corresponding homogeneous equation is

$$k^2 \frac{d^2y}{dx^2} - y = 0$$

Assume a solution of the form $y = e^{mx}$.
Substituting,

$$k^2m^2e^{mx} - e^{mx} = 0$$

$$k^2m^2 - 1 = 0$$

$$m = \pm\frac{1}{k}$$

Hence, the complementary function is

$$y_c = C_1e^{x/k} + C_2e^{-x/k}$$

Since the forcing term $-y_2$ is a constant, assume a constant particular solution $y_p = A$.
Substituting into the differential equation,

$$0 - A = -y_2 \Rightarrow A = y_2$$

Therefore,

$$y = \text{C.F.} + \text{P.I.} = (C_1e^{x/k} + C_2e^{-x/k}) + y_2$$

Using the boundary condition $y(\infty) = y_2$,

$$C_1 e^{\infty/k} + C_2 e^{-\infty/k} + y_2 = y_2$$

$$C_1 = 0$$

Using $y(0) = y_1$,

$$y_1 = C_2 e^{-0/k} + y_2$$

$$y_1 = C_2 + y_2$$

$$C_2 = y_1 - y_2$$

Hence,

$$y = (y_1 - y_2)e^{-x/k} + y_2.$$

Method 2

Define $z = y - y_2$.
Then

$$\frac{d^2z}{dx^2} = \frac{d^2y}{dx^2}$$

Substituting $y = z + y_2$ into the differential equation,

$$k^2 \frac{d^2z}{dx^2} - (z + y_2) = -y_2$$

$$k^2 \frac{d^2z}{dx^2} - z = 0$$

The solution is

$$z = Ae^{x/k} + Be^{-x/k}.$$

Using $z(\infty) = y(\infty) - y_2 = 0$, we get $A = 0$.
Using $z(0) = y_1 - y_2$, we get $B = y_1 - y_2$.
Therefore,

$$y = z + y_2 = (y_1 - y_2)e^{-x/k} + y_2.$$

$$\boxed{\text{(D) } y = (y_1 - y_2)\, e^{-x/k} + y_2}$$

Key Points

- A constant forcing term generally suggests assuming a constant particular solution.

Mistakes to be avoided

- While finding the particular solution, remember that the derivatives of a constant are zero.
- Apply the boundary condition at $x = \infty$ before using the condition at $x = 0$.

Q6.06.1 MCQ 2006 1M Ans: A ● ○ ○

Given differential equation is

$$\dot{x}(t) + 2x(t) = \delta(t) \ \text{ with initial condition } \ x(0^-) = 0.$$

Method 1

Taking Laplace transform,

$$sX(s) - x(0^-) + 2X(s) = 1$$

$$(s + 2)X(s) = 1 \qquad (\because \ (x(0^-) = 0)\ )$$

$$X(s) = \frac{1}{s+2}$$

Taking inverse Laplace transform,

$$x(t) = e^{-2t}u(t).$$

Method 2

Using the impulse-jump property,
Integrating the differential equation across an infinitesimal interval around $t = 0$,

$$\int_{0^-}^{0^+} (\dot{x}(t) + 2x(t))\ dt = \int_{0^-}^{0^+} \delta(t)\, dt$$

As the interval shrinks to zero,

$$x(0^+) - x(0^-) = 1$$

Given $x(0^-) = 0$, we obtain $x(0^+) = 1$.
For $t > 0$, the impulse disappears and the equation becomes

$$\dot{x} + 2x = 0$$

Its solution is

$$x(t) = Ce^{-2t}$$

Using $x(0^+) = 1$, $C = 1$.
Hence,

$$x(t) = e^{-2t}u(t).$$

$$\boxed{\text{(A) } e^{-2t}u(t)}$$

Q6.06.2 MCQ 2006 2M Ans: A ● ● ○

Given differential equation is

$$\frac{d^2y}{dx^2} + k^2y = 0 \text{ with boundary conditions } y(0) = 0,\ y(a) = 0.$$

Assume a solution of the form $y = e^{mx}$.
Substituting into the differential equation,

$$m^2 e^{mx} + k^2 e^{mx} = 0$$

$$m^2 + k^2 = 0$$

$$m = \pm ik$$

Hence, the general solution is

$$y = C_1 \cos(kx) + C_2 \sin(kx)$$

Using the boundary condition $y(0) = 0$,

$$A \cos 0 + B \sin 0 = 0 \Rightarrow A = 0$$

Using the second boundary condition $y(a) = 0$,

$$B \sin(ka) = 0$$

For a non-zero solution ($B \neq 0$),

$$\sin(ka) = 0$$

$$ka = m\pi, \qquad m = 1, 2, 3, \ldots$$

$$k = \frac{m\pi}{a}$$

Hence, the admissible solutions are

$$y = A_m \sin\left(\frac{m\pi x}{a}\right)$$

By the principle of superposition, the most general non-zero solution is

$$y = \sum_m A_m \sin\left(\frac{m\pi x}{a}\right).$$

$$\boxed{\text{(A) } y = \sum_m A_m \sin\left(\frac{m\pi x}{a}\right)}$$

Mistakes to be avoided

- Do not use exponential solutions directly in the final answer; convert them to sine and cosine form.

Q6.05.1 MCQ | 2005 | 1M | Ans: B | ● ○ ○

Given differential equation is

$$3\left(\frac{d^2y}{dt^2}\right) + 4\left(\frac{dy}{dt}\right)^3 + y^2 + 2 = x$$

To determine the **order** and **degree**, identify the derivatives present in the equation.
The derivatives are

$$\frac{dy}{dt} \quad \text{and} \quad \frac{d^2y}{dt^2}$$

The highest-order derivative is

$$\frac{d^2y}{dt^2}$$

Therefore, the order of the highest-order derivative = **Order = 2** and
The power of the highest-order derivative = **Degree = 1**.

$$\boxed{\text{(B) degree=1, order=2}}$$

Q6.05.2 MCQ | 2005 | 1M | Ans: B | ● ○ ○

Given differential equation is

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$$

Replacing $\frac{d}{dx}$ by D,

$$(D^2 - 5D + 6)y = 0$$

The auxiliary equation is

$$m^2 - 5m + 6 = 0$$

$$(m-2)(m-3) = 0$$

Hence, the roots are $m_1 = 2,\ m_2 = 3$.
Since the roots are real and distinct, the general solution is

$$y = C_1 e^{2x} + C_2 e^{3x}.$$

Comparing with the options, the correct option is:

$$\boxed{\text{(B) } e^{2x} + e^{3x}}$$

Q6.94.1 MCQ | 1994 | 2M | Ans: A | ● ● ○

Given the classifications:

1. Non-linear differential equation
2. Linear differential equation with constant coefficients
3. Linear homogeneous differential equation
4. Non-linear homogeneous differential equation
5. Non-linear first-order differential equation

We classify each equation individually.
For (A),

$$a_1 \frac{d^2y}{dx^2} + a_2\, y \frac{dy}{dx} + a_3 y = a_4$$

The term $y \frac{dy}{dx}$ contains a product of the dependent variable and its derivative.
Hence, this is a **Non-linear differential equation**.
For (B),

$$a_1 \frac{d^3y}{dx^3} + a_2 y = a_3$$

All derivatives appear to the first power and are not multiplied together.
Also, a_1, a_2, a_3 are constants.
Therefore, it is a **Linear differential equation with constant coefficients**.
For (C),

$$a_1 \frac{d^2y}{dx^2} + a_2\, x \frac{dy}{dx} + a_3 x^2 y = 0$$

The equation is sometimes incorrectly grouped with "non-linear" types due to coefficients involve the independent variable (x) raised to powers (like x^2). Because it does not possess constant coefficients.
Also, the right-hand side is zero.
Therefore, it is a **Non-linear homogeneous differential equation**.
Therefore,

$$\boxed{\text{(A) A–1, B–2, C–4}}$$

Key Points

- A differential equation is linear if the dependent variable and all its derivatives appear only to the first power and are not multiplied together.
- Constant coefficients imply that the coefficients of y and its derivatives are constants.

Q6.26.1 MCQ | 2026 | 2M | Ans: A | ● ○ ○

For the given differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0, \quad \text{with initial conditions } \; y(0) = 1, \; \left.\frac{dy}{dx}\right|_{x=0} = 1$$

Solving the characteristic equation

$$m^2 + m + 1 = 0$$

$$m = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm j\sqrt{3}}{2}$$

Hence, the complementary solution is

$$y = e^{-x/2}\left(C_1 \cos\frac{\sqrt{3}x}{2} + C_2 \sin\frac{\sqrt{3}x}{2}\right)$$

Applying the initial condition $y(0) = 1$,

$$1 = e^0 \left(C_1 \cos 0 + C_2 \sin 0\right)$$

$$\Rightarrow \; C_1 = 1$$

Differentiating the complementary solution,

$$\frac{dy}{dx} = -\frac{1}{2}e^{-x/2}\left(\cos\frac{\sqrt{3}x}{2} + C_2 \sin\frac{\sqrt{3}x}{2}\right) + e^{-x/2}\left(-\frac{\sqrt{3}}{2}\sin\frac{\sqrt{3}x}{2} + \frac{\sqrt{3}}{2}C_2\cos\frac{\sqrt{3}x}{2}\right).$$

Using the condition $\left.\frac{dy}{dx}\right|_{x=0} = 1$,

$$1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}C_2$$

$$\frac{3}{2} = \frac{\sqrt{3}}{2}C_2$$

$$\Rightarrow \; C_2 = \sqrt{3}$$

Thus,

$$y(x) = e^{-x/2}\left(\cos\frac{\sqrt{3}x}{2} + \sqrt{3}\,\sin\frac{\sqrt{3}x}{2}\right)$$

$$\boxed{\text{(A) } y(x) = e^{-x/2}\left(\cos\frac{\sqrt{3}x}{2} + \sqrt{3}\,\sin\frac{\sqrt{3}x}{2}\right)}$$

Key Points

- Complex roots $m = a \pm jb$ lead to the solution $y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$.
- Initial conditions are used to determine the arbitrary constants uniquely.

Q6.25.1 NAT 2025 2M Ans: 2 ● ● ○

For the given system

$$\dot{x}_1(t) = 2x_2(t), \quad \text{with initial condition } x_1(0) = 1 \quad \ldots(1)$$

$$\dot{x}_2(t) = r(t), \quad \text{with initial condition } x_2(0) = 0 \quad \ldots(2)$$

where $r(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0, \end{cases}$ is the unit step function.

Taking Laplace transforms of (2),

$$sX_2(s) - x_2(0) = R(s)$$

$$sX_2(s) = \frac{1}{s} \qquad \left(\because x_2(0) = 0 \text{ and } R(s) = \frac{1}{s}\right)$$

$$X_2(s) = \frac{1}{s^2}$$

Now, taking Laplace transform of (1),

$$sX_1(s) - x_1(0) = 2X_2(s)$$

$$sX_1(s) - 1 = \frac{2}{s^2} \qquad \left(\because x_1(0) = 1 \text{ and } X_2(s) = \frac{1}{s^2}\right)$$

$$sX_1(s) = 1 + \frac{2}{s^2}$$

$$X_1(s) = \frac{1}{s} + \frac{2}{s^3}$$

Taking inverse Laplace transform,

$$x_1(t) = 1 + t^2$$

Therefore,

$$x_1(1) = 1 + 1^2 = 2$$

$$\boxed{2}$$

Key Points

- For state equations, use $\mathcal{L}\{\dot{x}(t)\} = sX(s) - x(0)$.
- The inverse Laplace transform $\mathcal{L}^{-1}\{1/s^3\} = t^2/2$ is useful in obtaining $x_1(t)$.

Q6.24.1 MSQ 2024 2M Ans: B,D ● ○ ○

A differential equation is nonlinear if the the dependent variable and its derivatives appear in nonlinear ways such as raised to a power greater than one (e.g., y^2, $(\frac{dy}{dx})^3$), or multiplied by each other (e.g., $y\frac{dy}{dx}$). Also, the dependent variable appears inside functions like sines, cosines, or exponentials (e.g., $\sin(y)$, e^y).

Option (A):

$$x(t) + \frac{dx(t)}{dt} = t^2 e^t$$

The dependent variable $x(t)$ and its derivative appear linearly, and the forcing function depends only on t. Hence, (A) is a linear differential equation.

Option (B):

$$\frac{1}{2}e^t + x(t)\frac{dx(t)}{dt} = 0$$

The term $x(t)\frac{dx(t)}{dt}$ is a product of the dependent variable and its derivative.
Hence, (B) is a non-linear differential equation.

Option (C):

$$x(t)\cos t - \frac{dx(t)}{dt}\sin t = 1$$

The coefficients $\cos t$ and $\sin t$ are functions only of the independent variable t.
Hence, (C) is a linear differential equation.
Option (D):

$$x(t) + e^{\left(\frac{dx(t)}{dt}\right)} = 1$$

Using the series expansion,

$$e^{\left(\frac{dx(t)}{dt}\right)} = 1 + \frac{\frac{dx(t)}{dt}}{1!} + \frac{\left(\frac{dx(t)}{dt}\right)^2}{2!} + \frac{\left(\frac{dx(t)}{dt}\right)^3}{3!} + \cdots$$

The higher powers of the derivative are present.
Hence, (D) is a nonlinear differential equation.
Therefore, the non-linear differential equations are

$$\boxed{\text{(B) } \frac{1}{2}e^t + x(t)\frac{dx(t)}{dt} = 0; \text{ (D) } x(t) + e^{\left(\frac{dx(t)}{dt}\right)} = 1}$$

Q6.20.1 NAT 2020 1M Ans: 0.886 ● ● ○

The given initial value problem is

$$\frac{dy}{dx} = 2x - y, \qquad y(0) = 1$$

Method 1

Integrating Factor Method:
The linear differential equation is of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \qquad \text{where } P(x) = 1 \text{ and } Q(x) = 2x$$

The integrating factor is

$$\text{I.F.} = e^{\int P(x)\,dx} = e^x$$

The solution of the differential equation is given by,

$$y(\text{I.F.}) = \int Q(x)(\text{I.F.})\,dx + C$$

$$y\,e^x = \int 2xe^x\,dx + C$$

Using integration by parts,

$$ye^x = 2\left(xe^x - \int e^x\,dx\right) + C = 2xe^x - 2e^x + C$$

$$y = 2x - 2 + Ce^{-x}$$

Applying the initial condition $y(0) = 1$,

$$1 = -2 + C \Rightarrow\ C = 3$$

Therefore,

$$y = 2x - 2 + 3e^{-x}.$$

At $x = \ln 2$,

$$y = 2(\ln 2) - 2 + 3e^{-(\ln 2)}$$
$$y = 2(\ln 2) - 2 + 3\left(\frac{1}{2}\right)$$
$$y \approx 2(0.693147) - 2 + 1.5 = 0.886294$$

Method 2

Operator Method: Consider the homogeneous equation

$$\frac{dy}{dx} + y = 0$$

Assuming $y = e^{mx}$, the auxiliary equation is

$$m + 1 = 0 \Rightarrow \ m = -1$$

Hence, the complementary function is

$$y_c = Ce^{-x}$$

For the particular integral,

$$y_p = \frac{1}{D+1}(2x), \ \text{where } D = \frac{d}{dx}$$
$$y_p = (1 - D + D^2 - D^3 + \cdots)(2x)$$

Since $D(2x) = 2, \ D^2(2x) = 0$,
we get

$$y_p = 2x - 2$$

Thus, the complete solution is

$$y = y_c + y_p = Ce^{-x} + 2x - 2.$$

Applying the initial condition $y(0) = 1$,

$$1 = C - 2 \Rightarrow \ C = 3$$

Hence,

$$y = 2x - 2 + 3e^{-x}.$$

At $x = \ln 2$,

$$y = 2(\ln 2) - 2 + 3e^{-(\ln 2)}$$
$$y = 2(\ln 2) - 2 + 3\left(\frac{1}{2}\right)$$
$$y \approx 2(0.693147) - 2 + 1.5 = 0.886294$$

Rounded to three decimal places,

$$\boxed{0.886}$$

Q6.19.1 | MCQ | 2019 | 1M | Ans: B | ● ○ ○

The given partial differential equation is

$$\frac{\partial^2 u}{\partial t^2} - c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 0, \qquad c \neq 0$$

- **Heat Equation**

$$\frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

  It contains a *first-order* time derivative.

- **Wave Equation**

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

  On rearranging, this is exactly same as the given equation.

- **Poisson's Equation**

$$f(x, y) = \nabla^2 u = \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

  There is no time derivative.

- **Laplace Equation**

$$0 = \nabla^2 u = \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

  This is a special case of Poisson's equation with zero source term and also contains no time derivative.

$$\boxed{\text{(B) Wave Equation}}$$

Q6.17.1 | MCQ | 2017 | 2M | Ans: A | ● ● ○

Consider the differential equation

$$(t^2 - 81)\frac{dy}{dt} + 5ty = \sin t, \quad \text{with initial condition} \;\; y(1) = 2\pi$$

To determine the interval in which a unique solution exists, we use the **standard existence and uniqueness theorem for first-order linear differential equations**.

Rewrite the equation in the standard linear form

$$\frac{dy}{dt} + \frac{5t}{t^2 - 81}y = \frac{\sin t}{t^2 - 81}$$

It is of the form

$$\frac{dy}{dt} + P(t)y = Q(t), \qquad \text{where } P(t) = \frac{5t}{t^2 - 81}, \; Q(t) = \frac{\sin t}{t^2 - 81}$$

For a unique solution to exist through the initial point $t = t_0$, both $P(t)$ and $Q(t)$ must be continuous on an open interval containing t_0.

The only points where continuity fails are the zeros of the denominator:

$$t^2 - 81 = 0 \implies t = \pm 9$$

Hence, the real line is divided into three intervals:

$$(-\infty, -9), \qquad (-9, 9), \qquad (9, \infty)$$

Since the initial condition is specified at $t_0 = 1$, the interval containing $t = 1$ is

$$(-9, 9)$$

Therefore, no solution interval can cross either of these points.
From the given options, all intervals except $(-2, 2)$ contain the singular points $t = \pm 9$.
Therefore,

$$\boxed{\text{(A) } (-2, 2)}$$

Key Points

- The required interval is the largest open interval containing the initial point that does not cross any singularity.
- Always locate the initial point first and then identify the nearest discontinuities on either side.

Q6.16.1 MCQ 2016 1M Ans: B ● ○ ○

The given differential equation is

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0, \quad \text{with the conditions} \quad y(0) = 1, \; y(1) = 3e^{-1}$$

We are required to find the value of $y(2)$.
Assume a solution of the form $y = e^{mt}$.
Substituting into the differential equation, we obtain auxiliary equation as

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0$$

Thus, the characteristic roots are repeated: $m_1 = m_2 = -1$.
For repeated roots, the complementary solution is

$$y = (C_1 + C_2 t)e^{-t}$$

Using $y(0) = 1$,

$$1 = (C_1 + 0)e^0 \implies C_1 = 1$$

Using $y(1) = 3e^{-1}$,

$$3e^{-1} = (1 + C_2)e^{-1} \implies C_2 = 2$$

Thus,

$$y = (1 + 2t)e^{-t}.$$

At $t = 2$,

$$y(2) = (1 + 4)e^{-2}$$

$$y(2) = 5e^{-2}.$$

Therefore,

$$\boxed{\text{(B) } 5e^{-2}}$$

Mistakes to be avoided

- For repeated roots, do not write the solution as $C_1 e^{-t} + C_2 e^{-t}$; the second independent solution must be te^{-t}.

Q6.16.2 MCQ 2016 1M Ans: A ● ○ ○

The differential equation is

$$y''(t) + 2y'(t) + y(t) = 0, \quad \text{with initial conditions } \; y(0) = 0, \; y'(0) = 1.$$

The output is required for $t > 0$, where $u(t)$ denotes the unit step function.

Method 1

Taking the Laplace transform of the differential equation,
Using

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0),$$
$$\mathcal{L}\{y'(t)\} = sY(s) - y(0),$$

we obtain

$$\left(s^2Y(s) - 1\right) + 2sY(s) + Y(s) = 0$$
$$(s^2 + 2s + 1)Y(s) = 1$$
$$Y(s) = \frac{1}{(s+1)^2}$$

Taking the inverse Laplace transform,

$$y(t) = te^{-t}$$

Since the solution is valid for $t > 0$,

$$y(t) = te^{-t}u(t).$$

Method 2

Assume a solution of the form $y = e^{mt}$.
Substituting into the differential equation,

$$m^2 + 2m + 1 = 0$$
$$(m+1)^2 = 0$$

Thus, the repeated root is $m = -1$.
Hence, the complementary solution is

$$y(t) = (C_1 + C_2 t)e^{-t}$$

Applying the condition $y(0) = 0$,

$$0 = (C_1 + 0)e^0 \Rightarrow \; C_1 = 0$$

Therefore,

$$y(t) = C_2 te^{-t}$$

Differentiating,

$$y'(t) = C_2\left(e^{-t} - te^{-t}\right)$$

Using $y'(0) = 1$,

$$1 = C_2(1 - 0) \Rightarrow \; C_2 = 1$$

Thus,

$$y(t) = te^{-t}.$$

For $t > 0$, this can be written as

$$y(t) = te^{-t}u(t).$$

Hence,

$$\boxed{\text{(A) } te^{-t}u(t)}$$

Key Points

- In Laplace-domain analysis,
$$\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2}\right\} = te^{-at}.$$
- For causal signals, the time-domain response is commonly expressed using the unit step function $u(t)$.

Q6.16.3 NAT 2016 2M Ans: 7.389 ● ● ○

The given differential equation is

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0 \text{ with initial conditions } y(0) = 0, \left.\frac{dy}{dx}\right|_{x=0} = 1$$

We are required to find the value of $y(1)$.

Method 1

Assume a solution of the form $y = e^{mx}$.
Substituting into the differential equation, we obtain auxiliary equation as

$$m^2 - 4m + 4 = 0$$

$$(m-2)^2 = 0$$

Thus, the characteristic roots are repeated: $m_1 = m_2 = 2$.
Hence, the complementary solution is

$$y = (C_1 + C_2x)e^{2x}$$

Applying the condition $y(0) = 0$,

$$0 = (C_1 + 0)e^0 \Rightarrow C_1 = 0$$

Therefore,

$$y = C_2x\,e^{2x}$$

Differentiating,

$$\frac{dy}{dx} = C_2\left(e^{2x} + 2xe^{2x}\right) = C_2e^{2x}(1+2x)$$

Using $\left.\frac{dy}{dx}\right|_{x=0} = 1$,

$$1 = C_2$$

Hence,

$$y = xe^{2x}.$$

At $x = 1$,

$$y(1) = e^2 \approx 7.389.$$

Method 2

Taking Laplace transform and using $y(0) = 0,\ y'(0) = 1$,
we obtain

$$[s^2Y(s) - 1] - [4sY(s)] + 4Y(s) = 0$$

$$Y(s)\left(s^2 - 4s + 4\right) = 1$$

$$Y(s) = \frac{1}{(s-2)^2}$$

Taking the inverse Laplace transform,

$$y(x) = xe^{2x}.$$

Therefore,

$$y(1) = e^2 \approx 7.389.$$

$$\boxed{7.389}$$

Q6.15.1 NAT 2015 2M Ans: -3 ● ● ○

The given differential equation is

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0 \text{ with the conditions } y(0) = 2,\ y(1) = -\frac{1-3e}{e^3}.$$

We are required to determine $\left.\frac{dy}{dt}\right|_{t=0}$.
Assume a solution of the form $y = e^{mt}$.
Substituting into the differential equation, we obtain auxiliary equation as

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

Hence, the characteristic roots are $m_1 = -2,\ m_2 = -3$.
Since the roots are real and distinct, the general solution is

$$y = C_1e^{-2t} + C_2e^{-3t}$$

Using $y(0) = 2$,

$$C_1 + C_2 = 2$$

Using $y(1) = -\frac{1-3e}{e^3} = -e^{-3} + 3e^{-2}$,

$$C_1e^{-2} + C_2e^{-3} = -e^{-3} + 3e^{-2}.$$

On comparing, $C_1 = 3,\ C_2 = -1$.
Therefore,

$$y = 3e^{-2t} - e^{-3t}.$$

Differentiating,

$$\frac{dy}{dt} = -6e^{-2t} + 3e^{-3t}$$

Hence,

$$\left.\frac{dy}{dt}\right|_{t=0} = -6 + 3 = -3.$$

Therefore,

$$\boxed{-3}$$

Key Points

- Distinct roots m_1 and m_2 give the solution $y = C_1e^{m_1t} + C_2e^{m_2t}$.

Q6.15.2 NAT | 2015 | 2M | Ans: 1.652 | ● ○ ○

The differential equation is

$$\frac{di}{dt} - 0.2\,i = 0, \quad -10 < t < 10, \quad \text{with the condition} \;\; i(4) = 10$$

We are required to find the value of $i(-5)$.

Method 1

The equation is separable:

$$\frac{di}{dt} = 0.2\,i$$
$$\frac{di}{i} = 0.2\,dt$$

Integrating both sides,

$$\int \frac{di}{i} = \int 0.2\,dt,$$
$$\ln|i| = 0.2t + C$$
$$\Rightarrow \; i = e^{0.2t} \times e^{C},$$

Using the condition $i(4) = 10$,

$$10 = e^{0.8} \times e^{C}$$
$$e^{C} = 10\,e^{-0.8}$$

Therefore,

$$i(t) = e^{0.2t} \times 10\,e^{-0.8} = 10\,e^{0.2(t-4)}.$$

At $t = -5$,

$$i(-5) = 10\,e^{0.2(-5-4)} = 10\,e^{-1.8}$$
$$i(-5) \approx 1.652.$$

Method 2

Using the exponential growth property,

$$i(t) = i(t_0)\,e^{0.2(t-t_0)}$$

Taking $t_0 = 4, \; i(t_0) = 10$

$$i(t) = 10\,e^{0.2(t-4)}.$$

Therefore,

$$i(-5) = 10\,e^{0.2(-9)} = 10\,e^{-1.8} \approx 1.652.$$

This method directly uses the known value at $t = 4$ and avoids finding the integration constant explicitly.
Thus,

$$\boxed{1.652}$$

Key Points

- This exponential growth form is frequently used in RC/RL circuits, population models, and first-order dynamic systems.

Q6.14.1 MCQ | 2014 | 1M | Ans: C | ● ○ ○

The given differential equation is

$$\frac{d^2x}{dt^2} = -9x \text{ with initial conditions } x(0) = 1, \left.\frac{dx}{dt}\right|_{t=0} = 1$$

Method 1

Assume a solution of the form $x = e^{mt}$.
Substituting into the differential equation, the auxiliary equation is

$$m^2 e^{mt} + 9e^{mt} = 0$$

Since $e^{mt} \neq 0$,

$$m^2 + 9 = 0$$

$$m = \pm 3j$$

For complex roots $m = \alpha \pm j\beta$, the solution is

$$x(t) = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$$

Here, $\alpha = 0$ and $\beta = 3$, hence

$$x(t) = C_1 \cos 3t + C_2 \sin 3t$$

Using $x(0) = 1$, $1 = C_1$
Therefore,

$$x(t) = \cos 3t + C_2 \sin 3t$$

Differentiating,

$$\frac{dx}{dt} = -3 \sin 3t + 3C_2 \cos 3t$$

Using $\left.\frac{dx}{dt}\right|_{t=0} = 1$,

$$1 = 3C_2 \Rightarrow C_2 = \frac{1}{3}$$

Hence,

$$x(t) = \cos 3t + \frac{1}{3} \sin 3t.$$

Method 2

Observe that

$$\frac{d^2x}{dt^2} = -9x$$

is the equation of a simple harmonic oscillator with angular frequency $\omega = 3$.
The standard solution is

$$x(t) = A \cos \omega t + B \sin \omega t$$

$$x(t) = A \cos 3t + B \sin 3t$$

Using the initial position, $x(0) = A = 1$.
Differentiating,

$$\frac{dx}{dt} = -3A \sin 3t + 3B \cos 3t$$

Applying $\left.\frac{dx}{dt}\right|_{t=0} = 1$,

$$3B = 1, \Rightarrow B = \frac{1}{3}$$

Therefore,

$$x(t) = \cos 3t + \frac{1}{3}\sin 3t.$$

$$\boxed{\text{(C) } \frac{1}{3}\sin 3t + \cos 3t}$$

Key Points

- The equation

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

represents simple harmonic motion (SHM).

Q6.14.2 MCQ 2014 1M Ans: C ● ○ ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$$

This is a **Cauchy–Euler (or equidimensional)** differential equation of the form

$$x^2y'' + axy' + by = 0$$

We are required to identify a function that satisfies the differential equation.
For a Cauchy–Euler equation, assume a trial solution $y = x^m$.
Substituting into the differential equation,

$$x^2\left[m(m-1)x^{m-2}\right] + x\left[mx^{m-1}\right] - x^m = 0$$

$$m(m-1)x^m + mx^m - x^m = 0$$

$$\left[m(m-1) + m - 1\right]x^m = 0$$

$$(m^2-1)x^m = 0$$

Hence, the auxiliary equation is

$$m^2 - 1 = 0 \Rightarrow \; m = \pm 1$$

The general solution is

$$y = C_1x + C_2x^{-1}$$

$$y = C_1x + \frac{C_2}{x}$$

Since $1/x$ is part of the general solution, it satisfies the differential equation.
Or verify the options by substituting in the given differential equation.

$$\boxed{\text{(C) } 1/x}$$

Key Points

- A Cauchy–Euler equation has the standard form $x^2y'' + axy' + by = 0$.

Mistakes to be avoided

- Do not use the characteristic equation $m^2 + am + b = 0$ meant for constant-coefficient differential equations.

Q6.11.1 | MCQ | 2011 | 1M | Ans: A | ● ○ ○

The given first-order differential equation is

$$\frac{dy}{dx} = e^{-3x}$$

We need to determine its general solution.
The equation is directly integrable.

$$\int dy = \int e^{-3x}\, dx$$

Therefore,

$$y = \int e^{-3x}\, dx + K$$

$$y = -\frac{1}{3}e^{-3x} + K$$

$$\boxed{\text{(A)} \; -\frac{1}{3}e^{-3x} + K}$$

Q6.10.1 | MCQ | 2010 | 2M | Ans: B | ● ○ ○

The given differential equation is

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = 0, \quad \text{with initial conditions } \; x(0) = 1, \; \left.\frac{dx}{dt}\right|_{t=0} = 0$$

We are required to determine $x(t)$.

Method 1

Assume a solution of the form $x = e^{mt}$.
Substituting into the differential equation,

$$m^2e^{mt} + 6me^{mt} + 8e^{mt} = 0$$

Since $e^{mt} \neq 0$,

$$m^2 + 6m + 8 = 0$$

$$(m + 2)(m + 4) = 0$$

Hence, the characteristic roots are $m_1 = -2, m_2 = -4$.
Since the roots are real and distinct, the general solution is

$$x(t) = C_1e^{-2t} + C_2e^{-4t}$$

Using $x(0) = 1$, $C_1 + C_2 = 1$.
Differentiating,

$$\frac{dx}{dt} = -2C_1e^{-2t} - 4C_2e^{-4t}$$

Applying $\left.\frac{dx}{dt}\right|_{t=0} = 0$,

$$-2C_1 - 4C_2 = 0 \Rightarrow C_1 + 2C_2 = 0$$

Solving the simultaneous equations,

$$C_2 = -1, \qquad C_1 = 2$$

Therefore,

$$x(t) = 2e^{-2t} - e^{-4t}.$$

Method 2

Taking Laplace transforms,

$$\mathcal{L}\left\{\frac{d^2x}{dt^2}\right\} + 6\mathcal{L}\left\{\frac{dx}{dt}\right\} + 8\mathcal{L}\{x\} = 0$$

Using $x(0) = 1,\ x'(0) = 0,$

$$(s^2X - s) + 6(sX - 1) + 8X = 0$$

$$X(s)(s^2 + 6s + 8) = s + 6$$

$$X(s) = \frac{s+6}{(s+2)(s+4)}$$

Using partial fractions,

$$\frac{s+6}{(s+2)(s+4)} = \frac{2}{s+2} - \frac{1}{s+4}$$

Taking the inverse Laplace transform,

$$x(t) = 2e^{-2t} - e^{-4t}.$$

$$\boxed{\text{(B) } x(t) = 2e^{-2t} - e^{-4t}}$$

Q6.05.1 MCQ | 2005 | 1M | Ans: A | ● ○ ○

The given first-order differential equation is

$$\dot{x}(t) = \frac{dx}{dt} = -3x(t), \quad \text{with the initial condition } x(0) = x_0$$

We are required to determine $x(t)$.

Method 1

The equation is separable. Hence, Integrating both sides,

$$\int \frac{dx}{x} = -3\int dt.$$

$$\ln|x| = -3t + C$$

$$x = e^{-3t} \times e^C$$

Applying the initial condition $x(0) = x_0 = e^C$,
Hence,

$$x(t) = x_0\, e^{-3t}.$$

Method 2

Recognize the standard exponential response

$$\frac{dx}{dt} = ax$$

The solution is

$$x(t) = x(0)e^{at}, \text{ where } a = -3$$

Therefore,

$$x(t) = x_0\, e^{-3t}$$

This is the standard natural response of a stable first-order system.

$$\boxed{\text{(A) } x(t) = x_0\, e^{-3t}}$$

Key Points

- Such equations commonly model RC circuits, RL circuits, population decay, and thermal cooling.

Q6.05.2 MCQ 2005 2M Ans: B ● ○ ○

The given differential equation is

$$\ddot{x}(t) + 3\dot{x}(t) + 2x(t) = 5$$

We are required to determine $\lim_{t\to\infty} x(t)$.

Method 1

The complete solution is the sum of the homogeneous solution and the particular solution:

$$x(t) = x_h(t) + x_p(t)$$

For the homogeneous part,

$$\ddot{x} + 3\dot{x} + 2x = 0$$

Assuming $x = e^{mt}$,

$$m^2 + 3m + 2 = 0$$

$$(m+1)(m+2) = 0$$

Hence,

$$m_1 = -1, \qquad m_2 = -2$$

Therefore,

$$x_h(t) = C_1e^{-t} + C_2e^{-2t}$$

Since the forcing function is a constant, assume a constant particular solution $x_p = A$.
Substituting into the differential equation,

$$0 + 0 + 2A = 5 \Rightarrow \ A = \frac{5}{2}$$

Thus,

$$x(t) = C_1e^{-t} + C_2e^{-2t} + \frac{5}{2}$$

As $t \to \infty$,

$$e^{-t} \to 0, \qquad e^{-2t} \to 0.$$

Therefore,

$$\lim_{t\to\infty} x(t) = \frac{5}{2}$$

Method 2

Apply the Final Value Theorem.
Taking Laplace transforms with arbitrary initial conditions,

$$(s^2 + 3s + 2)X(s) - (\text{initial-condition terms}) = \frac{5}{s}$$

The terms arising from initial conditions contain factors of s and vanish when the final value theorem is applied.

Thus, for steady-state analysis,

$$X(s) = \frac{5}{s(s^2 + 3s + 2)}$$

Using

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s),$$

we obtain

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} = \frac{5}{2}.$$

Method 3

At steady state,

$$\dot{x}(t) = 0, \qquad \ddot{x}(t) = 0$$

Substituting these conditions

$$\ddot{x} + 3\dot{x} + 2x = 5$$

$$2x_{ss} = 5$$

$$x_{ss} = \frac{5}{2}$$

Since the characteristic roots are -1, -2, the system is stable and all transient terms decay to zero. Hence,

$$\lim_{t\to\infty} x(t) = x_{ss} = \frac{5}{2}$$

Therefore,

$$\boxed{(\text{B})\ \frac{5}{2}}$$

Key Points

- For stable systems, the transient terms vanish as $t \to \infty$.
- A constant forcing function generally produces a constant steady-state response.

Mistakes to be avoided

- Do not assume the limit is zero simply because the homogeneous solution decays.
- The Final Value Theorem can be used only when all poles of $sX(s)$ lie in the left-half plane.

Q6.26.1 MCQ 2026 1M Ans: C ● ○ ○

For a differential equation to represent a linear dynamical system,

1. The dependent variable and all its derivatives must appear only to the first power $\left(\frac{dx}{dt}, \frac{d^2x}{dt^2}\right)$.
2. There should be no products among the dependent variable and its derivatives $\left(x \cdot \frac{dx}{dt}, \frac{d^2x}{dt^2} \cdot \frac{dx}{dt}\right)$.

Checking given differential equations:
(i)

$$\frac{d^2x}{dt^2} + 2t\frac{dx}{dt} + x(t) = u(t)$$

Here, $x(t)$ and its derivatives appear linearly. The coefficient $2t$ is a function of the independent variable t.
Hence, (i) is **linear**.
(ii)

$$\left(\frac{dx}{dt}\right)^2 + 2\frac{dx}{dt} + x(t) = u(t)$$

The term $\left(\frac{dx}{dt}\right)^2$ contains the derivative raised to the second power, violating the linearity condition.
Hence, (ii) is **nonlinear**.
(iii)

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x(t) = u(t)$$

Here, $x(t)$ and its derivatives appear linearly. The coefficient 2 is a constant.
Hence, (iii) is **linear**.
Therefore, only equation (ii) is not representative of a linear dynamical system.

$$\boxed{\text{(C) (ii) only}}$$

Q6.26.2 MSQ 2026 2M Ans: B ● ○ ○

Given the differential equation

$$\frac{d^2x}{dt^2} + x = 0, \qquad x(0) \neq 0$$

This equation represents 'Simple Harmonic Oscillator' with $\omega = 1$. Hence, the general solution is

$$x(t) = A\cos t + B\sin t,$$

where A and B are constants determined by the initial conditions.
Differentiating,

$$\frac{dx}{dt} = -A\sin t + B\cos t$$

Now evaluate the quantity appearing in the options:

$$\begin{aligned}
|x(t)|^2 + \left|\frac{dx}{dt}\right|^2 &= (A\cos t + B\sin t)^2 + (-A\sin t + B\cos t)^2 \\
&= A^2(\cos^2 t + \sin^2 t) + B^2(\sin^2 t + \cos^2 t) \\
&\quad + 2AB\sin t\cos t - 2AB\sin t\cos t \\
&= A^2 + B^2.
\end{aligned}$$

Thus,

$$|x(t)|^2 + \left|\frac{dx}{dt}\right|^2 = A^2 + B^2 = c,$$

which is a constant for all $t \geq 0$.
Since $x(0) = A \neq 0$, we have

$$A^2 + B^2 > 0$$

Therefore, the constant satisfies $c > 0$.
Hence,

$$|x(t)|^2 + \left|\frac{dx}{dt}\right|^2 = c, \ \forall t \geq 0, \ \text{for some real constant } c > 0$$

$$\boxed{\text{(B) } |x(t)|^2 + \left|\frac{dx}{dt}\right|^2 = c, \text{ for all } t \geq 0, \text{ for some real constant } c > 0}$$

Mistakes to be avoided

- Do not assume that B must be nonzero; only $A = x(0) \neq 0$ is guaranteed.

Q6.25.1 MCQ 2025 1M Ans: A ● ○ ○

Given the differential equation

$$\frac{dy}{dx} = \frac{9x}{y}$$

We separate the variables and Integrate:

$$\int y\,dy = \int 9x\,dx$$

$$\frac{y^2}{2} = \frac{9x^2}{2} + C$$

$$y^2 = 9x^2 + C_1, \ \text{where } C_1 \text{ is an arbitrary constant.}$$

$$y^2 - 9x^2 = C_1$$

For $C_1 \neq 0$, dividing by C_1 gives

$$\frac{y^2}{C_1} - \frac{x^2}{C_1/9} = 1$$

This is the standard form

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

which represents a hyperbola.

$$\boxed{\text{(A) a hyperbola}}$$

Mistakes to be avoided

- Do not forget the constant of integration after integrating.
- A sum of squared terms indicates an ellipse, whereas a difference of squared terms indicates a hyperbola.

Q6.24.1 | NAT | 2024 | 2M | Ans: 33 | ● ● ○

Let

$$y(t) = e^{-t}P(t), \quad \text{where } P(t) = c_0 + c_1 t - c_2 t^2 + c_3 t^3$$

Given

$$y''' + 3y'' + 3y' + y = (D+1)^3 y = 30e^{-t}$$

$$(D+1)^3 y = e^{-t}P'''(t) = 30e^{-t} \qquad \left(\because\ (D+1)\left(e^{-t}P(t)\right) = e^{-t}P'(t)\right)$$

$$P'''(t) = 30$$

$$6c_3 = 30 \Rightarrow c_3 = 5$$

Using the initial conditions:

$$y(0) = P(0) = c_0 = 3$$

Also,

$$y'(t) = e^{-t}\left(P'(t) - P(t)\right)$$

At $t = 0$,

$$y'(0) = c_1 - c_0 = -3 \Rightarrow \ c_1 = 0$$

Further,

$$y''(t) = e^{-t}\left(P''(t) - 2P'(t) + P(t)\right)$$

At $t = 0$,

$$y''(0) = P''(0) - 2P'(0) + P(0)$$

Substituting $P''(0) = -2c_2$, $P'(0) = c_1 = 0$, and $P(0) = c_0 = 3$,

$$-2c_2 + 3 = -47 \Rightarrow \ c_2 = 25$$

Therefore,

$$c_0 + c_1 + c_2 + c_3 = 3 + 0 + 25 + 5 = 33$$

$$\boxed{33}$$

Q6.23.1 | MCQ | 2023 | 1M | Ans: C | ● ○ ○

Given

$$\ddot{x} = -k\dot{x}, \quad k > 0 \ \text{ with initial conditions } \ x(0) = 1,\ \dot{x}(0) = 0$$

Method 1 (Characteristic Equation)

Assume a solution of the form $x(t) = e^{rt}$.
Substituting into the differential equation, the auxiliary equation is

$$r(r+k) = 0.$$

$$r = 0,\ -k$$

Therefore,

$$x(t) = C_1 + C_2 e^{-kt}$$

Applying the initial conditions, $x(0) = C_1 + C_2 = 1$.
Differentiating,

$$\dot{x}(t) = -k\, C_2 e^{-kt}$$

Using $\dot{x}(0) = 0$,

$$-k\, C_2 = 0 \Rightarrow C_2 = 0$$

$$\Rightarrow \ C_1 = 1$$

Hence,

$$x(t) = 1.$$

Method 2 (Laplace Transform)

Taking Laplace transform,

$$s^2 X(s) - sx(0) - \dot{x}(0) = -k[sX(s) - x(0)]$$

Substituting $x(0) = 1$, $\dot{x}(0) = 0$,

$$s^2 X(s) - s = -k[sX(s) - 1]$$

$$(s^2 + ks)X(s) = s + k$$

$$X(s) = \frac{s+k}{s(s+k)} = \frac{1}{s}$$

Taking inverse Laplace transform,

$$x(t) = 1$$

$$\boxed{\text{(C) } x(t) = 1}$$

Key Points

- The root $r = 0$ corresponds to a constant solution.
- Physically, the system starts at rest and remains at the equilibrium position $x = 1$.

Q6.22.1 MCQ | 2022 | 2M | Ans: C | ● ○ ○

Given the differential equation

$$\frac{dy}{dx} + y\ln(y) = 0, \text{ with the condition } \; y(0) = e$$

we need to determine $y(1)$.

Method 1 (Variable Transformation)

Let $z = \ln(y)$.
Then

$$\frac{dz}{dx} = \frac{1}{y}\frac{dy}{dx}$$

Using the given differential equation,

$$\frac{dy}{dx} = -y\ln(y) = -yz$$

Hence,

$$\frac{dz}{dx} = -z$$

Separating variables and Integrating,

$$\int \frac{dz}{z} = \int -dx$$

$$\ln|z| = -x + C$$

$$z = Ce^{-x}$$

Since $z = \ln(y)$,

$$\ln(y) = Ce^{-x}$$

Using $y(0) = e$,

$$\ln(e) = C = 1$$

Thus,

$$\ln(y) = e^{-x}$$
$$y = e^{e^{-x}}$$

At $x = 1$,

$$y(1) = e^{e^{-1}}$$

Method 2 (Recognizing a Bernoulli-Type Structure)

Dividing the equation by y,

$$\frac{1}{y}\frac{dy}{dx} + \ln(y) = 0$$

Since

$$\frac{d}{dx}[\ln(y)] = \frac{1}{y}\frac{dy}{dx},$$

let $u = \ln(y)$

$$\frac{du}{dx} + u = 0$$
$$u = Ce^{-x}$$

Applying $u(0) = \ln(e) = 1$,

$$u = e^{-x}$$

Therefore,

$$y = e^{u} = e^{e^{-x}}.$$

Thus,

$$y(1) = e^{e^{-1}}.$$

$$\boxed{\text{(C) } e^{(1/e)}}$$

Key Points

- The substitution converts the nonlinear differential equation into the first-order linear equation $\frac{du}{dx} + u = 0$.

Q6.20.1 | MCQ | 2020 | 2M | Ans: A | ● ○ ○

Given the differential equation

$$\frac{dx}{dt} = \sin(x), \quad \text{with the initial condition} \quad x(0) = 0$$

Separating variables and Integrating,

$$\int \frac{dx}{\sin(x)} = \int dt$$
$$\int \csc(x)\, dx = \int dt$$
$$\ln\left|\tan\frac{x}{2}\right| = t + C$$

Therefore,

$$\tan\frac{x}{2} = Ke^{t}$$

Applying the initial condition $x(0) = 0$,

$$\tan\frac{0}{2} = K = 0$$

Hence,

$$\tan\frac{x}{2} = 0$$

$$\Rightarrow\ x(t) = 0$$

$$\boxed{\text{(A) } x(t) = 0}$$

Key Points

- Constant solutions of a differential equation are called equilibrium solutions.
- For $\frac{dx}{dt} = \sin(x)$, all equilibrium points satisfy $x = n\pi$.
- By the uniqueness theorem for first-order ODEs, the solution passing through $x(0) = 0$ remains at the equilibrium point $x = 0$.

Q6.18.1 MCQ 2018 2M Ans: D ● ● ○

Given

$$\frac{\partial V(x,y)}{\partial x} = px^2 + y^2 + 2xy \tag{1}$$

$$\frac{\partial V(x,y)}{\partial y} = x^2 + qy^2 + 2xy \tag{2}$$

We need to determine $V(x, y)$.
Integrating (1) with respect to x,

$$V(x,y) = \int \left(px^2 + y^2 + 2xy\right)\, dx$$

$$V(x,y) = \frac{p}{3}x^3 + xy^2 + x^2y + c(y)$$

where $c(y)$ is an arbitrary function of y.
Differentiating with respect to y,

$$\frac{\partial V}{\partial y} = 0 + 2xy + x^2 + \frac{dc}{dy}$$

Comparing with (2),

$$2xy + x^2 + \frac{dc}{dy} = x^2 + qy^2 + 2xy$$

$$\frac{dc}{dy} = qy^2$$

$$c(y) = \frac{q}{3}y^3 + C$$

Substituting back,

$$V(x,y) = \frac{p}{3}x^3 + \frac{q}{3}y^3 + x^2y + xy^2 + C$$

Ignoring the additive constant (which does not affect partial derivatives),

$$V(x,y) = \frac{p}{3}x^3 + \frac{q}{3}y^3 + x^2y + xy^2$$

$$\boxed{\text{(D) } p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2}$$

Key Points

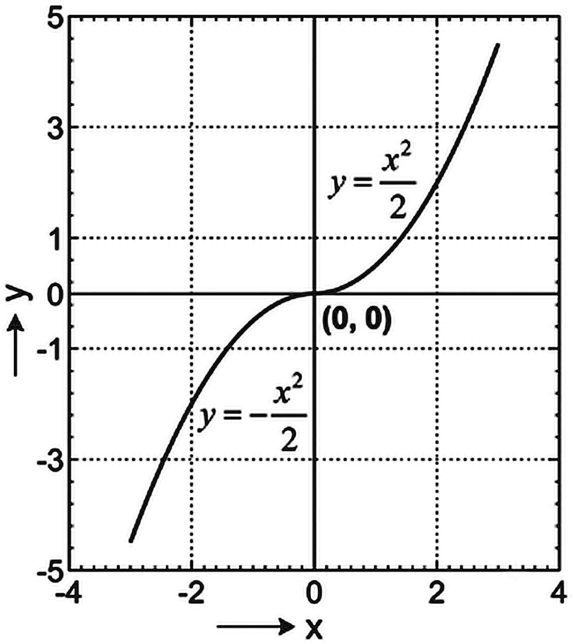
- When integrating a partial derivative with respect to one variable, the "constant of integration" becomes a function of the other variable.
- To determine this unknown function, differentiate the obtained expression with respect to the other variable and compare with the given partial derivative.
- Any additive constant in a potential function disappears after differentiation and is therefore irrelevant.

Mistakes to be avoided

- Do not use a numerical constant after partial integration; write $c(y)$ because y is treated as a constant during integration with respect to x.

Q6.14.1 MCQ 2014 1M Ans: D ● ○ ○

The solution must be inferred from the given graph.



From the graph, the curve is represented by

$$y = \begin{cases} \frac{x^2}{2}, & x \geq 0, \\ -\frac{x^2}{2}, & x < 0. \end{cases}$$

Method 1 (Piecewise Differentiation)

Differentiating each branch,

$$\frac{dy}{dx} = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

Since

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases}$$

we obtain

$$\frac{dy}{dx} = |x|.$$

Method 2 (Graph Interpretation)

Observe that the slope of the curve is always non-negative.
For $x > 0$,

$$y = \frac{x^2}{2} \quad \Rightarrow \quad \frac{dy}{dx} = x$$

For $x < 0$,

$$y = -\frac{x^2}{2} \quad \Rightarrow \quad \frac{dy}{dx} = -x$$

Combining both regions,

$$\frac{dy}{dx} = |x|.$$

$$\boxed{\text{(D) } \frac{dy}{dx} = |x|}$$

Key Points

- Differentiating piecewise yields the absolute value function.

Mistakes to be avoided

- Do not choose $\frac{dy}{dx} = x$ by looking only at the right half of the graph.

Q6.13.1 MCQ 2013 1M Ans: A ● ● ○

Given the partial differential equation

$$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$$

This can be rewritten as

$$\frac{\partial^2 f}{\partial x^2} - \frac{\partial f}{\partial t} = 0$$

Method 1 (Discriminant Test)

To classify a second-order linear PDE, we compare it with the general form

$$A\frac{\partial^2 f}{\partial x^2} + B\frac{\partial^2 f}{\partial x \partial t} + C\frac{\partial^2 f}{\partial t^2} + D\frac{\partial f}{\partial x} + E\frac{\partial f}{\partial t} + Ff = G$$

For the given equation, $A = 1,\ B = 0,\ C = 0$.
The classification depends on the discriminant

$$\Delta = B^2 - 4AC$$

$$\Delta = 0^2 - 4(1)(0) = 0$$

For a second-order PDE,

$$\begin{cases} \Delta > 0 & \text{Hyperbolic} \\ \Delta = 0 & \text{Parabolic} \\ \Delta < 0 & \text{Elliptic} \end{cases}$$

Since $\Delta = 0$, the PDE is **parabolic**.

Method 2 (Recognizing Standard PDE Forms)

The equation

$$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$$

is the one-dimensional *heat (diffusion) equation*.
Since the given PDE is the heat equation, it is **parabolic.**

$$\boxed{\text{(A) Parabolic}}$$

Key Points

Standard classifications of common PDEs are:

$$\text{Heat Equation} \implies \text{Parabolic}$$

$$\text{Wave Equation} \implies \text{Hyperbolic}$$

$$\text{Laplace Equation} \implies \text{Elliptic}$$

Q6.13.2 MCQ 2013 2M Ans: D ● ● ○

Given

$$y(t) + \ddot{y}(t) = 0, \quad \text{with initial conditions} \;\; y(0) = 1, \;\dot{y}(0) = 1$$

We are required to find the maximum value of $y(t)$ for $t \geq 0$.
The corresponding auxiliary equation is

$$m^2 + 1 = 0$$

$$m = \pm i$$

Therefore, the general solution is

$$y(t) = C_1 \cos t + C_2 \sin t$$

Differentiating,

$$\dot{y}(t) = -C_1 \sin t + C_2 \cos t$$

Using $y(0) = 1, C_1 = 1$,
and using $\dot{y}(0) = 1, C_2 = 1$.
Thus,

$$y(t) = \cos t + \sin t$$

Method 1 (Differential Equation Approach)

To locate extrema, setting $\dot{y}(t) = 0$,

$$\dot{y}(t) = -\sin t + \cos t = 0$$

$$\sin t = \cos t$$

$$\tan t = 1$$

For $t \geq 0$, the first stationary point occurs at

$$t = \frac{\pi}{4}$$

Now,

$$\ddot{y}(t) = -\cos t - \sin t$$

Evaluating at $t = \frac{\pi}{4}$,

$$\ddot{y}\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2} < 0$$

Hence, $t = \frac{\pi}{4}$ corresponds to a maximum.
Therefore,

$$y_{\max} = y\left(\frac{\pi}{4}\right) = \cos\frac{\pi}{4} + \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

Method 2 (Trigonometric Identity)

Using the identity

$$\sin t + \cos t = \sqrt{2}\sin\left(t + \frac{\pi}{4}\right),$$

we obtain

$$y(t) = \sqrt{2}\sin\left(t + \frac{\pi}{4}\right)$$

Since

$$-1 \le \sin\left(t + \frac{\pi}{4}\right) \le 1,$$

the maximum possible value of $y(t)$ is

$$y_{\max} = \sqrt{2}$$

$$\boxed{\text{(D) } \sqrt{2}}$$

Key Points

- A stationary point occurs when $\dot{y}(t) = 0$.
- The second derivative test determines whether a stationary point is a maximum or minimum.
- The identity
$$a\sin t + b\cos t = \sqrt{a^2 + b^2}\sin(t + \phi)$$
provides a quick way to find the maximum value of a sinusoidal expression.

Q6.11.1 MCQ 2011 2M Ans: C ● ○ ○

Given the differential equation

$$\ddot{y} + 2\dot{y} + y = 0 \text{ with boundary conditions } y(0) = 1,\ y(1) = 0$$

We are required to determine $y(2)$.
The auxiliary equation is

$$m^2 + 2m + 1 = 0$$
$$(m + 1)^2 = 0$$

Thus, the repeated root is $m = -1$.
For repeated roots, the general solution is

$$y(t) = (C_1 + C_2 t)\, e^{-t}$$

Applying the first boundary condition, $y(0) = 1$,

$$(C_1 + 0)e^0 = 1 \Rightarrow C_1 = 1$$

Using the second boundary condition, $y(1) = 0$,

$$(1 + C_2)e^{-1} = 0$$
$$1 + C_2 = 0, \qquad (\because e^{-1} \neq 0)$$
$$C_2 = -1$$

Therefore,

$$y(t) = (1 - t)e^{-t}.$$

Evaluating at $t = 2$,

$$y(2) = (1 - 2)e^{-2} = -e^{-2}.$$

$$\boxed{\text{(C) } -e^{-2}}$$

Q6.10.1 MCQ 2010 2M Ans: C ● ● ○

Given the differential equation

$$\frac{dy}{dx} + y = e^x, \quad \text{with the initial condition } \ y(0) = 1$$

We are required to determine $y(1)$.

Method 1 (Integrating Factor Method)

The given equation is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } \ P(x) = 1, \ Q(x) = e^x$$

The integrating factor is

$$\text{I.F.} = e^{\int P(x)\,dx} = e^x$$

The complete solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.} + C$$

$$y\,e^x = \int e^{2x}\,dx + C = \frac{e^{2x}}{2} + C$$

Therefore,

$$y = \frac{e^x}{2} + Ce^{-x}$$

Using the initial condition $y(0) = 1$,

$$1 = \frac{1}{2} + C \Rightarrow \ C = \frac{1}{2}$$

Hence,

$$y = \frac{e^x + e^{-x}}{2}$$

At $x = 1$,

$$y(1) = \frac{e + e^{-1}}{2}$$

Method 2 (Complementary Function + Particular Integral)

The complementary function is obtained from $(D + 1)y = 0$,

$$m + 1 = 0 \Rightarrow m = -1$$

Hence,

$$\text{C.F.} = Ce^{-x}$$

For the particular integral,

$$\text{P.I.} = \frac{1}{D + 1}e^x$$

Using the operator rule

$$\frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax},$$

we get

$$\text{P.I.} = \frac{1}{1+1}e^x = \frac{e^x}{2}$$

Thus,

$$y = Ce^{-x} + \frac{e^x}{2}.$$

Applying $y(0) = 1$,

$$C + \frac{1}{2} = 1 \Rightarrow C = \frac{1}{2}$$

Therefore,

$$y = \frac{e^{-x} + e^x}{2}$$

At $x = 1$,

$$y(1) = \frac{e + e^{-1}}{2}.$$

$$\boxed{\text{(C) } \frac{1}{2}\left(e + e^{-1}\right)}$$

Q6.08.1 MCQ 2008 2M Ans: A ● ○ ○

Given the differential equation

$$\frac{dy}{dx} = 1 + y^2$$

We are required to identify a particular solution from the given options.
Separating variables and Integrating,

$$\int \frac{dy}{1+y^2} = \int dx$$

Using

$$\int \frac{dy}{1+y^2} = \tan^{-1}(y),$$

we obtain

$$\tan^{-1}(y) = x + C$$

$$y = \tan(x + C)$$

Since no initial condition is provided, C remains arbitrary.
Among the given options,

$$y = \tan(x+3)$$

is of the required form and therefore is a valid particular solution.

$$\boxed{\text{(A) } y = \tan(x+3)}$$

Mistakes to be avoided

- Do not write the solution as $y = \tan(x) + C$; the constant appears inside the argument of the tangent function.

Q6.07.1 MCQ | 2007 | 2M | Ans: C | ● ● ○

Given the boundary value problem

$$y'' + \lambda y = 0,\ \text{with boundary conditions}\ \ y(0) = 0,\ y(\pi) = 0$$

We need to determine the values of λ for which non-zero solutions exist.

Method 1 (Characteristic Equation Method)

Assume a solution of the form $y = e^{mx}$
Substituting into the differential equation,

$$m^2 + \lambda = 0$$

For non-trivial solutions, we analyze the possible values of λ.
Case 1: $\lambda = 0$
The differential equation becomes

$$y'' = 0$$

Integrating twice,

$$y = C_1 + C_2 x$$

Applying $y(0) = 0,\ C_1 = 0$
and $y(\pi) = 0$. $C_2 = 0$.
Only the trivial solution exists.

Case 2: $\lambda < 0$
Let $\lambda = -\mu^2,\ \mu > 0$.

$$y'' - \mu^2 y = 0$$

The general solution is

$$y = C_1 e^{\mu x} + C_2 e^{-\mu x}$$

Applying the boundary conditions leads only to $C_1 = C_2 = 0$.
Hence no non-zero solution exists.

Case 3: $\lambda > 0$
Let $\lambda = \alpha^2$.
The roots are $m = \pm i\alpha$.
Therefore,

$$y = C_1 \cos(\alpha x) + C_2 \sin(\alpha x)$$

Using $y(0) = 0,\ C_1 = 0$.
Applying $y(\pi) = 0$, we obtain

$$C_2 \sin(\alpha\pi) = 0$$

For a non-zero solution, $C_2 \neq 0$,
therefore

$$\sin(\alpha\pi) = 0$$

Thus,

$$\alpha = n, \qquad n = 1, 2, 3, \ldots$$

and hence $\lambda = n^2$.
Therefore,

$$\lambda = 1, 4, 9, \ldots$$

Method 2 (Eigenvalue Interpretation)

The problem is a standard Sturm–Liouville eigenvalue problem.
For boundary conditions of the form $y(0) = 0,\ y(L) = 0$, the eigenvalues are

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad n = 1, 2, 3, \ldots$$

Here, $L = \pi$.
Therefore,

$$\lambda_n = \left(\frac{n\pi}{\pi}\right)^2 = n^2$$

Hence,

$$\lambda = 1, 4, 9, \ldots$$

$$\boxed{\text{(C) } 1, 4, 9, \ldots}$$

Key Points

- A boundary value problem admits a non-trivial solution only for specific values of the parameter λ, called eigenvalues.
- For $\lambda > 0$, the solutions are sinusoidal and can satisfy both boundary conditions simultaneously.
- For $\lambda \leq 0$, only the trivial solution satisfies the given boundary conditions.

Mistakes to be avoided

- Do not include $n = 0$; it corresponds to the trivial solution.

Q6.06.1 MSQ 2006 2M Ans: A ● ● ○

Given the differential equation

$$\ddot{y} + 2\dot{y} + 101y = 10.4e^x, \text{ with initial conditions } y(0) = 1.1, \; \dot{y}(0) = -0.9$$

We need to match:

P	General solution of homogeneous equation
Q	Particular integral
R	Total solution satisfying boundary conditions

Finding the Complementary Function:
The auxiliary equation is

$$m^2 + 2m + 101 = 0$$

$$m = \frac{-2 \pm \sqrt{4 - 404}}{2} = -1 \pm 10i.$$

Since the roots are complex conjugates, the complementary function is

$$\text{C.F.} = e^{-x}\left(C_1 \cos 10x + C_2 \sin 10x\right).$$

Comparing with Group 2,

$$P \longrightarrow 2$$

Finding the Particular Integral:
Using the operator method,

$$\text{P.I.} = \frac{1}{D^2 + 2D + 101}\left(10.4\, e^x\right)$$

For forcing functions of the form e^{ax},

$$\frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}$$

Therefore,

$$\text{P.I.} = 10.4 \frac{1}{1^2 + 2(1) + 101} e^x$$

$$\text{P.I.} = 10.4\,\frac{1}{104}e^{x}$$
$$\text{P.I.} = 0.1e^{x}.$$

Comparing with Group 2,

$$Q \longrightarrow 1$$

Finding the Complete Solution
Applying the initial conditions, the complete solution is

$$y = e^{-x}\left(C_1 \cos 10x + C_2 \sin 10x\right) + 0.1e^{x}$$

Using $y(0) = 1.1$,

$$1.1 = C_1 + 0.1 \Rightarrow\; C_1 = 1$$

Differentiating,

$$\dot{y} = e^{-x}\Big[(-C_1 + 10C_2)\cos 10x + (-C_2 - 10C_1)\sin 10x\Big] + 0.1e^{x}$$

Applying $\dot{y}(0) = -0.9$,

$$-0.9 = (-C_1 + 10C_2) + 0.1$$
$$-0.9 = -1 + 10C_2 + 0.1$$
$$10C_2 = 0 \Rightarrow\; C_2 = 0$$

Hence,

$$y = e^{-x}\cos 10x + 0.1e^{x}.$$

Comparing with Group 2,

$$R \longrightarrow 3$$

Therefore, the correct matching is

$$P-2, \qquad Q-1, \qquad R-3$$

$$\boxed{\text{(A) P–2, Q–1, R–3}}$$

Key Points

- Complex roots $a \pm bi$ produce solutions of the form $e^{ax}(C_1 \cos bx + C_2 \sin bx)$.
- For a forcing term e^{ax},

$$\text{P.I.} = \frac{1}{f(a)}e^{ax}$$

Q6.05.1 MCQ 2005 2M Ans: C ● ○ ○

Given the differential equation

$$(D^2 - 4D + 4)y = 0$$

or equivalently,

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

We need to determine the general solution.
The auxiliary equation is

$$m^2 - 4m + 4 = 0$$
$$(m-2)^2 = 0$$

Thus, the characteristic root is repeated: $m_1 = m_2 = 2$.

For a repeated root $m = 2$, the general solution is

$$y = (C_1 + C_2x)e^{2x}$$

$$y = C_1e^{2x} + C_2xe^{2x}.$$

$$\boxed{\text{(C) } C_1e^{2x} + C_2xe^{2x}}$$

Q6.26.1 | MCQ | 2026 | 1M | Ans: A | ● ○ ○

The given differential equation is

$$\frac{\partial^3 \varphi}{\partial x^3} + \frac{\partial^2 \varphi}{\partial y^2}\frac{\partial \varphi}{\partial x} + \left(\frac{\partial^2 \varphi}{\partial x^2}\right)^2 + \frac{\partial \varphi}{\partial y} = 0$$

To determine the **order**, identify the highest-order derivative present in the equation.
The derivatives appearing are:

$$\frac{\partial^3 \varphi}{\partial x^3}, \qquad \frac{\partial^2 \varphi}{\partial y^2}, \qquad \frac{\partial \varphi}{\partial x}, \qquad \frac{\partial^2 \varphi}{\partial x^2}, \qquad \frac{\partial \varphi}{\partial y}$$

The highest-order derivative is $\frac{\partial^3 \varphi}{\partial x^3}$.
Hence,

$$m = \text{Order} = 3$$

To determine the **degree**, the differential equation must be polynomial in its derivatives.
The highest-order derivative $\frac{\partial^3 y}{\partial x^3}$ appears to the first power and is not under any radical, fractional power, or transcendental function.
Therefore,

$$n = \text{Degree} = 1$$

Thus,

$$m - n = 3 - 1 = 2$$

$$\boxed{\text{(A) } 2}$$

Key Points

- **Order** of a differential equation is the order of the highest-order derivative present.
- **Degree** is the exponent of the highest-order derivative after expressing the equation as a polynomial in derivatives.
- Products or powers of lower-order derivatives do not affect the order.

Q6.26.2 | MCQ | 2026 | 2M | Ans: A | ● ● ○

The given partial differential equation is

$$\frac{\partial y}{\partial x} = 3\frac{\partial y}{\partial t} + y \quad \text{with the condition } y(x, 0) = 10e^{-2x}$$

Rearranging,

$$\frac{\partial y}{\partial x} - 3\frac{\partial y}{\partial t} - y = 0$$

Assume a separable solution

$$y(x, t) = X(x)T(t)$$

Then

$$\frac{\partial y}{\partial x} = X'T, \qquad \frac{\partial y}{\partial t} = XT'$$

Substituting into the PDE,

$$X'T - 3XT' - XT = 0$$

$$\frac{X'}{X} - 3\frac{T'}{T} - 1 = 0$$

$$\frac{X'}{X} = 1 + 3\frac{T'}{T} = K, \quad \text{where } K \text{ is a separation constant.}$$

From $\dfrac{X'}{X} = K$, we obtain

$$X = c_1 e^{Kx}$$

Also, $1 + 3\dfrac{T'}{T} = K$ gives

$$3\frac{T'}{T} = K - 1$$

$$\frac{T'}{T} = \frac{K-1}{3}$$

Integrating,

$$T = c_2 e^{\frac{K-1}{3}t}$$

Therefore,

$$y(x,t) = Ce^{Kx + \frac{K-1}{3}t} \quad \text{where } C = c_1 c_2.$$

Applying the initial condition $y(x,0) = 10e^{-2x}$,

$$Ce^{Kx} = 10e^{-2x}$$

$$\Rightarrow \; C = 10, \; K = -2$$

Substituting these values,

$$y(x,t) = 10e^{-2x + \frac{-2-1}{3}t} = 10e^{-2x-t}$$

Hence,

$$\boxed{\text{(A) } y(x,t) = 10\, e^{-2x-t}}$$

Key Points

- For linear homogeneous PDEs with constant coefficients, separation of variables is often an effective technique.
- Assume $y(x,t) = X(x)T(t)$ and reduce the PDE to two ordinary differential equations.

Q6.25.1 MCQ 2025 1M Ans: B ● ○ ○

The given options are to be checked against the two-dimensional Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

A function satisfying the above equation is called a *harmonic function*.

Method 1

Direct substitution into the Laplace equation.
Option (A): $u = e^{x+y}$

$$\frac{\partial^2 u}{\partial x^2} = e^{x+y}, \qquad \frac{\partial^2 u}{\partial y^2} = e^{x+y}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2e^{x+y} \neq 0.$$

Hence, option (A) is incorrect.

Option (B): $u = e^x \sin y$

$$\frac{\partial u}{\partial x} = e^x \sin y, \qquad \frac{\partial u}{\partial y} = e^x \cos y$$

$$\frac{\partial^2 u}{\partial x^2} = e^x \sin y, \qquad \frac{\partial^2 u}{\partial y^2} = -e^x \sin y$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \sin y - e^x \sin y = 0.$$

Thus, $u = e^x \sin y$ satisfies Laplace's equation.

Option (C): $u = \sin x \sin y$

$$\frac{\partial^2 u}{\partial x^2} = -\sin x \sin y, \qquad \frac{\partial^2 u}{\partial y^2} = -\sin x \sin y$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -2\sin x \sin y \neq 0.$$

Hence, option (C) is incorrect.

Option (D): $u = \cos x \cos y$

$$\frac{\partial^2 u}{\partial x^2} = -\cos x \cos y, \qquad \frac{\partial^2 u}{\partial y^2} = -\cos x \cos y$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -2\cos x \cos y \neq 0.$$

Hence, option (D) is incorrect.

Method 2

Use the standard harmonic-function result: $u = e^{ax}\sin(ay)$ always satisfies Laplace's equation because

$$\frac{\partial^2 u}{\partial x^2} = a^2 e^{ax}\sin(ay), \qquad \frac{\partial^2 u}{\partial y^2} = -a^2 e^{ax}\sin(ay)$$

Thus,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Option (B) corresponds to the special case $a = 1$.

$$\boxed{\text{(B) } u = e^x \sin y \text{ is a solution for all } x \text{ and } y}$$

Q6.25.2 NAT | 2025 | 2M | Ans: 0.50 | ● ● ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0 \text{ with the initial conditions } y(1) = 0, \left.\frac{dy}{dx}\right|_{x=1} = 1$$

This is a **Cauchy–Euler differential equation**. Using the standard transformation

$$x = e^z \qquad (z = \ln x),$$

Let $D = \dfrac{d}{dz}$. Then,

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

Substituting into the differential equation,

$$D(D-1)y + 3Dy + y = 0$$

$$(D^2 - D + 3D + 1)y = 0$$

$$(D^2 + 2D + 1)y = 0$$

$$(D+1)^2y = 0$$

The auxiliary equation is

$$(m+1)^2 = 0$$

Hence, the repeated root is $m = -1$.
Therefore, the complementary solution is

$$y = (C_1 + C_2z)e^{-z}$$

Since $z = \ln x$ and $e^{-z} = \frac{1}{x}$,

$$y = \frac{C_1 + C_2 \ln x}{x}$$

Applying the first initial condition $y(1) = 0$,

$$0 = \frac{C_1 + C_2 \ln 1}{1}$$

$$\Rightarrow \; C_1 = 0$$

Thus,

$$y = \frac{C_2 \ln x}{x}$$

Differentiating,

$$\frac{dy}{dx} = C_2 \frac{d}{dx}\left(\frac{\ln x}{x}\right) = C_2\left(\frac{1-\ln x}{x^2}\right)$$

Using $\left.\frac{dy}{dx}\right|_{x=1} = 1$, we get

$$1 = C_2\left(\frac{1-\ln 1}{1^2}\right)$$

$$\Rightarrow \; C_2 = 1$$

Hence,

$$y = \frac{\ln x}{x}$$

Given that $y(2) = A \ln 2$.
Substituting $x = 2$,

$$y(2) = \frac{\ln 2}{2}$$

Comparing with $A \ln 2$,

$$A = \frac{1}{2}$$

Rounding off to two decimal places,

$$\boxed{0.50}$$

Key Points

- A Cauchy–Euler equation of the form $ax^2y'' + bxy' + cy = 0$ can be converted into a constant-coefficient differential equation using $x = e^z$.
- The substitutions

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

are standard results for Euler equations.

Q6.24.1 NAT | 2024 | 2M | Ans: -1.38 | ● ● ○

The given differential equation is

$$t\frac{dx}{dt} + (t - x) = 0, \quad \text{with the initial condition } x(1) = 0$$

Rearranging the equation,

$$t\frac{dx}{dt} - x = -t$$

$$\frac{dx}{dt} - \frac{x}{t} = -1$$

This is a first-order linear differential equation of the form

$$\frac{dx}{dt} + P(t)x = Q(t), \text{ where } P(t) = -\frac{1}{t}, \ Q(t) = -1$$

$$\text{I.F.} = e^{\int P(t)\,dt} = e^{\int -dt/t} = e^{-\ln t} = \frac{1}{t}.$$

The solution is given by

$$x \times \text{I.F.} = \int Q \times \text{I.F.}\, dt + C$$

$$x \times \frac{1}{t} = \int (-1) \times \frac{1}{t}\, dt + C$$

$$\frac{x}{t} = -\ln t + C$$

Hence,

$$x = t(C - \ln t)$$

Using the condition $x(1) = 0$,

$$0 = 1\big(C - \ln 1\big) \Rightarrow C = 0$$

Therefore,

$$x = -t\ln t.$$

Substituting $t = 2$,

$$x(2) = -2\ln 2 \approx -2(0.6931) = -1.3862$$

Rounded to two decimal places,

$$\boxed{-1.38}$$

Q6.23.1 NAT | 2023 | 2M | Ans: 9 | ● ● ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0, \quad x \geq 1 \text{ with the initial conditions } y(1) = 6, \ \left.\frac{dy}{dx}\right|_{x=1} = 2$$

This is a **Cauchy–Euler differential equation**. Using the standard substitution

$$x = e^t, \qquad t = \ln x$$

and defining $D = \frac{d}{dt}$, we have

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

Substituting into the given equation,

$$D(D-1)y + Dy - y = 0$$

$$(D^2 - D + D - 1)y = 0$$

$$(D^2 - 1)y = 0$$

Hence, the auxiliary equation is

$$m^2 - 1 = 0$$

Therefore, the distinct roots are $m = \pm 1$.
The complementary solution is

$$y = C_1 e^{-t} + C_2 e^{t}$$

Since $t = \ln x$,

$$e^{-t} = \frac{1}{x}, \qquad e^{t} = x$$

Thus,

$$y = \frac{C_1}{x} + C_2 x.$$

Differentiating,

$$\frac{dy}{dx} = -\frac{C_1}{x^2} + C_2$$

Applying the condition $y(1) = 6$,

$$6 = C_1 + C_2$$

Applying $\left.\frac{dy}{dx}\right|_{x=1} = 2$, gives

$$2 = -C_1 + C_2$$

Solving the two equations,

$$C_1 = 2, \qquad C_2 = 4$$

Hence,

$$y = \frac{2}{x} + 4x$$

Substituting $x = 2$,

$$y(2) = \frac{2}{2} + 4(2) = 1 + 8 = 9.$$

$$\boxed{9}$$

Mistakes to be avoided

- Do not forget the factor $D(D-1)$ while transforming $x^2 y''$; using D^2 directly is incorrect.

Q6.22.1 MCQ 2022 1M Ans: B ● ○ ○

The temperature distribution $T(x, y)$ satisfies the two-dimensional Laplace equation

$$\nabla^2 T = 0, \ 0 < x < 1, \ 0 < y < 1,$$

with the boundary conditions

$$T(x,0) = x, \qquad T(0,y) = y, \qquad T(x,1) = 1 + x, \qquad T(1,y) = 1 + y$$

Method 1

Observe that all boundary conditions are linear functions of x and y.

Therefore, assume a trial solution of the form

$$T(x,y) = ax + by + c$$

Clearly,

$$\frac{\partial^2 T}{\partial x^2} = 0, \qquad \frac{\partial^2 T}{\partial y^2} = 0$$

$$\Rightarrow \nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Now applying the boundary conditions.
From $T(x,0) = ax + c = x$, we get

$$a = 1, \qquad c = 0$$

Thus,

$$T(x,y) = x + by$$

Using $T(0,y) = by = y$, gives

$$b = 1$$

Hence,

$$T(x,y) = x + y$$

Method 2

Using the uniqueness property of Laplace's equation:
If a harmonic function satisfies all boundary conditions of a boundary value problem, then that solution is unique.
Consider

$$T(x,y) = x + y$$

Checking Laplace's equation,

$$\frac{\partial^2 T}{\partial x^2} = 0, \qquad \frac{\partial^2 T}{\partial y^2} = 0$$

Therefore,

$$\nabla^2 T = 0$$

Also,

$$T(x,0) = x, \qquad T(0,y) = y, \qquad T(x,1) = x + 1, \qquad T(1,y) = 1 + y$$

Since all four boundary conditions are satisfied, $T = x + y$ is the required solution.

$$\boxed{\text{(B) } T(x,y) = x + y}$$

Key Points

- Any linear function of the form $ax + by + c$ is harmonic because all second derivatives vanish.

Q6.22.2 MCQ 2022 2M Ans: C ● ● ○

The given differential equation is

$$\frac{du}{dx} = -\frac{xu^2}{2 + x^2 u}$$

Rearranging,

$$(2 + x^2 u)\,du = -xu^2\,dx$$

$$(2 + x^2 u)\,du + xu^2\,dx = 0$$

Comparing with the standard differential form

$$M(x,u)\,dx + N(x,u)\,du = 0, \quad \text{with } M = xu^2,\ N = 2 + x^2u$$

Check whether the equation is exact.
For exactness,

$$\frac{\partial M}{\partial u} = \frac{\partial N}{\partial x}$$

Now,

$$\frac{\partial M}{\partial u} = \frac{\partial}{\partial u}(xu^2) = 2xu,$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(2 + x^2u) = 2xu.$$

The differential equation is exact.
Therefore, there exists a function $F(x,u)$ such that

$$dF = M\,dx + N\,du$$

Integrating M with respect to x,

$$F(x,u) = \int xu^2\,dx = \frac{x^2u^2}{2} + \phi(u)$$

where $\phi(u)$ is a function of u.
Differentiating with respect to u,

$$\frac{\partial F}{\partial u} = x^2u + \phi'(u)$$

Since $\frac{\partial F}{\partial u} = N = 2 + x^2u$, we obtain

$$\phi'(u) = 2$$

Integrating,

$$\phi(u) = 2u$$

Hence,

$$F(x,u) = \frac{x^2u^2}{2} + 2u$$

The solution is

$$\frac{x^2u^2}{2} + 2u = C.$$

$$\boxed{\text{(C) } \frac{1}{2}x^2u^2 + 2u = \text{Constant}}$$

Key Points

- An equation of the form $M(x,u)\,dx + N(x,u)\,du = 0$ is exact if

$$\frac{\partial M}{\partial u} = \frac{\partial N}{\partial x}$$

- For an exact equation, the solution is obtained from a potential function $F(x,u)$ satisfying $dF = M\,dx + N\,du$.

Mistakes to be avoided

- Remember to add the unknown function $\phi(u)$ after integration and determine it using N.
- A common error is writing $\phi(x)$ instead of $\phi(u)$; the correction term must depend on the variable held constant during integration.

Q6.21.1 MCQ | 2021 | 1M | Ans: C | ● ○ ○

The given differential equation is

$$(\sin x)\frac{dy}{dx} + y\cos x = 1, \quad x > 1, \text{ with the condition } y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$$

$$\frac{dy}{dx} + y\cot x = \csc x$$

It is first order linear differential equation

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{with } P(x) = \cot x,\ Q(x) = \csc x$$

The integrating factor is

$$\text{I.F.} = e^{\int \cot x\, dx} = e^{\ln(\sin x)} = \sin x$$

The solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.}\, dx + C$$

$$y \times \sin x = \int \csc x \times \sin x\, dx + C$$

$$y\sin x = x + C$$

$$y = (x + C)\csc x$$

Using the condition $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, we get

$$\frac{\pi}{2} = \left(\frac{\pi}{2} + C\right)$$

$$\Rightarrow\ C = 0$$

Thus,

$$y = x\csc x$$

Evaluating at $x = \frac{\pi}{6}$,

$$y\left(\frac{\pi}{6}\right) = \frac{\pi}{6}\csc\left(\frac{\pi}{6}\right)$$

$$y\left(\frac{\pi}{6}\right) = \frac{\pi}{6} \times 2 = \frac{\pi}{3}$$

$$\boxed{(\text{C})\ \frac{\pi}{3}}$$

Mistakes to be avoided

- A common mistake is writing $\int \cot x\, dx = \ln(\cos x)$, whereas the correct result is

$$\int \cot x\, dx = \ln(\sin x).$$

Q6.21.2 | MCQ | 2021 | 2M | Ans: C | ● ○ ○

The given differential equation is

$$(1+y)\frac{dy}{dx} = y \text{ with the initial condition } y(1) = 1$$

Separating the variables and Integrating,

$$\frac{1+y}{y}\,dy = dx$$

$$\int \left(1 + \frac{1}{y}\right) dy = \int dx$$

$$y + \ln y = x + C$$

Applying the initial condition $y(1) = 1$,

$$1 + \ln 1 = 1 + C$$

$$\Rightarrow \; C = 0$$

Therefore,

$$y + \ln y = x$$

$$e^{y+\ln y} = e^x$$

$$e^y \cdot e^{\ln y} = e^x$$

$$ye^y = e^x.$$

$$\boxed{\text{(C) } ye^y = e^x}$$

Q6.19.1 | MCQ | 2019 | 1M | Ans: C | ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + 7x^2 y = 0, \text{ with the condition } y(0) = \frac{3}{7}$$

Method 1

Separating the variables and Integrating on both sides,

$$\int \frac{1}{y}\,dy = \int -7x^2\,dx$$

$$\ln|y| = -\frac{7x^3}{3} + C$$

$$y = e^{-\frac{7x^3}{3}+C} = Ke^{-\frac{7x^3}{3}},$$

where $K = e^C$.

Applying the initial condition $y(0) = \frac{3}{7}$,

$$\frac{3}{7} = Ke^0 \Rightarrow \; K = \frac{3}{7}$$

Hence,

$$y = \frac{3}{7}e^{-\frac{7x^3}{3}}.$$

At $x = 1$,

$$y(1) = \frac{3}{7}e^{-7/3}.$$

Method 2

Recognize that the equation is a first-order linear homogeneous differential equation:

$$\frac{dy}{dx} + P(x)y = 0, \;\; P(x) = 7x^2$$

The standard solution is

$$y = Ce^{-\int P(x)\,dx}$$

Therefore,

$$y = Ce^{-7\int x^2 dx} = Ce^{-\frac{7x^3}{3}}.$$

Using the initial condition $y(0) = \frac{3}{7}$, $C = \frac{3}{7}$.
Thus,

$$y(1) = \frac{3}{7}e^{-7/3}.$$

$$\boxed{\text{(C) } \frac{3}{7}e^{-7/3}}$$

Q6.19.2 MCQ 2019 1M Ans: B ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + 4y = 5, \;\; 0 \le x \le 1, \text{ with the initial condition } y(0) = 2.25$$

Method 1

The equation is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \text{ where } P(x) = 4, \; Q(x) = 5$$

The integrating factor is

$$\text{I.F.} = e^{\int 4\,dx} = e^{4x}$$

The solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.}\, dx + C$$

$$ye^{4x} = 5\int e^{4x} \cdot dx + C$$

$$ye^{4x} = 5\frac{e^{4x}}{4} + C$$

Therefore,

$$y = \frac{5}{4} + Ce^{-4x}$$

Applying the initial condition $y(0) = 2.25$,

$$2.25 = \frac{5}{4} + C \Rightarrow \;\; C = 1$$

Hence,

$$y = 1.25 + e^{-4x}.$$

Method 2

Find the complementary function (CF) and particular integral (PI).
The homogeneous equation is

$$\frac{dy}{dx} + 4y = 0$$

Its solution is

$$y_c = Ce^{-4x}.$$

For the particular solution, since the forcing term is a constant, assume $y_p = A$.
Substituting into the differential equation,

$$0 + 4A = 5$$

$$A = \frac{5}{4} = 1.25$$

Therefore,

$$y = Ce^{-4x} + 1.25$$

Using $y(0) = 2.25$,

$$2.25 = C + 1.25 \Rightarrow C = 1$$

Hence,

$$y = e^{-4x} + 1.25$$

$$\boxed{\text{(B) } y = e^{-4x} + 1.25}$$

Mistakes to be avoided

- A common mistake is writing the complementary function as Ce^{4x} instead of Ce^{-4x}.

Q6.19.3 MCQ 2019 2M Ans: A ● ● ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 4$$

This is a non-homogeneous **Cauchy–Euler differential equation**.
Using the standard transformation

$$x = e^t, \qquad D = \frac{d}{dt},$$

the Cauchy–Euler operators become

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y.$$

Substituting into the differential equation,

$$D(D-1)y - 2Dy + 2y = 4$$

$$(D^2 - D - 2D + 2)y = 4$$

$$(D^2 - 3D + 2)y = 4$$

The auxiliary equation is

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0$$

Hence, the roots are $m_1 = 2,\ m_2 = 1$.
Therefore, the complementary function is

$$y_c = C_1 e^{2t} + C_2 e^t$$

Since $e^t = x$,

$$y_c = C_1 x^2 + C_2 x$$

For the particular integral,

$$y_p = \frac{4}{D^2 - 3D + 2}$$

Since the forcing term is a constant,

$$y_p = \frac{4}{0 - 0 + 2} = 2$$

Hence,

$$y = C_1 x^2 + C_2 x^2 + 2$$

$$\boxed{\text{(A) } y = C_1 x^2 + C_2 x^2 + 2}$$

Mistakes to be avoided

- Do not forget that
$$x^2 y'' \to D(D-1)y,$$
not $D^2 y$, under the transformation $x = e^t$.
- Check whether a constant particular solution satisfies the non-homogeneous equation before using more complicated methods.

Q6.18.1 MCQ 2018 1M Ans: C ● ○ ○

The given differential equation is

$$y^3 \frac{dy}{dx} + x^3 = 0, \text{ with the initial condition } \ y(0) = 1$$

Separating variables and Integrating on both sides,

$$\int y^3\, dy = -\int x^3\, dx$$

$$\frac{y^4}{4} = -\frac{x^4}{4} + C$$

$$x^4 + y^4 = C_1$$

where $C_1 = 4C$.
Applying the initial condition $y(0) = 1$,

$$0^4 + 1^4 = C_1 \Rightarrow \ C_1 = 1$$

Therefore,

$$y^4 = 1 - x^4$$

Since $y(0) = 1 > 0$, the required solution is

$$y = (1 - x^4)^{1/4}.$$

At $x = -1$,

$$y(-1) = \left(1 - (-1)^4\right)^{1/4} = (1-1)^{1/4} = 0.$$

$$\boxed{\text{(C) } 0}$$

Key Points

- When taking the fourth root, use the initial condition $y(0) = 1$ to select the positive branch near $x = 0$.

Q6.18.2 NAT 2018 2M Ans: 1.47 ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0, \text{ with the initial conditions } y(0) = 0, \left.\frac{dy}{dx}\right|_{x=0} = 1.$$

Since the equation is linear with constant coefficients, assume $y = e^{mx}$.
Substituting into the differential equation,

$$m^2e^{mx} + me^{mx} - 6e^{mx} = 0$$

Dividing by $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 + m - 6 = 0$$

$$(m + 3)(m - 2) = 0$$

Hence, the roots are $m_1 = 2,\ m_2 = -3$.
Since the roots are real and distinct, the complementary function is

$$y = C_1e^{2x} + C_2e^{-3x}$$

Apply the initial conditions directly.
Using $y(0) = 0$,

$$C_1 + C_2 = 0$$

Differentiating,

$$\frac{dy}{dx} = 2C_1e^{2x} - 3C_2e^{-3x}$$

Applying $\left.\frac{dy}{dx}\right|_{x=0} = 1$,

$$2C_1 - 3C_2 = 1$$

Om solving these equations,

$$C_1 = \frac{1}{5}, \qquad C_2 = -\frac{1}{5}$$

Therefore,

$$y = \frac{1}{5}e^{2x} - \frac{1}{5}e^{-3x}$$

Evaluating at $x = 1$,

$$y(1) = \frac{e^2 - e^{-3}}{5} \approx \frac{7.3891 - 0.0498}{5} = 1.4679.$$

Correct to two decimal places,

$$\boxed{1.47}$$

Q6.17.1 MCQ 2017 1M Ans: B ● ○ ○

The given partial differential equation is

$$\frac{\partial u}{\partial y} + c\frac{\partial u}{\partial x} = 0, \ c > 1$$

We need to identify the family of functions that satisfies this equation.

Method 1

Verify the options directly.
Consider option (B):

$$u(x, y) = f(x - cy),$$

where f is an arbitrary differentiable function.
Using the chain rule,

$$\frac{\partial u}{\partial x} = f'(x - cy)\frac{\partial (x - cy)}{\partial x} = f'(x - cy),$$
$$\frac{\partial u}{\partial y} = f'(x - cy)\frac{\partial (x - cy)}{\partial y} = -c\, f'(x - cy).$$

Substituting into the PDE,

$$\frac{\partial u}{\partial y} + c\frac{\partial u}{\partial x} = -cf'(x - cy) + cf'(x - cy) = 0$$

Hence, $u(x, y) = f(x - cy)$ satisfies the given PDE for *any* differentiable function f.
Therefore, option (B) is correct.

Method 2

Solve the PDE using the characteristic method.
The given equation can be written as

$$c\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$$

The characteristic equations are

$$\frac{dx}{c} = \frac{dy}{1} = \frac{du}{0}$$

From $\frac{dx}{c} = dy$, we get

$$dx = c\, dy$$

Integrating,

$$x - cy = C_1$$

Also, $\frac{du}{0}$ implies that u remains constant along each characteristic curve.
Therefore, u can depend only on the characteristic invariant $x - cy$:

$$u = f(x - cy)$$

Thus, the general solution is

$$u(x, y) = f(x - cy)$$

$$\boxed{\text{(B) } u(x, y) = f(x - cy)}$$

Key Points

- For a PDE of the form

$$a\frac{\partial u}{\partial x} + b\frac{\partial u}{\partial y} = 0,$$

the solution remains constant along characteristic curves.

- The characteristic equation is $\frac{dx}{a} = \frac{dy}{b}$.

Q6.17.2 MCQ 2017 1M Ans: A ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + 16y = 0, \text{ with the boundary conditions } \left.\frac{dy}{dx}\right|_{x=0} = 1, \left.\frac{dy}{dx}\right|_{x=\pi/2} = -1$$

We need to determine whether these boundary conditions admit a solution.
Solve the differential equation and check the boundary conditions.
The auxiliary equation is

$$m^2 + 16 = 0$$

$$m = \pm 4i$$

Therefore, the general solution is

$$y = C_1 \cos 4x + C_2 \sin 4x$$

Differentiating,

$$\frac{dy}{dx} = -4C_1 \sin 4x + 4C_2 \cos 4x.$$

Applying

$$\left.\frac{dy}{dx}\right|_{x=0} = 4C_2 = 1,$$

$$C_2 = \frac{1}{4}$$

Now applying

$$\left.\frac{dy}{dx}\right|_{x=\pi/2} = -4C_1 \sin(2\pi) + 4C_2 \cos(2\pi)$$

$$\left.\frac{dy}{dx}\right|_{x=\pi/2} = 4C_2$$

Since $C_2 = \frac{1}{4}$, $\left.\frac{dy}{dx}\right|_{x=\pi/2} = 1$.

However, the boundary condition requires $\left.\frac{dy}{dx}\right|_{x=\pi/2} = -1$, which is impossible.
Hence, the boundary conditions are inconsistent and no function can satisfy both simultaneously.

$$\boxed{\text{(A) no solution}}$$

Key Points

- Boundary-value problems may have zero, one, or multiple solutions depending on the consistency of the boundary conditions.

Mistakes to be avoided

- Remember that failure to determine a constant does *not* imply no solution; inconsistency of conditions must be demonstrated explicitly.

Q6.17.3 NAT | 2017 | 2M | Ans: 94.08 | ● ○ ○

The given differential equation is

$$3y''(x) + 27y(x) = 0, \text{ with the initial conditions } y(0) = 0,\ y'(0) = 2000$$

The equation can be rewritten as,

$$y'' + 9y = 0$$

This is a second-order linear homogeneous differential equation with constant coefficients.

Method 1

Assume a solution of the form $y = e^{mx}$.
Substituting into the differential equation,

$$m^2 e^{mx} + 9e^{mx} = 0$$

Dividing by $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 + 9 = 0$$

$$m = \pm 3i$$

Therefore, the general solution is

$$y = C_1 \cos 3x + C_2 \sin 3x$$

Differentiating,

$$y' = -3C_1 \sin 3x + 3C_2 \cos 3x$$

Applying $y(0) = 0$, $C_1 = 0$.
Using $y'(0) = 2000$,

$$2000 = 3C_2 \Rightarrow C_2 = \frac{2000}{3}$$

Therefore,

$$y = \frac{2000}{3} \sin 3x$$

Evaluating at $x = 1$,

$$y(1) = \frac{2000}{3} \sin 3 \approx \frac{2000}{3}(0.14112) = 94.08$$

Method 2

The equation

$$y'' + \omega^2 y = 0$$

represents simple harmonic motion with $\omega = 3$.
The standard solution is

$$y = A \cos 3x + B \sin 3x$$

Using $y(0) = 0,\ A = 0$.
Differentiating,

$$y' = 3B \cos 3x$$

Applying $y'(0) = 2000$,

$$3B = 2000 \Rightarrow B = \frac{2000}{3}$$

Thus,

$$y = \frac{2000}{3} \sin 3x$$

Evaluating at $x = 1$,

$$y(1) = \frac{2000}{3} \sin 3 \approx \frac{2000}{3}(0.14112) = 94.08$$

$$\boxed{94.08}$$

Key Points

- A differential equation of the form $y'' + \omega^2 y = 0$ has oscillatory solutions involving sine and cosine functions.

Mistakes to be avoided

- Evaluate $\sin 3$ in radians. Using degrees would produce an incorrect answer.

Q6.16.1 NAT 2016 2M Ans: -1 ● ○ ○

The given boundary value problem is

$$y'' + 9y = 0, \text{ with the boundary conditions } y(0) = 0,\ y\left(\frac{\pi}{2}\right) = \sqrt{2}$$

Assume a solution of the form $y = e^{mx}$.
Substituting into the differential equation, the auxiliary equation is

$$m^2 + 9 = 0$$

$$m = \pm 3i$$

Therefore, the general solution is

$$y = C_1 \cos 3x + C_2 \sin 3x$$

Applying $y(0) = 0$,

$$0 = C_1 \cos 0 + C_2 \sin 0 \Rightarrow C_1 = 0$$

Using $y\left(\frac{\pi}{2}\right) = \sqrt{2}$,

$$\sqrt{2} = C_2 \sin\left(\frac{3\pi}{2}\right) \Rightarrow C_2 = -\sqrt{2}$$

Thus,

$$y = -\sqrt{2} \sin 3x$$

Now evaluate at $x = \frac{\pi}{4}$.

$$y\left(\frac{\pi}{4}\right) = -\sqrt{2} \sin\left(\frac{3\pi}{4}\right)$$

$$y\left(\frac{\pi}{4}\right) = -\sqrt{2} \cdot \frac{\sqrt{2}}{2} = -1$$

$$\boxed{-1}$$

Q6.15.1 MCQ 2015 2M Ans: C ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} = y,$$

and the solution passes through the points $(0, 0)$ and $\left(\ln 2, \frac{3}{4}\right)$.

Method 1

Assume a trial solution $y = e^{mx}$.

$$m^2 e^{mx} - e^{mx} = 0$$

Dividing by $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 - 1 = 0$$

$$m = 1, -1$$

Hence, the general solution is

$$y = C_1 e^x + C_2 e^{-x}$$

Using the point $(0, 0)$,

$$C_1 + C_2 = 0 \Rightarrow \; C_1 = -C_2$$

Substituting into the general solution,

$$y = C_1 \left(e^x - e^{-x}\right)$$

Now using the point $\left(\ln 2, \frac{3}{4}\right)$,

$$\frac{3}{4} = C_1 \left(e^{\ln 2} - e^{-\ln 2}\right)$$

$$\frac{3}{4} = C_1 \left(2 - \frac{1}{2}\right) = C_1 \frac{3}{2}$$

$$C_1 = \frac{1}{2}$$

Hence,

$$y = \frac{1}{2}\left(e^x - e^{-x}\right).$$

Method 2

Observe that option (C)

$$y = \frac{1}{2}\left(e^x - e^{-x}\right)$$

Differentiating,

$$y' = \frac{1}{2}\left(e^x + e^{-x}\right)$$

$$y'' = \frac{1}{2}\left(e^x - e^{-x}\right) = y$$

Thus, it satisfies the differential equation.
Also,

$$y(0) = \frac{1}{2}(1 - 1) = 0$$

$$y(\ln 2) = \frac{1}{2}\left(2 - \frac{1}{2}\right) = \frac{3}{4}$$

Therefore, option (C) satisfies both the differential equation and the given points.

$$\boxed{\text{(C) } y = \frac{1}{2}\left(e^x - e^{-x}\right)}$$

Key Points

- The expression $\frac{1}{2}\left(e^x - e^{-x}\right)$ is the hyperbolic sine function $\sinh x$.

Q6.15.2 | MCQ | 2015 | 2M | Ans: C | ● ○ ○

The given differential equation is

$$\frac{dy}{dt} = -5y \text{ with the initial condition } y(0) = 2$$

We are required to find the value of y at $t = 3$.
Separating variables and Integrating both sides,

$$\int \frac{1}{y}\, dy = \int -5\, dt$$

$$\ln|y| = -5t + C$$

$$y = e^{-5t+C} = Ce^{-5t},$$

where $C = e^C$ is a constant.
Applying the initial condition $y(0) = 2$,

$$2 = Ce^0 \Rightarrow\ C = 2$$

Therefore,

$$y(t) = 2e^{-5t}.$$

At $t = 3$,

$$y(3) = 2e^{-15}.$$

$$\boxed{\text{(C) } 2e^{-15}}$$

Key Points

- A differential equation of the form $\frac{dy}{dt} = ky$ models exponential growth ($k > 0$) or exponential decay ($k < 0$).

Q6.14.1 | MCQ | 2014 | 1M | Ans: A | ● ○ ○

The given system of differential equations is

$$\frac{dx}{dt} = 3x - 5y,$$

$$\frac{dy}{dt} = 4x + 8y.$$

We wish to express the system in the matrix form.

Method 1

Write the left-hand side as a vector of derivatives:

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix}$$

Now compare each differential equation with the standard matrix multiplication form.
From

$$\frac{dx}{dt} = 3x - 5y,$$

the coefficients of x and y form the first row: $[\,3.\ -5\,]$.

Similarly, from

$$\frac{dy}{dt} = 4x + 8y,$$

the coefficients form the second row: $[\,4,\ 8\,]$.
Therefore,

$$A = \begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix}$$

Hence,

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}.$$

Method 2

Verify option (A) directly.
Multiplying,

$$\begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x - 5y \\ 4x + 8y \end{bmatrix}$$

Thus,

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x - 5y \\ 4x + 8y \end{bmatrix},$$

which exactly reproduces the given system of equations.
Therefore, option (A) is correct.

$$\boxed{\text{(A) } \frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}}$$

Key Points

- A system of linear differential equations can be written compactly as $\dot{\mathbf{x}} = A\mathbf{x}$, where A is the coefficient matrix.
- Each row of the coefficient matrix is obtained from the coefficients of one differential equation.
- The order of variables must remain consistent throughout the vector and matrix representation.
- Matrix form is useful for eigenvalue-based solution techniques and state-space analysis.

Mistakes to be avoided

- Do not interchange rows and columns while forming the coefficient matrix.

Q6.14.2 NAT 2014 2M Ans: 35 ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} = 0 \text{ with the boundary conditions } y(0) = 5,\ \left.\frac{dy}{dx}\right|_{x=10} = 2$$

Integrating the differential equation once,

$$\frac{dy}{dx} = C_1,$$

where C_1 is a constant.
Integrating again,

$$y = C_1 x + C_2,$$

where C_2 is another constant.

Using the boundary condition $y(0) = 5$,

$$5 = C_1(0) + C_2 \Rightarrow \; C_2 = 5$$

Applying $\left.\dfrac{dy}{dx}\right|_{x=10} = 2$,

$$C_1 = 2$$

Therefore,

$$y = 2x + 5.$$

Evaluating at $x = 15$,

$$y(15) = 2(15) + 5 = 35.$$

$$\boxed{35}$$

Key Points

- Geometrically, zero second derivative implies zero curvature, i.e., a straight line.

Q6.14.3 MCQ 2014 2M Ans: D ● ● ○

The given differential equation is

$$\frac{dy}{dx} = \cos(x + y)$$

We are required to obtain its general solution.
Introduce the substitution $t = x + y$.
Differentiating with respect to x,

$$\frac{dt}{dx} = 1 + \frac{dy}{dx}$$

Using the given differential equation,

$$\frac{dt}{dx} = 1 + \cos t$$

$$\frac{dt}{1 + \cos t} = dx$$

Using the trigonometric identity $1 + \cos t = 2\cos^2\left(\frac{t}{2}\right)$,

$$\frac{dt}{2\cos^2(t/2)} = dx$$

$$\frac{1}{2}\sec^2\left(\frac{t}{2}\right)\, dt = dx$$

Integrating both sides,

$$\frac{1}{2}\int \sec^2\left(\frac{t}{2}\right)\, dt = \int dx$$

Let $u = \dfrac{t}{2}$, $dt = 2\,du$
Therefore,

$$\int \sec^2 u \, du = x + C$$

$$\tan u = x + C$$

Substituting $u = \dfrac{t}{2}$,

$$\tan\left(\frac{t}{2}\right) = x + C$$

Finally, using $t = x + y$,

$$\tan\left(\frac{x + y}{2}\right) = x + C$$

$$\boxed{\text{(D)} \;\; \tan\left(\frac{x + y}{2}\right) = x + C}$$

Q6.14.4 MCQ | 2014 | 2M | Ans: A | ● ● ○

The differential equation is

$$\frac{d^2x(t)}{dt^2} + x(t) = 0, \; t > 0,$$

with two solutions $x_1(t)$ and $x_2(t)$ satisfying

$$x_1(0) = 1, \; x_1'(0) = 0, \; \text{and } x_2(0) = 0, \; x_2'(0) = 1$$

The Wronskian is defined as

$$W(t) = \begin{vmatrix} x_1(t) & x_2(t) \\ x_1'(t) & x_2'(t) \end{vmatrix}.$$

We are required to find $W\left(\frac{\pi}{2}\right)$.

Method 1

The auxiliary equation corresponding to $x'' + x = 0$ is

$$m^2 + 1 = 0$$

$$m = \pm i$$

Therefore, the general solution is

$$x(t) = C_1 \cos t + C_2 \sin t$$

Finding $x_1(t)$:
Using $x_1(0) = 1, \; C_1 = 1$.
Differentiating,

$$x_1'(t) = -C_1 \sin t + C_2 \cos t$$

Using

$$x_1'(0) = 0, \; C_2 = 0.$$

Hence,

$$x_1(t) = \cos t, \qquad x_1'(t) = -\sin t.$$

Finding $x_2(t)$:
Using $x_2(0) = 0, \; C_1 = 0$.
Using $x_2'(0) = 1, \; C_2 = 1$.
Hence,

$$x_2(t) = \sin t, \qquad x_2'(t) = \cos t.$$

Therefore,

$$W(t) = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix}.$$

Evaluating the determinant,

$$W(t) = \cos^2 t + \sin^2 t = 1.$$

Thus,

$$W\left(\frac{\pi}{2}\right) = 1.$$

Method 2

Use Abel's identity.
For a second-order linear differential equation $x'' + P(t)x' + Q(t)x = 0$, the Wronskian satisfies

$$W(t) = C\, e^{-\int P(t)\, dt}$$

Here, $P(t) = 0$.

Hence,

$$W(t) = C$$

Thus, the Wronskian is constant for all t.
Evaluate it at $t = 0$:

$$W(0) = \begin{vmatrix} x_1(0) & x_2(0) \\ x_1'(0) & x_2'(0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

Therefore, $W(t) = 1$ for every t, and in particular,

$$W\left(\frac{\pi}{2}\right) = 1.$$

$$\boxed{\text{(A) } 1}$$

Key Points

- If $W(t) \neq 0$, the solutions are linearly independent.

Q6.14.5 | MCQ | 2014 | 1M | Ans: B | ● ○ ○

The given initial value problem is

$$\frac{dy}{dx} = -2xy, \text{ with } y(0) = 2$$

We are required to determine $y(x)$.
Separating variables and Integrating both sides,

$$\int \frac{1}{y}\,dy = -2\int x\,dx,$$

$$\ln|y| = -x^2 + C$$

$$y = e^{-x^2+C} = Ce^{-x^2}$$

where $C = e^C$ is a constant.
Applying the initial condition $y(0) = 2$,

$$2 = Ce^0 \implies C = 2$$

Hence,

$$y = 2e^{-x^2}.$$

$$\boxed{\text{(B) } 2e^{-x^2}}$$

Q6.13.1 | MCQ | 2013 | 1M | Ans: D | ● ○ ○

The given partial differential equation is

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2}$$

To classify a differential equation, we determine:

1. Whether it is linear or non-linear.
2. Its order, i.e., the order of the highest derivative present.

Checking linearity:

A differential equation is linear if the dependent variable and all its derivatives appear only to the first power and are not multiplied together.

In the given equation, the term $u\dfrac{\partial u}{\partial x}$ contains the dependent variable u multiplied by its derivative $\dfrac{\partial u}{\partial x}$.
Therefore, the equation is **non-linear**.

Checking the order:

The highest-order derivative is $\dfrac{\partial^2 u}{\partial x^2}$, which is a second-order derivative.

Therefore, the given equation is a **Non-linear partial differential equation of order 2**

$$\boxed{\text{(D) Non-linear equation of order 2}}$$

Key Points

- The given equation is a form of the well-known Burgers' equation, which is a non-linear second-order PDE.

Q6.13.2 | MCQ | 2013 | 2M | Ans: B | ● ● ○

The given differential equation is

$$\frac{d^2u}{dx^2} - k\frac{du}{dx} = 0,$$

where k is a constant, with boundary conditions

$$u(0) = 0, \qquad u(L) = U$$

We are required to determine the function $u(x)$.
Assume a solution of the form $u = e^{mx}$.
Substituting into the differential equation,

$$m^2e^{mx} - kme^{mx} = 0$$

Dividing by $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 - km = 0$$

$$m(m-k) = 0$$

The roots are $m_1 = 0,\ m_2 = k$.
Therefore, the general solution is

$$u = C_1 + C_2e^{kx}$$

Applying the boundary condition $u(0) = 0$,

$$0 = C_1 + C_2 \Rightarrow\ C_2 = -C_1$$

Now use the condition $u(L) = U$,

$$U = C_1(1 - e^{kL})$$

$$C_1 = \frac{U}{1 - e^{kL}}$$

Therefore,

$$u = U\left(\frac{1 - e^{kx}}{1 - e^{kL}}\right).$$

$$\boxed{\text{(B) } u = U\left(\frac{1 - e^{kx}}{1 - e^{kL}}\right)}$$

Q6.12.1 MCQ | 2012 | 2M | Ans: A | ● ● ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 4y = 0, \text{ with the boundary conditions } y(0) = 0,\ y(1) = 1$$

This is a **Cauchy–Euler differential equation**.

Method 1

For a Cauchy–Euler equation, assume a trial solution $y = x^m$.
Then

$$\frac{dy}{dx} = mx^{m-1}, \qquad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Substituting into the differential equation,

$$x^2[m(m-1)x^{m-2}] + x[mx^{m-1}] - 4x^m = 0$$

$$[m(m-1) + m - 4]\,x^m = 0$$

$$(m^2 - 4)x^m = 0$$

Hence, the indicial (auxiliary) equation is

$$m^2 - 4 = 0$$

$$m = \pm 2$$

The general solution is

$$y = C_1x^2 + C_2x^{-2}$$

Applying the boundary condition $y(0) = 0$:
Since $x^{-2} \to \infty$ as $x \to 0$, the solution can remain finite at $x = 0$ only if $C_2 = 0$
Using $y(1) = 1$. $1 = C_1$.
Hence,

$$y = x^2.$$

Method 2

Using the standard transformation

$$x = e^t, \qquad t = \ln x,$$

with $D = \frac{d}{dt}$, the Euler equation transforms as

$$x^2y'' = D(D-1)y, \qquad xy' = Dy$$

Substituting,

$$D(D-1)y + Dy - 4y = 0$$

$$(D^2 - 4)y = 0$$

The auxiliary equation is

$$m^2 - 4 = 0$$

$$m = \pm 2$$

Hence,

$$y = C_1e^{2t} + C_2e^{-2t} = C_1x^2 + C_2x^{-2}.$$

Applying the boundary conditions again yields

$$y = x^2.$$

$$\boxed{\text{(A) } x^2}$$

Q6.11.1 MCQ 2011 2M Ans: D ● ○ ○

The given differential equation is

$$\frac{dy}{dx} = x(1 + y^2)$$

We are required to obtain the general solution.
Separating variables and Integrating both sides,

$$\int \frac{dy}{1 + y^2} = \int x\,dx$$

Using the standard result $\int \frac{dy}{1 + y^2} = \tan^{-1} y$, we obtain

$$\tan^{-1} y = \frac{x^2}{2} + C$$

$$y = \tan\left(\frac{x^2}{2} + C\right)$$

$$\boxed{\text{(D) } y = \tan\left(\frac{x^2}{2} + C\right)}$$

Mistakes to be avoided

- Do not write $\int \frac{dy}{1+y^2} = \ln(1 + y^2)$, which is incorrect.

Q6.10.1 MCQ 2010 1M Ans: B ● ○ ○

The given Blasius equation is

$$\frac{d^3 f}{d\eta^3} + \frac{f}{2}\frac{d^2 f}{d\eta^2} = 0$$

We need to classify the differential equation based on its order and linearity.
Determining the order:
The highest-order derivative is $\frac{d^3 f}{d\eta^3}$.
Hence, the order of the differential equation is 3
Since derivatives are taken with respect to a single independent variable η,
It is an ordinary differential equation (ODE).
Determining linearity:
In the given equation, $\frac{f}{2}\frac{d^2 f}{d\eta^2}$ contains a product of the dependent variable f and its second derivative.
Therefore, the equation is **non-linear**.
Thus, the equation is a **third-order non-linear ordinary differential equation**

$\boxed{\text{(B) third order nonlinear ordinary differential equation}}$.

Key Points

- The Blasius equation arises in boundary-layer theory in fluid mechanics and is a famous non-linear third-order ODE.

Q6.09.1 MCQ | 2009 | 2M | Ans: A | ● ● ○

The given differential equation is

$$x\frac{dy}{dx} + y = x^4 \text{ with the condition } y(1) = \frac{6}{5}$$

We are required to determine the solution $y(x)$.

$$\frac{dy}{dx} + \frac{1}{x}y = x^3$$

This is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P(x) = \frac{1}{x},\ Q(x) = x^3$$

The integrating factor is

$$\text{I.F.} = e^{\int (1/x)\,dx} = e^{\ln x} = x$$

The complete solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.}\, dx + C$$

$$y \times x = \int x^3 \times x\, dx + C$$

$$xy = \int x^4\, dx + C$$

$$xy = \frac{x^5}{5} + C$$

$$y = \frac{x^4}{5} + \frac{C}{x}$$

Applying the condition $y(1) = \frac{6}{5}$,

$$\frac{6}{5} = \frac{1}{5} + C \Rightarrow \quad C = 1$$

Hence,

$$y = \frac{x^4}{5} + \frac{1}{x}$$

$$\boxed{\text{(A) } y = \frac{x^4}{5} + \frac{1}{x}}$$

Q6.08.1 MCQ | 2008 | 1M | Ans: B | ● ○ ○

The given differential equation is

$$\ddot{x} + 3x = 0 \text{ with the initial conditions } x(0) = 1,\ \dot{x}(0) = 0$$

We are required to find the value of $x(1)$.

Method 1

Assume a solution of the form $x = e^{mt}$.
Substituting into the differential equation,

$$m^2 e^{mt} + 3e^{mt} = 0$$

Dividing by $e^{mt} \neq 0$, the auxiliary equation is

$$m^2 + 3 = 0$$

$$m = \pm i\sqrt{3}$$

Therefore, the general solution is

$$x(t) = C_1 \cos(\sqrt{3}\,t) + C_2 \sin(\sqrt{3}\,t)$$

Differentiating,

$$\dot{x}(t) = -\sqrt{3}\,C_1 \sin(\sqrt{3}\,t) + \sqrt{3}\,C_2 \cos(\sqrt{3}\,t)$$

Applying $x(0) = 1,\ 1 = C_1$.
Applying $\dot{x}(0) = 0$,

$$0 = \sqrt{3}\,C_2 \Rightarrow\ C_2 = 0$$

Hence,

$$x(t) = \cos(\sqrt{3}\,t)$$

At $t = 1$,

$$x(1) = \cos(\sqrt{3}) \approx \cos(1.732) \approx -0.16.$$

Method 2

Recognize that

$$\ddot{x} + \omega^2 x = 0$$

represents simple harmonic motion with $\omega = \sqrt{3}$.
The standard solution satisfying $x(0) = A = 1,\ \dot{x}(0) = 0$ is

$$x(t) = A\cos(\omega t)$$

Since $A = 1$,

$$x(t) = \cos(\sqrt{3}\,t)$$

Therefore,

$$x(1) = \cos(\sqrt{3}) \approx -0.16.$$

$$\boxed{(\mathrm{B})\ -0.16}$$

Key Points

- The equation $\ddot{x} + \omega^2 x = 0$ has oscillatory solutions involving sine and cosine functions.
- This equation models undamped simple harmonic motion.

Q6.08.2 MCQ 2008 2M Ans: C ● ● ○

The given function is $f = y^x$.
We are required to evaluate

$$\frac{\partial^2 f}{\partial x\,\partial y} \text{ at } x = 2,\ y = 1$$

Method 1

First differentiate $f = y^x$ partially with respect to y.

$$\frac{\partial f}{\partial y} = xy^{x-1}$$

Now differentiate this result with respect to x:

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(xy^{x-1}\right)$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = y^{x-1} + x\frac{\partial}{\partial x}\left(y^{x-1}\right)$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = y^{x-1} + xy^{x-1}\ln y$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = y^{x-1}(1 + x\ln y)$$

Substituting $x = 2$ and $y = 1$,

$$\frac{\partial^2 f}{\partial x\,\partial y} = 1^1\,(1 + 2\ln 1)$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = 1$$

Method 2

Differentiate first with respect to x.

$$\frac{\partial f}{\partial x} = y^x\ln y$$

Now differentiate with respect to y:

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(y^x\ln y\right)$$

$$\frac{\partial^2 f}{\partial y\,\partial x} = xy^{x-1}\ln y + y^x\frac{1}{y}$$

$$\frac{\partial^2 f}{\partial y\,\partial x} = xy^{x-1}\ln y + y^{x-1}$$

$$\frac{\partial^2 f}{\partial y\,\partial x} = y^{x-1}(1 + x\ln y)$$

Substituting $x = 2$ and $y = 1$,

$$\frac{\partial^2 f}{\partial y\,\partial x} = 1$$

$$\boxed{\text{(C) } 1}$$

Key Points

- For variable exponents, it is often convenient to write $y^x = e^{x\ln y}$.

Mistakes to be avoided

- Keep track of which variable is treated as constant during each partial differentiation step.

Q6.08.3 | MCQ | 2008 | 2M | Ans: A | ● ● ○

The given differential equation is

$$y'' + 2y' + y = 0 \text{ with the boundary conditions } y(0) = 0,\ y(1) = 0$$

We are required to find the value of $y(0.5)$.
Assume a solution of the form $y = e^{mt}$.
Substituting into the differential equation,

$$m^2 e^{mt} + 2me^{mt} + e^{mt} = 0$$

Dividing by $e^{mt} \neq 0$, the auxiliary equation is

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0$$

$$m = -1$$

For repeated roots, the general solution is

$$y = (C_1 + C_2 t)e^{-t}$$

Applying the first boundary condition, $y(0) = 0$,

$$0 = (C_1 + 0)e^0 \Rightarrow\ C_1 = 0$$

Applying the second boundary condition, $y(1) = 0$,

$$0 = C_2(1)e^{-1} \Rightarrow\ C_2 = 0$$

Therefore,

$$y(t) \equiv 0.$$

Hence,

$$y(0.5) = 0$$

$$\boxed{\text{(A) } 0}$$

Key Points

- The only solution satisfying both boundary conditions is the trivial solution.
- For homogeneous linear differential equations, boundary conditions can sometimes force the solution to vanish identically.

Q6.07.1 | MCQ | 2007 | 1M | Ans: A | ● ○ ○

The given partial differential equation is

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial \varphi}{\partial x} + \frac{\partial \varphi}{\partial y} = 0$$

To determine the **order** and **degree**, follow two simple steps.
Step 1: Find the order
The highest-order derivatives are

$$\frac{\partial^2 \varphi}{\partial x^2} \quad \text{and} \quad \frac{\partial^2 \varphi}{\partial y^2},$$

which are second-order derivatives.
Therefore, Order = 2.

Step 2: Find the degree
Degree is the power of the highest-order derivative after the equation is expressed as a polynomial in derivatives.
The highest-order derivatives appear only to the first power.
Hence, Degree = 1.

$$\boxed{\text{(A) degree 1 order 2}}$$

Q6.07.2 | MCQ | 2007 | 2M | Ans: C | ● ● ○

The given differential equation is

$$\frac{dy}{dx} = y^2 \text{ with the initial condition } y(0) = 1$$

We are required to determine the interval in which the solution exists and remains finite.
Separating variables and Integrating both sides,

$$\int y^{-2}\, dy = \int dx$$

$$-\frac{1}{y} = x + C$$

$$y = -\frac{1}{x + C}$$

Using the initial condition $y(0) = 1$,

$$1 = -\frac{1}{C} \Rightarrow C = -1$$

Substituting back,

$$y = -\frac{1}{x-1} = \frac{1}{1-x}$$

Observe that the denominator becomes zero at $x = 1$.
Therefore, $y(x) \to \infty$ as $x \to 1$.
So either $x < 1$ or $x > 1$

$$\boxed{\text{(C) } x < 1,\ x > 1}$$

Key Points

- The solution of an initial value problem must contain the initial point.
- A vertical asymptote occurs when the denominator becomes zero.

Q6.06.1 | MCQ | 2006 | 1M | Ans: B | ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + 2xy = e^{-x^2} \text{ with the initial condition } y(0) = 1$$

We are required to determine $y(x)$.
The equation is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \text{ where } P(x) = 2x,\ Q(x) = e^{-x^2}$$

The integrating factor is

$$\text{I.F.} = e^{\int 2x\, dx} = e^{x^2}$$

The complete solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.}\, dx + C$$

$$y \times e^{x^2} = \int e^{-x^2} \cdot e^{x^2} \cdot dx + C$$

$$ye^{x^2} = x + C$$

$$y = (x + C)e^{-x^2}$$

Applying the initial condition $y(0) = 1$,

$$1 = (0 + C)e^0 \Rightarrow \ C = 1$$

Hence,

$$y = (1 + x)e^{-x^2}.$$

$$\boxed{\text{(B) } (1 + x)\, e^{-x^2}}$$

Q6.06.2 MCQ 2006 2M Ans: B ● ● ○

The given differential equation is

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 3e^{2x}$$

We are asked to find only the **Particular Integral (P.I.)**.

Method 1

Using the differential operator $D = \dfrac{d}{dx}$,

$$(D^2 + 4D + 3)y = 3e^{2x}$$

The particular integral is

$$\text{P.I.} = \frac{1}{D^2 + 4D + 3}\left(3e^{2x}\right)$$

For an exponential forcing term e^{ax}, use the shortcut

$$\frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax},$$

provided $f(a) \neq 0$.
Substituting $D = 2$,

$$f(2) = 2^2 + 4(2) + 3 = 4 + 8 + 3 = 15$$

Therefore,

$$\text{P.I.} = \frac{3}{15}e^{2x} = \frac{e^{2x}}{5}.$$

Method 2

Assume a particular solution of the form $y_p = Ae^{2x}$.
Then

$$y_p' = 2Ae^{2x}, \qquad y_p'' = 4Ae^{2x}$$

Substituting into the differential equation,

$$4Ae^{2x} + 4(2Ae^{2x}) + 3(Ae^{2x}) = 3e^{2x}$$

$$(4A + 8A + 3A)e^{2x} = 3e^{2x}$$
$$15Ae^{2x} = 3e^{2x}$$
$$A = \frac{3}{15} = \frac{1}{5}$$

Hence,

$$y_p = \frac{e^{2x}}{5}.$$

$$\boxed{\text{(B) } \frac{1}{5}e^{2x}}$$

Key Points

- For $f(D)y = e^{ax}$,
$$\text{P.I.} = \frac{1}{f(a)}e^{ax}.$$
- The Particular Integral represents only one specific solution of the non-homogeneous equation.

Mistakes to be avoided

- Do not solve the entire differential equation when only the P.I. is asked.
- Always check whether $f(a) = 0$. If it were zero, the standard shortcut would fail and a modified P.I. would be required.

Q6.05.1 | MCQ | 2005 | 2M | Ans: D | ● ● ○

The given differential equation is

$$x^2\frac{dy}{dx} + 2xy = \frac{2\ln x}{x} \text{ with the initial condition } y(1) = 0$$

We are required to find $y(e)$.

Method 1

First divide the equation by x^2:
$$\frac{dy}{dx} + \frac{2}{x}y = \frac{2\ln x}{x^3}$$
This is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P(x) = \frac{2}{x},\ Q(x) = \frac{2\ln x}{x^3}$$

The integrating factor is
$$\text{I.F.} = e^{\int 2/x\,dx} = e^{2\ln x} = x^2$$

The complete solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.}\,dx + C$$

$$y \times x^2 = \int \frac{2\ln x}{x^3} \cdot x^2 \cdot dx + C$$

$$x^2y = \int \frac{2\ln x}{x}\,dx + C$$

Let $t = \ln x$. Then $dt = \dfrac{dx}{x}$.

Therefore,

$$x^2y = 2\int t\,dt + C$$

$$x^2y = t^2 + C$$

Substituting back $t = \ln x$,

$$x^2y = (\ln x)^2 + C$$

Applying the condition $y(1) = 0$,

$$0 = (\ln 1)^2 + C \Rightarrow \ C = 0$$

Thus,

$$x^2y = (\ln x)^2$$

$$y = \frac{(\ln x)^2}{x^2}.$$

Now evaluate at $x = e$.

$$y(e) = \frac{1^2}{e^2} = \frac{1}{e^2}.$$

Method 2

Notice that

$$\frac{d}{dx}(x^2y) = x^2\frac{dy}{dx} + 2xy$$

Hence the given equation can be written directly as

$$\frac{d}{dx}(x^2y) = \frac{2\ln x}{x}$$

Integrating,

$$x^2y = (\ln x)^2 + C$$

Using $y(1) = 0$ gives $C = 0$, and therefore

$$y = \frac{(\ln x)^2}{x^2}$$

Now evaluate at $x = e$.

$$y(e) = \frac{1^2}{e^2} = \frac{1}{e^2}.$$

$$\boxed{\text{(D) } \frac{1}{e^2}}$$

Key Points

- For linear differential equations, recognizing a product derivative can save time compared to the full integrating-factor method.

Mistakes to be avoided

- A common error is writing $\int \frac{\ln x}{x}\,dx = \ln(\ln x)$. The correct result is

$$\int \frac{\ln x}{x}\,dx = \frac{(\ln x)^2}{2}.$$

Q6.05.2 MCQ 2005 2M Ans: C ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = 0,$$

and its complete solution is

$$y = C_1e^{-x} + C_2e^{-3x}.$$

We are required to determine the values of p and q.

Method 1

For the given second-order linear homogeneous differential equation, the auxiliary equation is

$$m^2 + pm + q = 0$$

From the given solution $y = C_1e^{-x} + C_2e^{-3x}$, the roots are $m_1 = -1,\ m_2 = -3$.
Hence, the auxiliary equation is

$$(m+1)(m+3) = 0$$

$$m^2 + 4m + 3 = 0$$

Comparing with $m^2 + pm + q = 0$, we obtain

$$p = 4, \qquad q = 3.$$

Method 2

Use the relationship between roots and coefficients.
For $m^2 + pm + q = 0$,

$$m_1 + m_2 = -p,$$

$$m_1m_2 = q$$

Given $m_1 = -1,\ m_2 = -3$,
Sum of roots:

$$-1 + (-3) = -4 = -p \Rightarrow\ p = 4$$

Product of roots:

$$(-1)(-3) = 3 \Rightarrow\ q = 3$$

$$\boxed{\text{(C) } p = 4,\ q = 3}$$

Mistakes to be avoided

- Do not take the roots as 1 and 3; the roots are -1 and -3 because the exponents are negative.

Q6.05.3 MCQ 2005 2M Ans: C ● ● ○

From the previous question, we obtained $p = 4,\ q = 3$.
The given differential equation is

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + (q+1)y = 0$$

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$$

We need to determine which option is a solution.
Form the auxiliary equation:

$$m^2 + 4m + 4 = 0$$

$$(m+2)^2 = 0$$

We have a repeated root $m = -2$.
For repeated roots, the general solution is

$$y = (C_1 + C_2x)e^{-2x}$$

Now compare with the options:
A: e^{-3x} not present in the general solution.
B: contains e^{-x}, not e^{-2x}.
C: xe^{-2x} is obtained by choosing $C_1 = 0,\ C_2 = 1$.
Hence it is a valid solution.
D: not of the form $(C_1 + C_2x)e^{-2x}$ due to x^2e^{-2x}
Therefore, only option (C) is a solution.

$$\boxed{\text{(C) } x^2e^{-2x}}$$

Mistakes to be avoided

- Do not use the previous equation $y'' + 4y' + 3y = 0$. The current equation is $y'' + 4y' + 4y = 0$.

Q6.03.1 MCQ | 2003 | 2M | Ans: A | ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + y^2 = 0$$

We are required to find its general solution.
Separating variables and Integrating both sides

$$\int \frac{dy}{y^2} = -dx$$

$$\int y^{-2}\,dy = -\int dx.$$

$$-\frac{1}{y} = -x + C$$

$$\frac{1}{y} = x + C_1$$

Therefore,

$$y = \frac{1}{x + C_1}$$

$$\boxed{\text{(A) } y = \frac{1}{x + C}}$$

Key Points

- Nonlinear equations are not necessarily unsolvable.

Q6.01.1 MCQ | 2001 | 5M | Ans: C | ● ● ○

The given differential equation is

$$\frac{d^2y}{dx^2} + y = x \text{ with the conditions } y(0) = 1,\ y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$$

We are required to find the solution $y(x)$.
This is a second-order linear non-homogeneous differential equation.
Step 1: Find the Complementary Function (CF)

The associated homogeneous equation is

$$y'' + y = 0$$

Assume $y = e^{mx}$.
Substituting, the auxiliary equation becomes

$$m^2 + 1 = 0$$

$$m = \pm i$$

Therefore,

$$\text{CF} = C_1 \cos x + C_2 \sin x$$

Step 2: Find the Particular Integral (PI)
Since the forcing term is x, try

$$y_p = Ax + B$$

Substituting

$$y_p'' + y_p = x$$

$$0 + (Ax + B) = x$$

Comparing coefficients, $A = 1,\ B = 0$
Hence,

$$\text{PI} = x$$

Step 3: Form the complete solution

$$y = C_1 \cos x + C_2 \sin x + x$$

Apply $y(0) = 1$:

$$1 = C_1 + 0 + 0 \Rightarrow C_1 = 1$$

Apply $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$,

$$\frac{\pi}{2} = \cos\frac{\pi}{2} + C_2 \sin\frac{\pi}{2} + \frac{\pi}{2}$$

$$\frac{\pi}{2} = 0 + C_2 + \frac{\pi}{2}$$

$$C_2 = 0$$

Therefore,

$$y = \cos x + x.$$

$$\boxed{\text{(C) } y = \cos x + x}$$

Q6.00.1 MCQ 2000 1M Ans: D ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

We are required to find its general solution.
Assume a trial solution $y = e^{mx}$.
Substituting into the differential equation,

$$m^2 e^{mx} + m e^{mx} + e^{mx} = 0$$

Dividing by $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 + m + 1 = 0$$

$$m = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{-3}}{2}$$

$$m = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}.$$

The roots are complex conjugates of the form $m = \alpha \pm i\beta$.
For complex roots, the complementary function is

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$$

Substituting $\alpha = -\frac{1}{2}$ and $\beta = \frac{\sqrt{3}}{2}$,

$$y = e^{-x/2}\left[C_1 \cos\left(\frac{\sqrt{3}}{2}x\right) + C_2 \sin\left(\frac{\sqrt{3}}{2}x\right)\right]$$

$$\boxed{\text{(D) } e^{-x/2}\left\{A \cos\left(\frac{\sqrt{3}}{2}\right)x + B \sin\left(\frac{\sqrt{3}}{2}\right)x\right\}}$$

Q6.99.1 MCQ | 1999 | 2M | Ans: C | ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + (x^2 + 4x)\frac{dy}{dx} + y = x^8 - 8$$

We need to identify the type of differential equation.
From the given differential equation, it is identified that:

1. There are no partial derivatives $\left(\text{such as } \frac{\partial y}{\partial x}, \frac{\partial y}{\partial t}\right)$.
2. All derivatives are of first degree $\left(\frac{d^2y}{dx^2}, \frac{dy}{dx}, y\right)$.
3. The constant terms are non-zero $(x^8 - 8)$.

Hence, the equation is a **Non-homogeneous Ordinary Differential Equation**.

$$\boxed{\text{(C) non-homogeneous differential equation}}$$

Mistakes to be avoided

- The term $(x^2 + 4x)\frac{dy}{dx}$ does not make the equation nonlinear because the coefficient depends only on x, not on y.

Q6.98.1 MCQ | 1998 | 2M | Ans: D | ● ● ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$$

This is a **Cauchy–Euler (Equidimensional) differential equation**.
We are required to find its general solution.
For a Cauchy–Euler equation, assume $y = x^m$.
Then

$$\frac{dy}{dx} = mx^{m-1}, \qquad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Substituting into the differential equation,

$$x^2[m(m-1)x^{m-2}] - x[mx^{m-1}] + x^m = 0$$

$$\Big(m(m-1) - m + 1\Big)x^m = 0$$

$$(m^2 - 2m + 1)x^m = 0$$

Hence, the auxiliary (indicial) equation is

$$m^2 - 2m + 1 = 0$$

$$(m-1)^2 = 0$$

Thus, we have a repeated root $m = 1$.
For repeated roots in a Cauchy–Euler equation,

$$y = x^m\Big(A + B\ln x\Big)$$

Substituting $m = 1$,

$$y = x\Big(A + B\ln x\Big).$$

$$y = Ax + Bx\ln x, \text{ where } A \text{ and } B \text{ are arbitrary constants.}$$

$$\boxed{\text{(D) } Ax + Bx\ln x, \text{ where } A \text{ and } B \text{ are arbitrary constants.}}$$

Key Points

- For Cauchy–Euler equations, repeated roots lead to $\ln x$, unlike constant-coefficient equations where repeated roots lead to an extra factor of x.

Q6.98.2 NAT 1998 5M Ans: 0.625 ● ● ●

The radial displacement $u(x)$ is governed by

$$\frac{d^2u}{dx^2} + \frac{1}{x}\frac{du}{dx} - \frac{u}{x^2} = 8x \text{ with the boundary conditions } u(0) = 0,\ u(1) = 2$$

We are required to find $u\left(\frac{1}{2}\right)$.
Multiply the equation by x^2:

$$x^2u'' + xu' - u = 8x^3$$

This is a Cauchy–Euler differential equation.
Assume $u = x^m$.
Then

$$u' = mx^{m-1}, \qquad u'' = m(m-1)x^{m-2}$$

Substituting into the homogeneous equation, the auxiliary equation becomes

$$m(m-1) + m - 1 = 0$$

$$m^2 - 1 = 0$$

$$(m-1)(m+1) = 0$$

$$m = 1, -1$$

Therefore, the complementary function is

$$u_c = C_1x + \frac{C_2}{x}$$

Since the RHS is $8x^3$, try $u_p = Ax^3$.
Then

$$u'_p = 3Ax^2, \qquad u''_p = 6Ax$$

Substituting into the differential equation

$$x^2(6Ax) + x(3Ax^2) - Ax^3 = 8x^3$$

$$(6A + 3A - A)x^3 = 8x^3$$

$$8Ax^3 = 8x^3$$

$$A = 1$$

Hence,

$$u_p = x^3$$

Therefore,

$$u = C_1 x + \frac{C_2}{x} + x^3$$

Applying the boundary conditions:
From $u(0) = 0$, the term $\frac{C_2}{x}$ would become infinite unless $C_2 = 0$.
Using $u(1) = 2$,

$$2 = C_1 + 1 \Rightarrow C_1 = 1$$

Thus,

$$u = x + x^3$$

Now evaluate at $x = \frac{1}{2}$.

$$u\left(\frac{1}{2}\right) = \frac{1}{2} + \left(\frac{1}{2}\right)^3$$

$$u\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$

$$u\left(\frac{1}{2}\right) = 0.625.$$

$$\boxed{0.625}$$

Key Points

- For polynomial forcing terms, a polynomial trial PI is often the fastest approach.

Q6.96.1 MCQ 1996 2M Ans: A ●●○

The given differential equation is

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 5\cos x$$

We are required to find only the **Particular Integral (P.I.)**.

Method 1

Using the differential operator $D = \frac{d}{dx}$, the equation becomes

$$(D^2 + 3D + 2)y = 5\cos x$$

Hence,

$$\text{P.I.} = \frac{1}{D^2 + 3D + 2}(5\cos x)$$

For trigonometric functions $\cos(ax)$ or $\sin(ax)$, use $D^2 \to -a^2$.
Therefore,

$$\text{P.I.} = \frac{5}{(-1) + 3D + 2} \cos x = \frac{5}{3D + 1} \cos x$$

Multiply numerator and denominator by the conjugate operator $(3D - 1)$:

$$\text{P.I.} = \frac{5(3D - 1)}{(3D + 1)(3D - 1)} \cos x$$

$$\text{P.I.} = \frac{5(3D - 1)}{9D^2 - 1} \cos x$$

Again substitute $D^2 = -1$.

$$\text{P.I.} = \frac{5(3D - 1)}{-9 - 1} \cos x$$

$$\text{P.I.} = -\frac{1}{2}(3D - 1) \cos x$$

$$\text{P.I.} = -\frac{1}{2}[-3 \sin x - \cos x]$$

$$\text{P.I.} = \frac{3}{2} \sin x + \frac{1}{2} \cos x$$

Method 2

Assume the particular solution in the form

$$y_p = A \cos x + B \sin x$$

Then

$$y_p' = -A \sin x + B \cos x, \qquad y_p'' = -A \cos x - B \sin x$$

Substituting into differential equation,

$$(-A \cos x - B \sin x) + 3(-A \sin x + B \cos x) + 2(A \cos x + B \sin x) = 5 \cos x$$

$$(A + 3B) \cos x + (B - 3A) \sin x = 5 \cos x$$

Equating coefficients:

$$A + 3B = 5, \qquad B - 3A = 0$$

Solving the equations gives

$$A = \frac{1}{2}, \qquad B = \frac{3}{2}$$

Therefore,

$$y_p = \frac{1}{2} \cos x + \frac{3}{2} \sin x.$$

Therefore, the correct option is

$$\boxed{\text{(A) } 0.5 \cos x + 1.5 \sin x}$$

Key Points

- For RHS containing $\sin ax$ or $\cos ax$, assume $y_p = A \cos ax + B \sin ax$.
- In operator methods, $D^2 \to -a^2$ for trigonometric functions.

Q6.95.1 MCQ 1995 1M Ans: A ● ○ ○

The statement given is: $\frac{dy}{dx} = f(x, y)$ is a homogeneous differential equation if $f(x, y)$ depends only on $\frac{y}{x}$ or $\frac{x}{y}$.

We need to determine whether this statement is true or false.

Recall the definition of a first-order homogeneous differential equation.

A differential equation is called homogeneous if it can be expressed as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right) \text{ or equivalently as } \frac{dy}{dx} = G\left(\frac{x}{y}\right).$$

That is, the function on the right-hand side depends only on the ratio of the variables and not separately.

Therefore, the given statement is exactly the definition of a homogeneous first-order differential equation.

Hence, the statement is **True**.

$$\boxed{\text{(A) True}}$$

Key Points

- Such equations are solved using the substitution $y = vx$ or $v = \frac{y}{x}$.

Mistakes to be avoided

- Do not confuse a homogeneous differential equation with a homogeneous linear differential equation.

Q6.95.2 MCQ 1995 1M Ans: C ● ○ ○

The given differential equation is

$$f''(x) + 4f'(x) + 4f(x) = 0$$

We need to determine which of the given functions is a solution.

This is a second-order linear homogeneous differential equation with constant coefficients.

Assume a trial solution $f(x) = e^{mx}$.

Then

$$f'(x) = me^{mx}, \qquad f''(x) = m^2e^{mx}$$

Substituting into the differential equation,

$$m^2e^{mx} + 4me^{mx} + 4e^{mx} = 0$$

Dividing by $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 + 4m + 4 = 0$$

$$(m + 2)^2 = 0$$

Thus, the repeated root is $m = -2$.

For repeated roots, the general solution is

$$f(x) = (C_1 + C_2x)\,e^{-2x}$$

Now compare with the options:

A: $f(x) = e^{-2x}$ obtained by choosing $C_1 = 1,\ C_2 = 0$.

Hence, it is a solution.

B: $f_1(x) = e^{-2x},\ f_2(x) = e^{-2x}$

Both functions are identical, so they are not two distinct solutions.

C: $f_1(x) = e^{-2x},\ f_2(x) = xe^{-2x}$

Choosing $(C_1, C_2) = (1, 0)$ gives e^{-2x}, and choosing $(C_1, C_2) = (0, 1)$ gives xe^{-2x}.

Hence, both are valid and linearly independent solutions.
D: $f_1(x) = e^{-2x}$, $f_2(x) = e^{-x}$
Since e^{-x} is not of the form $(C_1 + C_2x)e^{-2x}$, it is not a solution.
Therefore, the correct option is

$$\boxed{\text{(C) } f_1(x) = e^{-2x},\ f_2(x) = xe^{-2x}}.$$

Q6.94.1 MCQ 1994 1M Ans: B ● ○ ○

The given differential equation is

$$\frac{dy}{dt} + 5y = 0, \text{ with the initial condition } y(0) = 1$$

We are required to find the solution $y(t)$.

Method 1

This is a first-order linear homogeneous differential equation.
Separating the variables,

$$\frac{dy}{y} = -5\,dt$$

Integrating both sides,

$$\int \frac{dy}{y} = -5 \int dt$$

$$\ln|y| = -5t + C$$

$$|y| = e^{-5t+C} = Ce^{-5t},$$

where C is a new arbitrary constant.
Thus,

$$y = Ce^{-5t}$$

Using the initial condition, $y(0) = 1$,

$$1 = Ce^0 \Rightarrow C = 1$$

Hence,

$$y = e^{-5t}.$$

Method 2

Assume a trial solution $y = e^{mt}$.
Substituting into the differential equation,

$$me^{mt} + 5e^{mt} = 0$$

Dividing by $e^{mt} \neq 0$,

$$m + 5 = 0$$

$$m = -5$$

Therefore, the general solution is

$$y = Ce^{-5t}$$

Applying $y(0) = 1$, gives $C = 1$.
Thus,

$$y = e^{-5t}.$$

$$\boxed{\text{(B) } e^{-5t}}$$

Key Points

- The solution represents exponential decay because the coefficient of y is positive.

Q6.94.2 | MCQ | 1994 | 1M | Ans: A | ● ○ ○

The statement given is:

If $f(x, y)$ is a homogeneous function of degree n, then

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf.$$

We need to determine whether the statement is true or false.

The given statement is exactly **Euler's Theorem for Homogeneous Functions**.

If a function $f(x, y)$ is homogeneous of degree n, then

$$f(\lambda x, \lambda y) = \lambda^n f(x, y),$$

for every scalar λ.

Differentiating this relation with respect to λ and then putting $\lambda = 1$, gives

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf(x, y)$$

Therefore, the given statement is **True**.

$$\boxed{\text{(A) True}}$$

Mistakes to be avoided

- Do not apply Euler's theorem to non-homogeneous functions.
- The degree n is the degree of the function, not the number of variables.

Q6.94.3 | MCQ | 1994 | 2M | Ans: D | ● ● ○

The given differential equation is

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0 \text{ with the initial conditions } y(0) = 1,\ y'(0) = 2.$$

We are required to determine the solution $y(t)$.

This is a second-order linear homogeneous differential equation with constant coefficients.

Assume a trail solution $y = e^{mt}$.

Substituting into the differential equation,

$$m^2e^{mt} + 2me^{mt} + e^{mt} = 0$$

Dividing by $e^{mt} \neq 0$, the auxiliary equation is

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0$$

Therefore, the repeated root is $m = -1$.

For repeated roots, the general solution is

$$y = (C_1 + C_2t)e^{-t}$$

Differentiate using the product rule,

$$y' = C_2e^{-t} - (C_1 + C_2t)e^{-t}$$

$$y' = (C_2 - C_1 - C_2 t)e^{-t}$$

Using the initial condition $y(0) = 1$,

$$1 = (C_1 + 0)e^0 \Rightarrow C_1 = 1$$

Now applying $y'(0) = 2$,

$$2 = (C_2 - C_1)e^0$$

$$2 = C_2 - 1 \Rightarrow C_2 = 3$$

Therefore,

$$y = (1 + 3t)e^{-t} = e^{-t} + 3te^{-t}.$$

$$\boxed{\text{(D) } e^{-t} + 3te^{-t}}$$

Q6.93.1 MCQ 1993 1M Ans: B ● ○ ○

The given differential equation is

$$y'' + y = 0, \text{ with the boundary conditions } y(0) = 0,\ y(\lambda) = 0$$

We are required to find the values of λ for which the boundary value problem has a **non-trivial solution** (i.e., a solution other than $y \equiv 0$).

Assume a trail solution $y = e^{mx}$.

Substituting into the differential equation,

$$m^2 e^{mx} + e^{mx} = 0$$

Dividing by $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 + 1 = 0$$

$$m = \pm i$$

Therefore, the general solution is

$$y = C_1 \cos x + C_2 \sin x$$

Using the boundary condition $y(0) = 0$,

$$0 = C_1 \cos 0 + C_2 \sin 0 \Rightarrow C_1 = 0$$

Now applying $y(\lambda) = 0$,

$$0 = C_2 \sin \lambda$$

For a **non-trivial solution**, $C_2 \neq 0$.

Therefore,

$$\sin \lambda = 0$$

This occurs when

$$\lambda = n\pi, \qquad n = 1, 2, 3, \ldots$$

Therefore, the correct option is

$$\boxed{\text{(B) } m\pi}.$$

Mistakes to be avoided

- Do not conclude $C_2 = 0$ from $C_2 \sin \lambda = 0$. Since the question asks for a non-trivial solution, we require $C_2 \neq 0$.

Q6.93.2 MCQ 1993 1M Ans: B ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + \sin y = 0$$

We are required to classify the differential equation.
A differential equation is **linear** if:

- The dependent variable y and its derivatives appear only to the first power.
- The dependent variable and its derivatives are not inside nonlinear functions such as

$$\sin y, \quad e^y, \quad y^2, \quad \ln y, \quad y\frac{dy}{dx}, \quad \left(\frac{dy}{dx}\right)^2.$$

In the given equation, $\sin y$ is a nonlinear function of the dependent variable y.
Therefore, the equation is **Non-linear**.

$$\boxed{\text{(B) Non-Linear}}$$

Q6.26.1 | MCQ | 2026 | 2M | Ans: A | ● ○ ○

The given differential equation is a Cauchy–Euler equation:

$$x^2\frac{d^2y}{dx^2} - 6y = 0$$

Let $x = e^z$. Then $z = \ln x$.
Using the standard transformations,

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y,$$

where $D = \dfrac{d}{dz}$.
Hence, the given equation becomes

$$(D^2 - D - 6)y = 0$$

The auxiliary equation is

$$m^2 - m - 6 = 0$$

$$(m-3)(m+2) = 0$$

Therefore, the roots are $m = 3, -2$.
Hence,

$$y = C_1e^{-2z} + C_2e^{3z}$$

Since $e^z = x$,

$$y = C_1x^{-2} + C_2x^3 = \frac{C_1}{x^2} + C_2x^3.$$

Thus, the general solution is

$$y = ax^3 + \frac{b}{x^2}.$$

$$\boxed{\text{(A) } y(x) = ax^3 + \frac{b}{x^2}}$$

Key Points

- A Cauchy–Euler differential equation can be transformed into a constant-coefficient differential equation by substituting $x = e^z$.
- The identities $x\dfrac{dy}{dx} = Dy$ and $x^2\dfrac{d^2y}{dx^2} = D(D-1)y$ simplify the solution process.

Q6.26.2 | MSQ | 2026 | 1M | Ans: A,C | ● ○ ○

The given partial differential equation is

$$\frac{\partial^2u}{\partial x^2} - \frac{\partial^2u}{\partial y^2} = 0$$

Verify each option by computing the second-order partial derivatives.
For **Option A**, $u = (x+y)^5$.

$$\frac{\partial u}{\partial x} = 5(x+y)^4 \Rightarrow \frac{\partial^2u}{\partial x^2} = 20(x+y)^3$$

Similarly,

$$\frac{\partial u}{\partial y} = 5(x+y)^4 \Rightarrow \frac{\partial^2u}{\partial y^2} = 20(x+y)^3$$

Hence,

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 20(x+y)^3 - 20(x+y)^3 = 0$$

Hence, Option A is a solution.
For **Option B**, $u = (x-2y)^3$.

$$\frac{\partial^2 u}{\partial x^2} = 6(x-2y)$$

$$\frac{\partial^2 u}{\partial y^2} = 24(x-2y)$$

Thus,

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 6(x-2y) - 24(x-2y) = -18(x-2y) \neq 0$$

Hence, Option B is not a solution.
For **Option C**, $u = \cos(x+y)$.

$$\frac{\partial^2 u}{\partial x^2} = -\cos(x+y)$$

$$\frac{\partial^2 u}{\partial y^2} = -\cos(x+y)$$

Since

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = -\cos(x+y) + \cos(x+y) = 0$$

Hence, Option C is also a solution.
For **Option D**, $u = \sin(x-2y)$.

$$\frac{\partial^2 u}{\partial x^2} = -\sin(x-2y)$$

$$\frac{\partial^2 u}{\partial y^2} = -4\sin(x-2y)$$

Therefore,

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = -\sin(x-2y) + 4\sin(x-2y) = 3\sin(x-2y) \neq 0.$$

Hence, Option D is not a solution.

$$\boxed{\text{(A) } (x+y)^5 \text{ ; (C) } \cos(x+y)}$$

Q6.26.3 | NAT | 2026 | 2M | Ans: 0.39 | ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + xy = x \text{ with the initial condition } y(0) = 0$$

Comparing with the standard linear differential equation

$$\frac{dy}{dx} + Py = Q, \quad \text{we have } P = x, \; Q = x$$

The integrating factor (I.F.) is

$$\text{I.F.} = e^{\int P\,dx} = e^{\int x\,dx} = e^{x^2/2}$$

The complete solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.}\, dx + C$$

$$y\, e^{x^2/2} = \int x\, e^{x^2/2}\, dx + C$$

Let $t = \dfrac{x^2}{2}$, so that $dt = x\, dx$.
Therefore,

$$y\, e^{x^2/2} = \int e^t\, dt + C = e^t + C = e^{x^2/2} + C$$

$$y = 1 + Ce^{-x^2/2}$$

Using the initial condition $y(0) = 0$,

$$0 = 1 + C \implies C = -1$$

Hence,

$$y = 1 - e^{-x^2/2}.$$

At $x = 1$,

$$y(1) = 1 - e^{-1/2} = 1 - \frac{1}{\sqrt{e}} \approx 1 - \frac{1}{1.6487} \approx 0.39347.$$

Rounded off to two decimal places,

$$\boxed{0.39}$$

Key Points

- A first-order linear differential equation of the form $\dfrac{dy}{dx} + Py = Q$ is solved using the integrating factor $e^{\int P\, dx}$.

Q6.25.1 MSQ 2025 1M Ans: B ● ○ ○

From the given PDEs, we need to find the equation(s) that is/are second-order, linear, and homogeneous.
Examine each option individually.
For **Option A**,

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + xy$$

The presence of the forcing term xy makes it *non-homogeneous*.
Hence, Option A is incorrect.
For **Option B**,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

The equation is: second order, linear, and homogeneous.
Hence, Option B is correct.
For **Option C**,

$$\frac{\partial u}{\partial t} = c\frac{\partial u}{\partial x}$$

The equation is first order. Therefore, it cannot belong to the required class.
Hence, Option C is incorrect.
For **Option D**,

$$\left(\frac{\partial^2 u}{\partial t^2}\right)^2 = c^2\frac{\partial^2 u}{\partial x^2}$$

Since the highest-order derivative appears squared, the equation is *non-linear*.
Hence, Option D is incorrect.

$$\boxed{\text{(B) } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0}$$

Key Points

- The order of a PDE is determined by the highest-order derivative present.
- A PDE is linear if the dependent variable and all its derivatives appear only to the first power and are not multiplied together.
- A linear PDE is homogeneous when all terms involving the dependent variable and its derivatives are on one side and the other side is zero.

Q6.25.2 | NAT | 2025 | 2M | Ans: 5.83 | ● ○ ○

The given initial value problem is

$$y'' + 0.8y' + 0.16y = 0 \text{ with initial conditions } y(0) = 3,\ y'(0) = 4.5$$

The auxiliary equation is given by

$$m^2 + 0.8m + 0.16 = 0$$

$$(m + 0.4)^2 = 0$$

Hence, the repeated root is $m = -0.4$.
Therefore, the complementary function is

$$y = e^{-0.4x}(C_1 + C_2 x)$$

Differentiating,

$$y' = e^{-0.4x}\left[C_2 - 0.4(C_1 + C_2 x)\right]$$

Using the initial condition $y(0) = 3,\ C_1 = 3$
Also, using $y'(0) = 4.5$,

$$4.5 = C_2 - 0.4(3) \Rightarrow\ C_2 = 5.7$$

Hence,

$$y = e^{-0.4x}(3 + 5.7x).$$

At $x = 1$,

$$y(1) = e^{-0.4}(3 + 5.7) = e^{-0.4}(8.7) \approx 0.6703 \times 8.7 \approx 5.83.$$

$$\boxed{5.83}$$

Mistakes to be avoided

- Do not forget the extra factor of x in the complementary function for repeated roots.

Q6.25.3 | NAT | 2025 | 1M | Ans: 3 | ● ○ ○

The given differential equation is

$$\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^6 + \left(\frac{dy}{dx}\right)^4 + y = 0$$

The order of a differential equation is the order of the highest-order derivative present in the equation.
Here, the highest-order derivative is $\dfrac{d^3y}{dx^3}$.
Hence, the order of the given differential equation is **3.**

$$\boxed{3}$$

Key Points

- The order of a differential equation is determined solely by the highest-order derivative present.
- The powers of derivatives do not affect the order of the differential equation.

Q6.25.4 MCQ 2025 2M Ans: A ● ○ ○

The given differential equation is

$$\frac{dy}{dx} = e^{x-y}$$

We need to find its solution y.
Separating the variables,

$$\frac{dy}{dx} = e^{x-y} = e^x e^{-y}$$
$$e^y\,dy = e^x\,dx$$

Integrating both sides,

$$\int e^y\,dy = \int e^x\,dx$$
$$e^y = e^x + C$$

Taking the natural logarithm,

$$y = \ln(e^x + C)$$

Hence, the correct option is

$$\boxed{\text{(A) } y = \ln(e^x + \text{Constant})}$$

Mistakes to be avoided

- Do not write $\ln(e^x) + C$ after taking logarithms; $\ln(a+b) \neq \ln a + \ln b$.

Q6.24.1 MCQ 2024 1M Ans: B ● ○ ○

The given second-order differential equation is

$$\frac{\partial^2 u}{\partial x^2} = 2$$

where u is function of both x and y.

Method 1

Integrate the given partial differential equation successively with respect to x.

$$\frac{\partial u}{\partial x} = 2x + f(y),$$

where $f(y)$ is an arbitrary function of y.
Integrating again with respect to x,

$$u = \int (2x + f(y))\,dx = x^2 + xf(y) + h(y),$$

where $h(y)$ is another arbitrary function of y.

Method 2

Verify the options by differentiation.
For $u = x^2 + xf(y) + h(y)$, we obtain

$$\frac{\partial u}{\partial x} = 2x + f(y),$$

$$\frac{\partial^2 u}{\partial x^2} = 2$$

which satisfies the given partial differential equation exactly.
Hence, this is the required general solution.
Therefore,

$$\boxed{\text{(B) } u = x^2 + xf(y) + h(y)}$$

Key Points

- While integrating a PDE with respect to one variable, the constant of integration becomes an arbitrary function of the remaining independent variables.
- Each successive integration introduces a new arbitrary function.

Q6.24.2 MSQ 2024 1M Ans: A,D ● ● ○

The given PDE is

$$x\frac{\partial^2 f}{\partial x^2} + y\frac{\partial^2 f}{\partial y^2} = \frac{x^2 + y^2}{2}$$

Write the given equation in the standard second-order linear PDE form.

$$x\frac{\partial^2 f}{\partial x^2} + 0 \cdot \frac{\partial^2 f}{\partial x \partial y} + y\frac{\partial^2 f}{\partial y^2} = \frac{x^2 + y^2}{2}$$

Comparing with

$$a\frac{\partial^2 u}{\partial x^2} + b\frac{\partial^2 u}{\partial x \partial y} + c\frac{\partial^2 u}{\partial y^2} + d\frac{\partial u}{\partial x} + e\frac{\partial u}{\partial y} + f = g,$$

we obtain $a = x,\ b = 0,\ c = y$.
The discriminant is

$$b^2 - 4ac = 0^2 - 4xy = -4xy$$

Use the discriminant criterion.

- If $b^2 - 4ac < 0$, the PDE is elliptic.
- If $b^2 - 4ac = 0$, the PDE is parabolic.
- If $b^2 - 4ac > 0$, the PDE is hyperbolic.

Checking the Options,
A and B: For $x > 0$ and $y > 0,\ -4xy < 0$.
So the PDE is **elliptic**. Hence, Option A is correct.
C: For $x = 0$ and $y > 0,\ -4xy = 0$.
So the PDE is **parabolic**. Hence, Option C is incorrect.
D: For $x < 0$ and $y > 0,\ -4xy > 0$.
So the PDE is **hyperbolic**. Hence, Option D is correct.

$$\boxed{\text{(A) elliptic for } x > 0 \text{ and } y > 0, \text{ (D) hyperbolic for } x < 0 \text{ and } y > 0}$$

Key Points

- Classification criteria:

$$\begin{cases} b^2 - 4ac < 0 & \Rightarrow \text{Elliptic,} \\ b^2 - 4ac = 0 & \Rightarrow \text{Parabolic,} \\ b^2 - 4ac > 0 & \Rightarrow \text{Hyperbolic.} \end{cases}$$

Q6.24.3 MCQ | 2024 | 1M | Ans: B | ● ● ○

Given two ODEs are:

$$P : \frac{dy}{dx} = \frac{x^4 + 3x^2y^2 + 2y^4}{x^3y},$$

$$Q : \frac{dy}{dx} = -\frac{y^2}{x^2}$$

We need to find whether P is a homogeneous ODE and Q is an exact ODE.
Classify equation P:
A first-order differential equation of the form

$$\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$$

is homogeneous if both $f(x,y)$ and $g(x,y)$ are homogeneous functions of the same degree.
Here,

$$f(x,y) = x^4 + 3x^2y^2 + 2y^4, \qquad g(x,y) = x^3y$$

Now,

$$f(kx,ky) = (kx)^4 + 3(kx)^2(ky)^2 + 2(ky)^4 = k^4 f(x,y),$$

$$g(kx,ky) = (kx)^3(ky) = k^4 g(x,y)$$

Since both are homogeneous functions of degree 4, P is a homogeneous ODE.
Test equation Q for exactness:
Rearranging,

$$y^2\,dx + x^2\,dy = 0$$

Comparing with

$$M\,dx + N\,dy = 0,$$

we have $M = y^2,\ N = x^2$.
Now,

$$\frac{\partial M}{\partial y} = 2y, \qquad \frac{\partial N}{\partial x} = 2x$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, Q is not an exact ODE.

$$\boxed{\text{(B) P is a homogeneous ODE and Q is not an exact ODE}}$$

Key Points

- A first-order ODE of the form $\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$ is homogeneous if f and g are homogeneous functions of the same degree.
- A differential equation $M\,dx + N\,dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Q6.23.1 NAT | 2023 | 1M | Ans: 0.446 | ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + \alpha xy = 0 \text{ with } y(0) = 1,\ y(1) = 0.8$$

Separating the variables,

$$\frac{dy}{dx} = -\alpha xy$$

$$\frac{dy}{y} = -\alpha x\, dx$$

Integrating both sides,

$$\int \frac{dy}{y} = -\alpha \int x\, dx$$

$$\ln y = -\frac{\alpha x^2}{2} + C$$

Using the initial condition $y(0) = 1$,

$$\ln 1 = C = 0$$

Using $y(1) = 0.8$,

$$\ln(0.8) = -\frac{\alpha}{2}$$

$$\alpha = -2\ln(0.8) = -\ln(0.8)^2 \approx 0.446287$$

Rounded to three decimal places,

$$\boxed{0.446}$$

Q6.23.2 MCQ | 2023 | 2M | Ans: A | ● ● ○

Consider the given differential equation:

$$\frac{d^3y}{dx^3} - 5.5\frac{d^2y}{dx^2} + 9.5\frac{dy}{dx} - 5y = 0$$

Express the differential equation in operator form by letting $D = \frac{d}{dx}$.
Then,

$$(D^3 - 5.5D^2 + 9.5D - 5)y = 0$$

The auxiliary equation is

$$m^3 - 5.5m^2 + 9.5m - 5 = 0$$

Using the Rational Root Theorem, test the possible roots.
For $m = 1$,

$$1 - 5.5 + 9.5 - 5 = 0$$

Hence, $(m-1)$ is a factor.
Dividing the polynomial by $(m-1)$,

$$m^3 - 5.5m^2 + 9.5m - 5 = (m-1)(m^2 - 4.5m + 5)$$

Now solve the quadratic equation: $m^2 - 4.5m + 5 = 0$.
Using the quadratic formula,

$$m = \frac{4.5 \pm \sqrt{(4.5)^2 - 4(1)(5)}}{2} = \frac{4.5 \pm 0.5}{2}$$

$$m = 2, \qquad m = 2.5$$

Hence, the complementary function is

$$y = C_1 e^{x} + C_2 e^{2x} + C_3 e^{2.5x}$$

Comparing with given solution

$$y = C_1 e^{2.5x} + C_2 e^{\alpha x} + C_3 e^{\beta x},$$

the remaining exponents are $\alpha = 1,\ \beta = 2$.
Therefore,

$$\boxed{\text{(A) 1 and 2}}$$

Key Points

- Each distinct real root m_i contributes a term $C_i e^{m_i x}$ to the complementary function.

Q6.22.1 MCQ 2022 1M Ans: A ● ● ○

Consider

$$z = \sin(y + it) + \cos(y - it),$$

where $i = \sqrt{-1}$.

Differentiate the given expression directly with respect to t.

$$\frac{\partial z}{\partial t} = i\cos(y + it) + i\sin(y - it).$$

Differentiating once again,

$$\frac{\partial^2 z}{\partial t^2} = i\,[-\sin(y + it)]\,(i) + i\,[\cos(y - it)]\,(-i)$$

$$\frac{\partial^2 z}{\partial t^2} = \sin(y + it) + \cos(y - it)$$

Now differentiate with respect to y:

$$\frac{\partial z}{\partial y} = \cos(y + it) - \sin(y - it)$$

$$\frac{\partial^2 z}{\partial y^2} = -\sin(y + it) - \cos(y - it)$$

Therefore,

$$\frac{\partial^2 z}{\partial t^2} + \frac{\partial^2 z}{\partial y^2} = [\sin(y + it) + \cos(y - it)] + [-\sin(y + it) - \cos(y - it)] = 0$$

Therefore, the correct option is

$$\boxed{\text{(A) } \frac{\partial^2 z}{\partial t^2} + \frac{\partial^2 z}{\partial y^2} = 0}$$

Mistakes to be avoided

- Be careful with the sign while differentiating $(y - it)$, since

$$\frac{\partial}{\partial t}(y - it) = -i.$$

Q6.22.2 MCQ 2022 1M Ans: A ● ○ ○

The given differential equation is

$$\frac{d^3 y}{dx^3} + x\left(\frac{dy}{dx}\right)^{3/2} + x^2 y = 0$$

To determine the *order*, identify the highest-order derivative present.

The highest-order derivative is $\frac{d^3 y}{dx^3}$.

Hence, Order = 3.

To determine the *degree*, first remove the fractional power in the equation.

Rearranging and Squaring on both sides,

$$\frac{d^3y}{dx^3} + x^2y = -x\left(\frac{dy}{dx}\right)^{3/2}$$

$$\left(\frac{d^3y}{dx^3} + x^2y\right)^2 = x^2\left(\frac{dy}{dx}\right)^3$$

Now the equation is polynomial in all derivatives, and the highest-order derivative $\frac{d^3y}{dx^3}$ appears with power 2.
Therefore, Degree = 2.
Therefore, the correct option is

$$\boxed{\text{(A) an ordinary differential equation of order 3 and degree 2}}$$

Key Points

- The degree is defined only after expressing the equation as a polynomial in the derivatives.
- Fractional or radical powers of derivatives must be removed before determining the degree.

Mistakes to be avoided

- Do not determine the degree before eliminating fractional powers of derivatives.
- Do not confuse the exponent of the first derivative with the degree of the equation.

Q6.21.1 | MCQ | 2021 | 2M | Ans: A | ● ○ ○

Consider the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0 \text{ with the boundary conditions } y(0) = 1,\ y(1) = 3$$

Using the differential operator $D = \frac{d}{dx}$,

$$(D^2 + 2D + 1)y = 0$$

The auxiliary equation is

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0$$

Thus, the repeated root is $m = -1$.
Hence, the complementary function is

$$y = (C_1 + C_2x)e^{-x}$$

Applying the boundary conditions,
At $x = 0$,

$$1 = (C_1 + 0)e^0 \Rightarrow\ C_1 = 1$$

At $x = 1$,

$$3 = (1 + C_2)e^{-1}$$

$$1 + C_2 = 3e \Rightarrow\ C_2 = 3e - 1$$

Therefore,

$$y(x) = [1 + (3e - 1)x]\, e^{-x} = e^{-x} + (3e - 1)xe^{-x}.$$

Hence, the correct option is

$$\boxed{\text{(A) } e^{-x} + (3e - 1)xe^{-x}}$$

Mistakes to be avoided

- While substituting $x = 1$, remember that $e^{-1} = \frac{1}{e}$.

Q6.21.2 MCQ 2021 2M Ans: A ● ○ ○

Consider the differential equation

$$\frac{dy}{dx} - \frac{y}{x} = 1$$

Compare the equation with the standard linear form

$$\frac{dy}{dx} + Py = Q \quad \text{where } P = -\frac{1}{x},\ Q = 1$$

The integrating factor is

$$\text{I.F.} = e^{\int P\,dx} = e^{\int -1/x\,dx} = e^{-\ln x} = \frac{1}{x}$$

The complete solution is given by

$$y \times \text{I.F.} = \int Q \times \text{I.F.}\,dx + C$$

$$\frac{y}{x} = \int \frac{1}{x}\,dx + C = \ln x + C$$

Hence,

$$y = x(\ln x + C)$$

Since $C = \ln k$ for an arbitrary positive constant k,

$$y = x\ln(kx).$$

Therefore, the correct option is

$$\boxed{\text{(A) } y = x\ln(kx)}$$

Key Points

- The identity

$$\frac{d}{dx}\left(\frac{y}{x}\right) = \frac{1}{x}\frac{dy}{dx} - \frac{y}{x^2}$$

simplifies the solution considerably.

Q6.20.1 MCQ 2020 1M Ans: B ● ○ ○

Consider the given partial differential equation:

$$D(\theta)\frac{\partial^2\theta}{\partial z^2} + \frac{\partial K(\theta)}{\partial z} - \frac{\partial\theta}{\partial t} = 0,$$

where $D(\theta)$ and $K(\theta)$ are functions of θ.
Determine the order of the equation.
The highest-order derivative present is $\frac{\partial^2\theta}{\partial z^2}$.
Hence, the order of the PDE is 2.
Determine linearity of the equation.
Observe that the coefficient of the highest-order derivative $D(\theta)$, is a function of the dependent variable θ.
Hence, the given PDE is non-linear.
Therefore, the correct description is

$$\boxed{\text{(B) a second order non-linear equation}}$$

Q6.20.2 MCQ | 2020 | 2M | Ans: A | ● ○ ○

Consider the differential equation

$$\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0 \text{ with the initial conditions } x(0) = 0,\ \frac{dx}{dt}(0) = 10$$

Using the differential operator $D = \dfrac{d}{dt}$,

$$(D^2 - 5D + 6)x = 0$$

The auxiliary equation is

$$m^2 - 5m + 6 = 0$$

$$(m-2)(m-3) = 0$$

Hence, the roots are $m = 2,\ 3$.
Therefore, the complementary function is

$$x = C_1 e^{2t} + C_2 e^{3t}$$

Using the initial condition, $x(0) = 0$,

$$C_1 + C_2 = 0 \Rightarrow\ C_1 = -C_2$$

Differentiating,

$$\frac{dx}{dt} = 2C_1 e^{2t} + 3C_2 e^{3t}$$

Using $\dfrac{dx}{dt}(0) = 10$,

$$2C_1 + 3C_2 = 10$$

Substituting $C_1 = -C_2$, gives

$$C_2 = 10, \qquad C_1 = -10$$

Hence,

$$x(t) = -10e^{2t} + 10e^{3t}.$$

Therefore, the correct option is

$$\boxed{\text{(A) } -10e^{2t} + 10e^{3t}}$$

Q6.20.3 MCQ | 2020 | 1M | Ans: D | ● ○ ○

Consider the differential equation

$$\frac{d^2u}{dx^2} - 2x^2u + \sin x = 0$$

Rearrange the equation as

$$\frac{d^2u}{dx^2} - 2x^2u = -\sin x$$

Comparing with the standard linear differential equation

$$\frac{d^2u}{dx^2} + P(x)\frac{du}{dx} + Q(x)u = R(x),$$

we identify $P(x) = 0,\ Q(x) = -2x^2,\ R(x) = -\sin x$.
The dependent variable u and its derivatives appear only to the first power and are not multiplied together.
Hence, the equation is *linear*.
Further, because $R(x) \neq 0$, the equation is *nonhomogeneous*.
Therefore, the correct option is

$$\boxed{\text{(D) linear and nonhomogeneous}}$$

Key Points

- A linear ODE has the general form

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = g(x),$$

where the coefficients depend only on the independent variable.

- If $g(x) = 0$, the equation is homogeneous; otherwise, it is nonhomogeneous.

Mistakes to be avoided

- Do not classify an equation as non-linear merely because its coefficients are functions of x.

Q6.20.4 MCQ 2020 1M Ans: A ● ○ ○

The given partial differential equation is

$$\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial x^2}, \quad y \geq 0, \; x_1 \leq x \leq x_2$$

Method 1

The equation is first order with respect to the variable y and second order with respect to the variable x.
Since y represents the evolution variable, one initial condition is required along the y-direction.
Also, because the equation is second order in x, two boundary conditions are required at the spatial boundaries

$$x = x_1 \text{ and } x = x_2$$

Hence, the problem requires one initial condition and two boundary conditions.

Method 2

Recognize that

$$\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial x^2}$$

is the one-dimensional heat (diffusion) equation.
For a well-posed heat conduction problem,

- an initial temperature distribution $u(x, 0)$ is specified, and
- boundary conditions are prescribed at $x = x_1$ and $x = x_2$.

Thus, one initial condition and two boundary conditions are necessary.
Therefore, the correct option is

$$\boxed{\text{(A) one initial condition and two boundary conditions}}$$

Key Points

- The number of initial conditions equals the order of the PDE with respect to the time (or evolution) variable.
- The number of boundary conditions equals the order of the PDE with respect to the spatial variable.

Mistakes to be avoided

- A second-order derivative with respect to x requires two independent boundary conditions, not two initial conditions.

Q6.20.5 MCQ 2020 2M Ans: A ● ○ ○

Consider the ordinary differential equation

$$6\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$$

Using the differential operator $D = \frac{d}{dx}$,

$$(6D^2 + D - 1)y = 0$$

The auxiliary equation is

$$6m^2 + m - 1 = 0$$

$$(2m + 1)(3m - 1) = 0$$

Hence, the roots are $m = -\frac{1}{2}$, $m = \frac{1}{3}$.

Since the roots are real and distinct, the complementary function is

$$y = C_1 e^{x/3} + C_2 e^{-x/2}$$

Therefore, the correct option is

$$\boxed{\text{(A) } y(x) = C_1\, e^{x/3} + C_2\, e^{-x/2}}$$

Q6.19.1 MCQ 2019 1M Ans: C ● ○ ○

Consider a two-dimensional flow through an isotropic soil in the x- and z-directions, where h denotes the hydraulic head.

The general three-dimensional Laplace equation is

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = 0$$

Since the flow is two-dimensional (x-z plane), there is no variation along the y-direction, i.e.,

$$\frac{\partial h}{\partial y} = 0 \quad \Rightarrow \quad \frac{\partial^2 h}{\partial y^2} = 0.$$

Therefore, the Laplace equation for two-dimensional flow reduces to

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial z^2} = 0$$

Therefore, the correct option is

$$\boxed{\text{(C) } \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial z^2} = 0}$$

Key Points

- Laplace's equation governs steady seepage through homogeneous and isotropic soils.
- The sum of the second partial derivatives of the hydraulic head with respect to the flow coordinates is zero.

Mistakes to be avoided

- Do not confuse the continuity equation with Laplace's equation.

Q6.19.2 NAT 2019 2M Ans: 6 ● ● ○

Consider the Cauchy–Euler differential equation

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0 \text{ with the conditions } y(1) = 0,\ y(2) = 2$$

Method 1

Use the standard transformation $x = e^t,\ t = \ln x$,
and let $D = \frac{d}{dt}$.

The Cauchy–Euler transformations are

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

Substituting into the given equation,

$$D(D-1)y - 2Dy + 2y = 0$$

$$(D^2 - 3D + 2)y = 0$$

The auxiliary equation is

$$m^2 - 3m + 2 = 0$$

$$m = 1,\ 2$$

Hence,

$$y = C_1e^t + C_2e^{2t}$$

Since $e^t = x$,

$$y = C_1x + C_2x^2$$

Using the given conditions,
At $x = 1$,

$$C_1 + C_2 = 0$$

At $x = 2$,

$$2C_1 + 4C_2 = 2$$

Solving,

$$C_1 = -1, \qquad C_2 = 1$$

Hence,

$$y = x^2 - x$$

Therefore,

$$y(3) = 3^2 - 3 = 9 - 3 = 6.$$

Method 2

Assume a trial solution $y = x^m$.
Then,

$$\frac{dy}{dx} = mx^{m-1}, \qquad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Substituting into the differential equation,

$$m(m-1)x^m - 2mx^m + 2x^m = 0$$

Since $x^m \neq 0$, the auxiliary equation is

$$m^2 - 3m + 2 = 0$$

$$m = 1,\ 2$$

Therefore,

$$y = C_1x + C_2x^2$$

Using the given conditions,
At $x = 1$,

$$C_1 + C_2 = 0$$

At $x = 2$,

$$2C_1 + 4C_2 = 2$$

Solving,

$$C_1 = -1, \qquad C_2 = 1$$

Hence,

$$y = x^2 - x$$

Therefore,

$$y(3) = 3^2 - 3 = 9 - 3 = 6.$$

$$\boxed{6}$$

Key Points

- A Cauchy–Euler differential equation can be solved either by the substitution $x = e^t$ or by assuming a trial solution $y = x^m$.

Q6.19.3 MCQ 2019 2M Ans: D ● ○ ○

Consider the differential equation

$$(x \ln x)\left(\frac{dy}{dx}\right) = y$$

Method 1

Separate the variables and Integrate on both sides

$$\int \frac{dy}{y} = \int \frac{dx}{x \ln x}$$

$$\ln|y| = \int \frac{dx}{x \ln x}$$

Let $t = \ln x$. Then, $dt = \frac{dx}{x}$.
Hence,

$$\ln|y| = \int \frac{dt}{t} = \ln|t| + C$$

Substituting $t = \ln x$,

$$\ln|y| = \ln|\ln x| + C$$

$$y = K \ln x,$$

where $K = e^C$ is an arbitrary constant.

Method 2

Rewrite the equation as

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x \ln x}$$

Recognize that

$$\frac{d}{dx}(\ln|y|) = \frac{1}{y}\frac{dy}{dx}, \quad \text{and} \quad \frac{d}{dx}(\ln|\ln x|) = \frac{1}{x\ln x}$$

Therefore,

$$\ln|y| = \ln|\ln x| + C$$
$$y = K\ln x,$$

where $K = e^C$ is an arbitrary constant.
Therefore, the correct option is

$$\boxed{\text{(D) } y = K\ln x}$$

Mistakes to be avoided

- Do not write $\int \frac{dx}{x\ln x} = \ln x$. The correct result is $\ln|\ln x|$.

Q6.18.1 MCQ 2018 2M Ans: D ● ● ●

Consider the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} = 25\frac{\partial^2 u}{\partial t^2} \text{ with the initial conditions } u(x,0) = 3x,\ \frac{\partial u}{\partial t}(x,0) = 3$$

Rewrite the given equation in the standard one-dimensional wave equation form:

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$$

We obtain

$$\frac{\partial^2 u}{\partial x^2} = 25\frac{\partial^2 u}{\partial t^2}$$
$$\frac{\partial^2 u}{\partial t^2} = \frac{1}{25}\frac{\partial^2 u}{\partial x^2}$$

so that $c = 1/5$.
Using D'Alembert's formula,

$$u(x,t) = \frac{1}{2}[f(x-ct) + f(x+ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds,$$

where $f(x) = u(x,0) = 3x,\ g(x) = \frac{\partial u}{\partial t}(x,0) = 3$
Substituting,

$$u(x,t) = \frac{1}{2}\left[3\left(x-\frac{t}{5}\right) + 3\left(x+\frac{t}{5}\right)\right] + \frac{5}{2}\int_{x-\frac{t}{5}}^{x+\frac{t}{5}} 3\,ds$$

The first term becomes $\frac{1}{2}(6x) = 3x$.
Since $g(s) = 3$,

$$\frac{5}{2}\int_{x-\frac{t}{5}}^{x+\frac{t}{5}} 3\,ds = 3\left[\left(x+\frac{t}{5}\right) - \left(x-\frac{t}{5}\right)\right]$$
$$= \frac{5}{2}\times\frac{6t}{5} = 3t$$

Therefore,

$$u(x,t) = 3x + 3t$$

At $x = 1,\ t = 1$,

$$u(1,1) = 3 + 3 = 6.$$

Therefore, the correct option is

$$\boxed{\text{(D) } 6}$$

Key Points

- In D'Alembert's solution,

$$f(x) = u(x, 0), \qquad g(x) = \frac{\partial u}{\partial t}(x, 0).$$

Q6.18.2 | NAT | 2018 | 2M | Ans: 0.368 | ● ● ○

Consider the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0 \text{ with the boundary conditions } y(0) = 1, \ \frac{dy}{dx}(0) = -1$$

Using the differential operator $D = \dfrac{d}{dx}$,

$$(D^2 + 2D + 1)y = 0$$

The auxiliary equation is

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0$$

Hence, the repeated root is $m = -1$.
Therefore, the complementary function is

$$y = (C_1 + C_2x)e^{-x}$$

Differentiating,

$$\frac{dy}{dx} = -(C_1 + C_2x)e^{-x} + C_2e^{-x} = [-C_1 + C_2 - C_2x]\, e^{-x}$$

Using the boundary conditions,
At $x = 0$, $C_1 = 1$.
Also,

$$-1 = -C_1 + C_2$$

$$-1 = -1 + C_2 \Rightarrow C_2 = 0$$

Hence,

$$y = e^{-x}.$$

At $x = 1$,

$$y(1) = e^{-1} = \frac{1}{e} \approx 0.3679.$$

Rounded to three decimal places,

$$\boxed{0.368}$$

Q6.18.3 | MCQ | 2018 | 1M | Ans: C | ● ○ ○

Consider the differential equation

$$x\frac{dy}{dx} + y = 0,$$

passing through the point $(1, 1)$.

Method 1

Separating the variables and Integrating

$$x\frac{dy}{dx} = -y$$

$$\frac{dy}{y} = -\frac{dx}{x}$$

$$\int \frac{dy}{y} = -\int \frac{dx}{x}$$

$$\ln|y| = -\ln|x| + C$$

Using logarithmic properties,

$$\ln|xy| = C$$

$$y = \frac{K}{x},$$

where K is an arbitrary constant.
Applying the condition $y(1) = 1$,

$$1 = \frac{K}{1} \Rightarrow K = 1$$

Hence,

$$y = x^{-1}.$$

Method 2

Observe that

$$x\frac{dy}{dx} + y = \frac{d}{dx}(xy)$$

Therefore,

$$\frac{d}{dx}(xy) = 0$$

Integrating directly,

$$xy = C.$$

Using the point $(1, 1)$, gives $C = 1$.
Hence,

$$y = \frac{1}{x} = x^{-1}.$$

Hence, the correct option is

$$\boxed{\text{(C) } x^{-1}}$$

Key Points

- Alternatively, recognize that

$$\frac{d}{dx}(xy) = x\frac{dy}{dx} + y,$$

which provides a much shorter solution.

Q6.17.1 NAT 2017 1M Ans: 36 ● ○ ○

Consider the partial differential equation

$$3\frac{\partial^2\phi}{\partial x^2} + B\frac{\partial^2\phi}{\partial x\partial y} + 3\frac{\partial^2\phi}{\partial y^2} + 4\phi = 0$$

Compare the given equation with the standard second-order linear PDE

$$a\frac{\partial^2 u}{\partial x^2} + b\frac{\partial^2 u}{\partial x\partial y} + c\frac{\partial^2 u}{\partial y^2} + \cdots = 0$$

Hence, $a = 3,\ b = B,\ c = 3$.
For the equation to be parabolic, $b^2 - 4ac = 0$.

Substituting the coefficients,

$$B^2 - 4(3)(3) = 0$$
$$B^2 = 36.$$

Hence,

$$\boxed{36}$$

Mistakes to be avoided

- Do not include lower-order terms while computing the discriminant.
- The question asks for B^2, not B.

Q6.17.2 MCQ 2017 2M Ans: D ● ○ ○

Consider the differential equation

$$\frac{dQ}{dt} + Q = 1 \text{ with the initial condition } Q(0) = 0$$

Method 1

Compare the equation with the standard linear differential equation

$$\frac{dQ}{dt} + PQ = R, \quad \text{where} \quad P = 1, \ R = 1$$

The integrating factor is

$$\text{I.F.} = e^{\int P\,dt} = e^t.$$

The complete solution is given by

$$Q \times \text{I.F.} = \int R \times \text{I.F.}\,dt + C$$
$$Qe^t = \int e^t \cdot dt + C$$
$$Qe^t = e^t + C$$
$$Q = 1 + Ce^{-t}$$

Using the initial condition, $Q(0) = 0$, we get

$$0 = 1 + C \Rightarrow C = -1$$

Therefore,

$$Q(t) = 1 - e^{-t}.$$

Method 2

The complementary function is

$$Q_c = Ce^{-t}$$

Assuming a constant particular integral, $Q_p = A$, and substituting gives

$$A = 1$$

Hence,

$$Q = Ce^{-t} + 1$$

Applying $Q(0) = 0$, yields

$$C = -1$$

Therefore,

$$Q(t) = 1 - e^{-t}.$$

Therefore, the correct option is

$$\boxed{\text{(D) } Q(t) = 1 - e^{-t}}$$

Key Points

- A first-order linear differential equation can be solved using either the integrating factor method or the complementary function plus particular integral approach.
- For a constant forcing term, the particular integral is usually a constant.

Q6.17.3 MCQ 2017 2M Ans: A ● ● ○

Consider the differential equation

$$y'' - 4y' + 3y = 2t - 3t^2$$

Method 1

Using the differential operator $D = \frac{d}{dt}$,

$$(D^2 - 4D + 3)y = 2t - 3t^2$$

The particular integral is

$$\text{P.I.} = \frac{1}{D^2 - 4D + 3}(2t - 3t^2)$$

Factor the operator as

$$\text{P.I.} = \frac{1}{3}\left(1 + \frac{D^2 - 4D}{3}\right)^{-1}(2t - 3t^2)$$

Using the binomial expansion, $(1 + x)^{-1} = 1 - x + x^2 - \cdots$,
Noting that the forcing function is a quadratic polynomial, only a finite number of terms survive.
Thus,

$$\text{P.I.} = \frac{1}{3}\left[1 - \frac{D^2 - 4D}{3} + \frac{(D^2 - 4D)^2}{9}\right](2t - 3t^2)$$

Now,

$$D(2t - 3t^2) = 2 - 6t,$$

$$D^2(2t - 3t^2) = -6,$$

$$D^3(2t - 3t^2) = 0.$$

Therefore,

$$(D^2 - 4D)(2t - 3t^2) = -6 - 4(2 - 6t) = 24t - 14,$$

$$(D^2 - 4D)^2(2t - 3t^2) = 16(-6) = -96$$

Substituting,

$$\text{P.I.} = \frac{1}{3}\left[2t - 3t^2 - \frac{1}{3}(24t - 14) + \frac{1}{9}(-96)\right]$$

$$\text{P.I.} = -t^2 - 2t - 2$$

Method 2

Assume a quadratic particular solution $y_p = At^2 + Bt + C$.
Then,

$$y'_p = 2At + B, \qquad y''_p = 2A$$

Substituting into given differential equation

$$2A - 4(2At + B) + 3(At^2 + Bt + C) = 2t - 3t^2$$

Equating coefficients of like powers of t, For t^2:

$$3A = -3 \Rightarrow \; A = -1.$$

For t:

$$-8A + 3B = 2 \Rightarrow \; 8 + 3B = 2 \Rightarrow \; B = -2$$

For the constant term:

$$2A - 4B + 3C = 0$$

$$-2 + 8 + 3C = 0 \Rightarrow \; C = -2$$

Hence,

$$y_p = -t^2 - 2t - 2.$$

Hence, the correct option is

$$\boxed{\text{(A)} \; -2 - 2t - t^2}$$

Key Points

- Since the forcing term is quadratic, assume a quadratic polynomial for the particular solution.

Q6.16.1 MCQ 2016 1M Ans: C ● ○ ○

Consider the partial differential equation

$$\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + 3\frac{\partial^2 P}{\partial x \partial y} + 2\frac{\partial P}{\partial x} - \frac{\partial P}{\partial y} = 0$$

Compare the given equation with the standard second-order linear PDE

$$a\frac{\partial^2 u}{\partial x^2} + b\frac{\partial^2 u}{\partial x \partial y} + c\frac{\partial^2 u}{\partial y^2} + d\frac{\partial u}{\partial x} + e\frac{\partial u}{\partial y} + fu = g$$

Hence, $a = 1, \; b = 3, \; c = 1$.
The discriminant is

$$b^2 - 4ac = 3^2 - 4(1)(1) = 9 - 4 = 5 > 0$$

Therefore, the given PDE is *hyperbolic*.
Therefore, the correct option is

$$\boxed{\text{(C) hyperbolic}}$$

Mistakes to be avoided

- Use the coefficient of the mixed derivative directly as b; do not divide it by 2.

Q6.16.2 MCQ 2016 1M Ans: B ● ● ○

Consider the partial differential equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

where α is a positive constant.
Use the method of separation of variables.
Assume $u(x, t) = X(x)T(t)$.
Substituting into the given PDE,

$$XT' = \alpha X''T$$

$$\frac{1}{\alpha}\frac{T'}{T} = \frac{X''}{X} = k,$$

where k is the separation constant.

Thus, the equations become

$$X'' - kX = 0, \qquad T' - \alpha kT = 0$$

The solution of the temporal equation is

$$T(t) = Ce^{\alpha kt}$$

The spatial equation has the solution

$$X(x) = C_1 e^{\sqrt{k}\,x} + C_2 e^{-\sqrt{k}\,x}$$

Replacing k by $\dfrac{k}{\alpha}$ to match the given options,

$$X(x) = C_1 e^{\sqrt{k/\alpha}\,x} + C_2 e^{-\sqrt{k/\alpha}\,x}$$

Hence,

$$u(x,t) = Ce^{kt}\left(C_1 e^{\sqrt{k/\alpha}\,x} + C_2 e^{-\sqrt{k/\alpha}\,x}\right).$$

Therefore, the correct option is

$$\boxed{\text{(B) } Ce^{kt}\left(C_1 e^{\sqrt{k/\alpha}\,x} + C_2 e^{-\sqrt{k/\alpha}\,x}\right)}$$

Key Points

- Separation converts the PDE into two ordinary differential equations, one in space and one in time.

Q6.16.3 MCQ 2016 2M Ans: A ● ● ○

Consider the differential equation

$$\frac{d^4y}{dx^4} + 3\frac{d^2y}{dx^2} = 108x^2$$

Using the differential operator $D = \dfrac{d}{dx}$,

$$(D^4 + 3D^2)y = 108x^2$$

The auxiliary equation is

$$m^4 + 3m^2 = 0$$

$$m^2(m^2 + 3) = 0$$

Hence, the roots are $m = 0,\ 0,\ \pm i\sqrt{3}$.
Therefore, the complementary function is

$$y_c = C_1 + C_2x + C_3\sin(\sqrt{3}\,x) + C_4\cos(\sqrt{3}\,x)$$

The particular integral is

$$y_p = \frac{1}{D^4 + 3D^2}(108x^2)$$

Factor the operator:

$$y_p = \frac{1}{3D^2}\left(1 + \frac{D^2}{3}\right)^{-1} 108x^2$$

Using the binomial expansion, $(1+x)^{-1} = 1 - x + x^2 - \cdots$,
Noting that higher derivatives of x^2 vanish,

$$y_p = \frac{1}{3D^2}\left[108x^2 - \frac{1}{3}D^2(108x^2)\right]$$

$$y_p = \frac{1}{3D^2}(108x^2 - 72)$$

Now integrate twice:

$$y_p = \frac{1}{3D^2}(108x^2 - 72) = \frac{1}{3}\int\int(108x^2 - 72)\,dx\,dx = 9x^4 - 36x^2$$

Therefore,

$$y_p = 3x^4 - 12x^2.$$

Hence, the correct option is

$$\boxed{\text{(A) } [c_1 + c_2x + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x] \text{ and } [3x^4 - 12x^2 + c]}$$

Key Points

- Imaginary roots $\pm i\beta$ produce the terms $C_3\sin(\beta x) + C_4\cos(\beta x)$

Q6.15.1 MCQ 2015 2M Ans: C ● ● ●

Consider the differential equation

$$x(y\,dx + x\,dy)\cos\left(\frac{y}{x}\right) = y(x\,dy - y\,dx)\sin\left(\frac{y}{x}\right)$$

Divide both sides by $xy(x\,dy - y\,dx)\cos\left(\frac{y}{x}\right)$, to obtain

$$\frac{y\,dx + x\,dy}{y(x\,dy - y\,dx)} = \frac{1}{x}\tan\left(\frac{y}{x}\right)$$

Now substitute $y = vx$, where $v = \frac{y}{x}$.
Differentiating,

$$dy = v\,dx + x\,dv$$

Hence,

$$y\,dx + x\,dy = 2vx\,dx + x^2\,dv,$$

$$x\,dy - y\,dx = x(v\,dx + x\,dv) - vx\,dx = x^2\,dv$$

On Substituting,

$$\frac{2vx\,dx + x^2\,dv}{vx^3\,dv} = \frac{1}{x}\tan v,$$

$$\frac{2\,dx}{x\,dv} + \frac{1}{v} = \tan v$$

$$\frac{2\,dx}{x} = \left(\tan v - \frac{1}{v}\right)dv$$

Integrating,

$$2\ln x = \int\tan v\,dv - \int\frac{dv}{v}$$

Using $\int\tan v\,dv = \ln(\sec v)$, we obtain

$$2\ln x = \ln(\sec v) - \ln v + \ln C$$

Hence,

$$x^2 = C\frac{\sec v}{v}$$

Substituting $v = \frac{y}{x}$, gives

$$x^2 = C\,\frac{\sec(y/x)}{y/x}$$

$$xy = C\sec\left(\frac{y}{x}\right)$$

Therefore,

$$xy\cos\left(\frac{y}{x}\right) = C.$$

Therefore, the correct option is

$$\boxed{\text{(C) } xy\cos\frac{y}{x} = c}$$

Key Points

- The appearance of the ratio $\frac{y}{x}$ suggests the substitution $y = vx$.
- The standard integral $\int \tan v\,dv = \ln(\sec v) + C$ is useful in solving homogeneous differential equations.

Mistakes to be avoided

- While differentiating $y = vx$, remember that $dy = v\,dx + x\,dv$.

Q6.15.2 NAT 2015 2M Ans: 18 ● ○ ○

Consider the second-order linear differential equation

$$\frac{d^2y}{dx^2} = -12x^2 + 24x - 20 \text{ with the boundary conditions } y(0) = 5,\ y(2) = 21$$

Integrating once with respect to x,

$$\frac{dy}{dx} = -4x^3 + 12x^2 - 20x + C_1$$

Integrating again,

$$y = -x^4 + 4x^3 - 10x^2 + C_1x + C_2$$

Using the boundary conditions,
At $x = 0$, $5 = C_2$.
At $x = 2$,

$$21 = -16 + 32 - 40 + 2C_1 + 5$$

$$2C_1 = 40 \Rightarrow\ C_1 = 20$$

Hence,

$$y = -x^4 + 4x^3 - 10x^2 + 20x + 5$$

At $x = 1$,

$$y(1) = -1 + 4 - 10 + 20 + 5 = 18.$$

Therefore,

$$\boxed{18}$$

Mistakes to be avoided

- Do not forget the constant of integration after each integration.

Q6.14.1 | MCQ | 2014 | 1M | Ans: D | ● ○ ○

Consider the differential equation

$$\frac{dP}{dt} + k_2 P = k_1 L_0 e^{-k_1 t}$$

Compare the given equation with the standard first-order linear differential equation

$$\frac{dP}{dt} + RP = S$$

Here, $R = k_2,\ S = k_1 L_0 e^{-k_1 t}$.
The integrating factor is

$$\text{I.F.} = e^{\int P\,dt} = e^{\int k_2\,dt} = e^{k_2 t}.$$

Therefore, the correct option is

$$\boxed{\text{(D) } e^{k_2 t}}$$

Q6.14.2 | NAT | 2014 | 2M | Ans: 3 | ● ○ ○

Consider the one-dimensional steady-state flow equation

$$\frac{d^2H}{dx^2} = 0,$$

where H is the hydraulic head.
The boundary conditions are

$$H(0) = 5 \text{ m}, \qquad H(10) = 1 \text{ m}$$

Method 1

Integrating the differential equation once,

$$\frac{dH}{dx} = C_1$$

Integrating again,

$$H = C_1 x + C_2$$

Using the boundary conditions,
At $x = 0,\ 5 = C_2$.
At $x = 10$,

$$1 = 10C_1 + 5 \Rightarrow C_1 = -\frac{2}{5}$$

Hence,

$$H = -\frac{2}{5}x + 5$$

At the middle of the strip, $x = \frac{10}{2} = 5$,

$$H = -\frac{2}{5}(5) + 5 = 3.$$

Method 2

Since

$$\frac{d^2H}{dx^2} = 0,$$

the hydraulic head varies linearly along the strip.
Hence, the hydraulic head at the midpoint is simply the average of the end values:

$$H\left(\frac{10}{2}\right) = \frac{5+1}{2} = 3.$$

Therefore,

$$\boxed{3.0}$$

Key Points

- For a linear variation, the head at the midpoint equals the average of the head values at the two ends.

Q6.13.1 NAT 2013 2M Ans: 3.8 ● ○ ○

Consider the Laplace equation for one-dimensional flow:

$$\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} + \frac{\partial^2 H}{\partial z^2} = 0$$

Since the hydraulic head H does not vary with y and z,

$$\frac{\partial^2 H}{\partial y^2} = \frac{\partial^2 H}{\partial z^2} = 0$$

Hence, the governing equation reduces to

$$\frac{d^2 H}{dx^2} = 0$$

The boundary conditions are

$$H(0) = 5, \qquad \frac{dH}{dx} = -1$$

Method 1

Integrating once,

$$\frac{dH}{dx} = C_1$$

Using $\frac{dH}{dx} = -1$, we obtain $C_1 = -1$.

Integrating again,

$$H = -x + C_2$$

Using $H(0) = 5$, gives $C_2 = 5$.
Therefore,

$$H = 5 - x$$

At $x = 1.2$,

$$H = 5 - 1.2 = 3.8.$$

Method 2

Since $\frac{d^2 H}{dx^2} = 0$, the hydraulic gradient $\frac{dH}{dx}$ is constant.

Given $\frac{dH}{dx} = -1$, the hydraulic head decreases linearly with distance.

Starting from $H(0) = 5$, the head at $x = 1.2$ is

$$H = 5 - (1)(1.2) = 3.8.$$

Therefore,

$$\boxed{3.8}$$

Key Points

- A zero second derivative implies a constant hydraulic gradient and a linear variation of hydraulic head.
- The boundary condition on $\frac{dH}{dx}$ directly specifies the slope of the hydraulic head profile.

Q6.12.1 MCQ 2012 2M Ans: D ● ○ ○

Consider the differential equation

$$\frac{dy}{dx} + 2y = 0 \text{ with the boundary condition } y(1) = 5$$

Method 1

Separate the variables and Integrate:

$$\frac{dy}{dx} = -2y$$
$$\frac{dy}{y} = -2\,dx$$
$$\int \frac{dy}{y} = \int -2\,dx$$
$$\ln|y| = -2x + C$$
$$y = C_1 e^{-2x},$$

where $C_1 = e^C$.
Using the boundary condition, $5 = C_1 e^{-2}$,

$$C_1 = 5e^2 \approx 36.95$$

Therefore,

$$y = 36.95\, e^{-2x}.$$

Method 2

Using the differential operator $D = \frac{d}{dx}$,

$$(D + 2)y = 0$$

The auxiliary equation is

$$m + 2 = 0$$
$$m = -2$$

Hence, the complementary function is

$$y = C_1 e^{-2x}$$

Applying the condition $y(1) = 5$, we obtain

$$C_1 = 5e^2 \approx 36.95$$

Thus,

$$y = 36.95\, e^{-2x}.$$

Hence, the correct option is

$$\boxed{\text{(D) } y = 36.95\, e^{-2x}}$$

Q6.11.1 MCQ 2011 2M Ans: D ● ○ ○

Consider the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = x \text{ with the condition } y(1) = 1$$

Compare the given equation with the standard first-order linear differential equation

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Here, $P(x) = 1/x,\ Q(x) = x$.
The integrating factor is

$$\text{I.F.} = e^{\int 1/x\, dx} = e^{\ln x} = x.$$

The complete solution is given by,

$$y \times \text{I.F.} = \int Q(x) \times \text{I.F.}\, dx + C$$

$$xy = \int x^2 + C$$

$$xy = \frac{x^3}{3} + C$$

Hence,

$$y = \frac{x^2}{3} + \frac{C}{x}$$

Using the condition $y(1) = 1$, we get

$$1 = \frac{1}{3} + C \Rightarrow\ C = \frac{2}{3}$$

Therefore,

$$y = \frac{x^2}{3} + \frac{2}{3x}.$$

Hence, the correct option is

$$\boxed{\text{(D) } y = \frac{2}{3x} + \frac{x^2}{3}}$$

Q6.10.1 MCQ 2010 1M Ans: A ● ● ○

Consider the differential equation

$$\frac{d^3y}{dx^3} + 4\sqrt{\left(\frac{dy}{dx}\right)^3 + y^2} = 0$$

The highest-order derivative present is $\frac{d^3y}{dx^3}$.
Hence, the order of the differential equation is 3.
To determine the degree, first express the equation as a polynomial in the derivatives.
Rearranging and Squaring on both sides,

$$\frac{d^3y}{dx^3} = -4\sqrt{\left(\frac{dy}{dx}\right)^3 + y^2}$$

$$\left(\frac{d^3y}{dx^3}\right)^2 = 16\left[\left(\frac{dy}{dx}\right)^3 + y^2\right]$$

Now the equation is polynomial in the derivatives.
The highest-order derivative appears with exponent 2.
Therefore, Degree = 2.
Therefore, the correct option is

$$\boxed{\text{(A) 3 and 2}}$$

Key Points

- Radicals involving derivatives must be eliminated before determining the degree.

Mistakes to be avoided

- Do not determine the degree directly from the original equation containing a square root.

Q6.10.2 MCQ 2010 2M Ans: C ● ○ ○

Consider the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

Using the differential operator $D = \frac{d}{dx}$,

$$(D^2 + D - 6)y = 0$$

The auxiliary equation is

$$m^2 + m - 6 = 0$$

$$(m + 3)(m - 2) = 0$$

Hence, the roots are $m = -3,\ m = 2$.
Since the roots are real and distinct, the complementary function is

$$y = C_1e^{-3x} + C_2e^{2x}$$

As the differential equation is homogeneous, the particular integral is zero.
Therefore,

$$y = C_1e^{-3x} + C_2e^{2x}.$$

Hence, the correct option is

$$\boxed{\text{(C) } C_1e^{-3x} + C_2e^{2x}}$$

Q6.10.3 MCQ 2010 2M Ans: C ● ○ ○

Consider the complete integral

$$z = ax + by + ab,$$

where a and b are arbitrary constants.
Differentiate the given equation partially with respect to x:

$$\frac{\partial z}{\partial x} = a$$

Since $p = \frac{\partial z}{\partial x}$, we obtain $a = p$.
Similarly, differentiating with respect to y,

$$\frac{\partial z}{\partial y} = b$$

Since $q = \frac{\partial z}{\partial y}$, we obtain $b = q$.

Substituting $a = p,\ b = q$ into the given equation,

$$z = px + qy + pq.$$

Hence, the correct option is

$$\boxed{\text{(C) } z = px + qy + pq}$$

Mistakes to be avoided

- Do not differentiate with respect to the arbitrary constants a or b; differentiate only with respect to the independent variables.

Q6.09.1 MCQ 2009 2M Ans: A ● ○ ○

Consider the differential equation

$$3y\frac{dy}{dx} + 2x = 0$$

Separating the variables and Integrating on both sides,

$$3y\,dy = -2x\,dx$$

$$3\int y\,dy = -2\int x\,dx$$

$$\frac{3}{2}y^2 = -x^2 + C$$

Rearranging,

$$2x^2 + 3y^2 = C_1,$$

where $C_1 = 2C$.
Dividing throughout by C_1,

$$\frac{x^2}{C_1/2} + \frac{y^2}{C_1/3} = 1.$$

This is the standard equation of an ellipse.

$$\boxed{\text{(A) ellipses}}$$

Key Points

- The standard form $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ represents an ellipse.

Q6.08.1 MCQ 2008 1M Ans: A ● ○ ○

Consider the differential equation

$$\frac{d^2y}{dx^2} + y = 0$$

Using the differential operator $D = \dfrac{d}{dx}$,

$$(D^2 + 1)y = 0$$

The auxiliary equation is

$$m^2 + 1 = 0$$

$$m = \pm i$$

Since the roots are a pair of complex conjugates, $m = \alpha \pm i\beta$, where $\alpha = 0,\ \beta = 1$, The complementary function is

$$y = e^{\alpha x}\left(C_1 \cos \beta x + C_2 \sin \beta x\right)$$

Therefore,

$$y = C_1 \cos x + C_2 \sin x.$$

Hence, the correct option is

$$\boxed{\text{(A) } y = P\cos x + Q\sin x}$$

Mistakes to be avoided

- Do not write the solution as $C_1 e^{ix} + C_2 e^{-ix}$; the real-valued general solution is expressed using sine and cosine.

Q6.08.2 MCQ 2008 2M Ans: D ● ● ○

Consider the partial differential equation

$$k_x \frac{\partial^2 h}{\partial x^2} + k_z \frac{\partial^2 h}{\partial z^2} = 0$$

Method 1

From Option (D), Take $x_1 = x\sqrt{\frac{k_z}{k_x}}$.

Using the chain rule,

$$\frac{\partial x_1}{\partial x} = \sqrt{\frac{k_z}{k_x}}$$

Hence,

$$\frac{\partial h}{\partial x} = \frac{\partial h}{\partial x_1}\frac{\partial x_1}{\partial x} = \sqrt{\frac{k_z}{k_x}}\frac{\partial h}{\partial x_1}$$

Differentiating once again,

$$\frac{\partial^2 h}{\partial x^2} = \frac{k_z}{k_x}\frac{\partial^2 h}{\partial x_1^2}$$

Substituting into the given equation,

$$k_x\left(\frac{k_z}{k_x}\frac{\partial^2 h}{\partial x_1^2}\right) + k_z\frac{\partial^2 h}{\partial z^2} = 0$$

$$k_z\left(\frac{\partial^2 h}{\partial x_1^2} + \frac{\partial^2 h}{\partial z^2}\right) = 0$$

Since $k_z \neq 0$, we obtain

$$\frac{\partial^2 h}{\partial x_1^2} + \frac{\partial^2 h}{\partial z^2} = 0$$

which is the required transformed equation.

Method 2

To convert

$$k_x \frac{\partial^2 h}{\partial x^2} + k_z \frac{\partial^2 h}{\partial z^2} = 0$$

into the standard Laplace equation, scale the spatial coordinate so that the coefficients of the second derivatives become equal.

Assume $x_1 = \lambda x$.

Then,

$$\frac{\partial^2 h}{\partial x^2} = \lambda^2 \frac{\partial^2 h}{\partial x_1^2}$$

Substituting,

$$k_x \lambda^2 \frac{\partial^2 h}{\partial x_1^2} + k_z \frac{\partial^2 h}{\partial z^2} = 0$$

For equal coefficients, $k_x \lambda^2 = k_z$.
Hence,

$$\lambda = \sqrt{\frac{k_z}{k_x}}$$

$$\Rightarrow x_1 = x\sqrt{\frac{k_z}{k_x}}.$$

Hence, the correct option is

$$\boxed{\text{(D) } x_1 = x\sqrt{\frac{k_z}{k_x}}}$$

Key Points

- Coordinate scaling is commonly used to transform anisotropic Laplace equations into the standard isotropic form.

Q6.08.3 MCQ | 2008 | 2M | Ans: D | ● ○ ○

Consider the differential equation

$$\frac{dy}{dx} = -\frac{x}{y} \text{ with the condition } y = \sqrt{3} \text{ at } x = 1$$

Separate the variables and Integrate on both sides:

$$y\,dy = -x\,dx$$

$$\int y\,dy = -\int x\,dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + C$$

$$x^2 + y^2 = C_1,$$

where $C_1 = 2C$.
Using the given condition,

$$1^2 + (\sqrt{3})^2 = 4 = C_1$$

Hence,

$$x^2 + y^2 = 4.$$

Hence, the correct option is

$$\boxed{\text{(D) } x^2 + y^2 = 4}$$

Q6.07.1 MCQ 2007 1M Ans: B ● ○ ○

Consider the differential equation

$$\frac{d^2x}{dt^2} + 2x^3 = 0$$

The highest-order derivative present is $\frac{d^2x}{dt^2}$.

The given equation is already polynomial in its derivatives.
The highest-order derivative appears to the first power.
Hence, the degree of the differential equation is 1.
Therefore, the correct option is

$$\boxed{\text{(B) } 1}$$

Mistakes to be avoided

- Do not confuse the order with the degree.

Q6.07.2 MCQ 2007 1M Ans: D ● ○ ○

Consider the differential equation

$$\frac{dy}{dx} = x^2y \text{ with the initial condition } y(0) = 1$$

Separate the variables and Integrate both sides:

$$\frac{dy}{y} = x^2\,dx$$

$$\int \frac{dy}{y} = \int x^2\,dx$$

$$\ln|y| = \frac{x^3}{3} + C$$

$$y = C_1 e^{x^3/3},$$

where $C_1 = e^C$.
Using the initial condition, $y(0) = 1$, we obtain

$$1 = C_1 e^0 \implies C_1 = 1$$

Hence,

$$y = e^{x^3/3}.$$

Hence, the correct option is

$$\boxed{\text{(D) } y = e^{x^3/3}}$$

Q6.06.1 MCQ 2006 2M Ans: A ● ○ ○

Consider the differential equation

$$x^2\frac{dy}{dx} + 2xy - x + 1 = 0 \text{ with the boundary condition } y(1) = 0$$

Method 1

Rearrange the equation as

$$x^2\frac{dy}{dx} + 2xy = x - 1$$

Dividing throughout by x^2,

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{x-1}{x^2}$$

This is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P(x) = \frac{2}{x},\ Q(x) = \frac{x-1}{x^2}$$

The integrating factor is

$$\text{I.F.} = e^{\int 2/x\,dx} = e^{2\ln x} = x^2$$

The complete solution is given by

$$y \times \text{I.F.} = \int Q(x) \times \text{I.F.}\,dx + C$$

$$x^2 y = \int \frac{x-1}{x^2} \times x^2 \cdot dx + C$$

$$x^2 y = \frac{x^2}{2} - x + C$$

Hence,

$$y = \frac{1}{2} - \frac{1}{x} + \frac{C}{x^2}$$

Using the boundary condition, $y(1) = 0$,

$$0 = \frac{1}{2} - 1 + C \Rightarrow C = \frac{1}{2}$$

Therefore,

$$y = \frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}.$$

Method 2

Observe that

$$\frac{d}{dx}(x^2 y) = x^2\frac{dy}{dx} + 2xy$$

Hence, the given equation becomes

$$\frac{d}{dx}(x^2 y) = x - 1$$

Integrating directly,

$$x^2 y = \frac{x^2}{2} - x + C$$

Applying $y(1) = 0$, gives

$$C = \frac{1}{2}$$

Thus,

$$y = \frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}.$$

Hence, the correct option is

$$\boxed{\text{(A) } \frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}}$$

Key Points

- Recognizing $\frac{d}{dx}(x^2 y) = x^2\frac{dy}{dx} + 2xy$ provides a much shorter solution.

Q6.05.1 MCQ | 2005 | 2M | Ans: A | ● ● ○

Consider the Bernoulli differential equation

$$\frac{dy}{dt} + p(t)\,y = q(t)\,y^n, \qquad n > 0,$$

Let $v = y^{1-n}$.
Differentiate $v = y^{1-n}$ with respect to t:

$$\frac{dv}{dt} = (1-n)\,y^{-n}\frac{dy}{dt}$$

$$\frac{dy}{dt} = \frac{y^n}{1-n}\frac{dv}{dt}$$

Substituting into the given differential equation,

$$\frac{y^n}{1-n}\frac{dv}{dt} + p(t)\,y = q(t)\,y^n$$

Since $y = y^n\,y^{1-n} = y^n v$, we obtain

$$\frac{1}{1-n}\frac{dv}{dt} + p(t)v = q(t)$$

Multiplying throughout by $(1-n)$,

$$\frac{dv}{dt} + (1-n)p(t)v = (1-n)q(t).$$

Therefore, the correct option is

$$\boxed{(\text{A})\ \frac{dv}{dt} + (1-n)pv = (1-n)q}$$

Key Points

- A Bernoulli differential equation has the standard form

$$\frac{dy}{dt} + P(t)y = Q(t)y^n, \qquad n \neq 0, 1.$$

- The substitution $v = y^{1-n}$ converts the nonlinear equation into a first-order linear differential equation.

Q6.05.2 MCQ | 2005 | 2M | Ans: A | ● ● ○

Consider the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 17y = 0 \text{ with the boundary conditions } y(0) = 1,\ \left.\frac{dy}{dx}\right|_{x=\pi/4} = 0$$

Using the differential operator $D = \dfrac{d}{dx}$,

$$(D^2 + 2D + 17)y = 0$$

The auxiliary equation is

$$m^2 + 2m + 17 = 0$$

$$m = \frac{-2 \pm \sqrt{4-68}}{2} = -1 \pm 4i$$

Since the roots are complex conjugates, the complementary function is

$$y = e^{-x}\left(C_1 \cos 4x + C_2 \sin 4x\right)$$

Using $y(0) = 1$, gives $1 = C_1$.
Differentiating,

$$\frac{dy}{dx} = e^{-x}\Big[(-4C_1 \sin 4x + 4C_2 \cos 4x) - (C_1 \cos 4x + C_2 \sin 4x)\Big]$$

$$\frac{dy}{dx} = e^{-x}\Big[(-4C_1 - C_2)\sin 4x + (4C_2 - C_1)\cos 4x\Big]$$

Using $\left.\frac{dy}{dx}\right|_{x=\pi/4} = 0$, we obtain

$$-(4C_2 - C_1)e^{-\pi/4} = 0$$

$$4C_2 = C_1 \Rightarrow C_2 = \frac{1}{4}$$

Hence,

$$y = e^{-x}\left(\cos 4x + \frac{1}{4}\sin 4x\right).$$

Therefore, the correct option is

$$\boxed{\text{(A) } e^{-x}\left(\cos 4x + \frac{1}{4}\sin 4x\right)}$$

Q6.04.1 MCQ | 2004 | 2M | Ans: B | ● ○ ○

Consider the differential equation

$$\frac{dx}{dt} + kx^2 = 0 \text{ with the initial condition } x(0) = a$$

where k is a reaction rate constant.
Separate the variables and Integrate on both sides:

$$\frac{dx}{x^2} = -k\,dt$$

$$\int x^{-2}\,dx = -k\int dt,$$

$$-\frac{1}{x} = -kt + C$$

$$\frac{1}{x} = kt + C_1.$$

Using the initial condition, $x = a$ at $t = 0$, we obtain

$$\frac{1}{a} = C_1$$

Hence,

$$\frac{1}{x} = \frac{1}{a} + kt.$$

Therefore, the correct option is

$$\boxed{\text{(B) } \frac{1}{x} = \frac{1}{a} + kt}$$

Q6.01.1 | MCQ | 2001 | 1M | Ans: C | ● ○ ○

Consider the Laplace equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

Integrate the equation successively with respect to the two independent variables.
Since the equation is second order in both x and y,

- two arbitrary functions arise from integrating with respect to x,
- two more arbitrary functions arise from integrating with respect to y.

Thus, the general solution contains four arbitrary constants.
Or equivalently, requires four independent boundary specifications.
Hence, four boundary conditions are required to obtain a unique solution.
Therefore, the correct option is

$$\boxed{\text{(C) } 4}$$

Key Points

- Laplace's equation is a second-order elliptic partial differential equation. And elliptic PDEs require boundary information over the complete boundary of the domain.

Mistakes to be avoided

- Do not confuse boundary conditions with initial conditions.
- Do not assume that only two boundary conditions are sufficient because the equation is second order.

Q6.01.2 | MCQ | 2001 | 2M | Ans: C | ● ○ ○

Consider the differential equation

$$\frac{d^2 y}{dx^2} = 3x - 2 \text{ with the boundary conditions } y(0) = 2,\ y'(1) = -3$$

Integrating once with respect to x,

$$\frac{dy}{dx} = \int (3x - 2)\, dx = \frac{3x^2}{2} - 2x + C_1$$

Integrating again,

$$y = \int \left(\frac{3x^2}{2} - 2x + C_1 \right) dx = \frac{x^3}{2} - x^2 + C_1 x + C_2$$

Using the boundary conditions, $x = 0,\ 2 = C_2$.
Using $y'(1) = -3$, gives

$$-3 = \frac{3}{2} - 2 + C_1 \Rightarrow\ C_1 = -\frac{5}{2}$$

Hence,

$$y = \frac{x^3}{2} - x^2 - \frac{5x}{2} + 2.$$

Therefore, the correct option is

$$\boxed{\text{(C) } y = \frac{x^3}{2} - x^2 - \frac{5x}{2} + 2}$$

Key Points

- Integrating a second-order differential equation twice introduces two arbitrary constants.
- Boundary (or initial) conditions determine these constants uniquely.

Q6.00.1 MCQ 2000 2M Ans: A ● ● ○

Consider the function

$$f(x, y, z) = \left(x^2 + y^2 + z^2\right)^{-1/2}$$

Method 1

Let $r^2 = x^2 + y^2 + z^2$.
Then, $f = r^{-1}$.
The first partial derivative is

$$\frac{\partial f}{\partial x} = -\frac{x}{r^3}$$

Differentiating again,

$$\frac{\partial^2 f}{\partial x^2} = -\frac{\partial}{\partial x}\left(xr^{-3}\right)$$

Applying the product rule,

$$\frac{\partial^2 f}{\partial x^2} = -r^{-3} + 3x^2 r^{-5}$$

Substituting r^2, this becomes

$$\frac{\partial^2 f}{\partial x^2} = \frac{2x^2 - y^2 - z^2}{r^5}$$

Similarly,

$$\frac{\partial^2 f}{\partial y^2} = \frac{-x^2 + 2y^2 - z^2}{r^5},$$

$$\frac{\partial^2 f}{\partial z^2} = \frac{-x^2 - y^2 + 2z^2}{r^5}$$

Adding the three expressions,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \frac{0}{r^5} = 0$$

Method 2

Recognize that

$$f = \frac{1}{r}, \qquad r = \sqrt{x^2 + y^2 + z^2},$$

is the fundamental solution of Laplace's equation in three dimensions.

A standard result is $\nabla^2\left(\frac{1}{r}\right) = 0, \ (r \neq 0)$, where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Hence,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

Hence, the correct option is

$$\boxed{\text{(A) Zero}}$$

Key Points

- The Laplacian operator in Cartesian coordinates is

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

- The function $\dfrac{1}{\sqrt{x^2 + y^2 + z^2}}$ is a harmonic function for $r \neq 0$.

Mistakes to be avoided

- The result is valid everywhere except at the singular point $(x, y, z) = (0, 0, 0)$.

Q6.99.1 | MCQ | 1999 | 2M | Ans: C | ● ○ ○

Consider the differential equation

$$\frac{dy}{dx} = 1 + y^2$$

Separate the variables and Integrate on both sides:

$$\frac{dy}{1 + y^2} = dx$$

$$\int \frac{dy}{1 + y^2} = \int dx$$

Using the standard integral $\int \frac{dy}{1 + y^2} = \tan^{-1} y$, we obtain

$$\tan^{-1} y = x + C$$

Therefore,

$$y = \tan(x + C).$$

Hence, the correct option is

$$\boxed{\text{(C) } y = \tan(x + c)}$$

Q6.98.1 | MCQ | 1998 | 5M | Ans: A | ● ● ○

Consider the differential equation

$$\frac{d^4 y}{dx^4} - y = 15 \cos 2x$$

Using the differential operator $D = \frac{d}{dx}$,

$$(D^4 - 1)y = 15 \cos 2x$$

The auxiliary equation is

$$m^4 - 1 = 0$$

$$(m^2 - 1)(m^2 + 1) = 0$$

Hence, the roots are $m = \pm 1, \ m = \pm i$.
Since the roots are two real and a pair of complex conjugates, the complementary function is

$$y_c = C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x$$

The particular integral is

$$y_p = \frac{1}{D^4 - 1}(15\cos 2x)$$

Using the operator rule $D^2(\cos ax) = -a^2 \cos ax$, we have

$$D^2(\cos 2x) = -4\cos 2x \Rightarrow D^4(\cos 2x) = 16\cos 2x$$

Therefore,

$$y_p = \frac{15}{16-1}\cos 2x = \cos 2x$$

Thus, the complete solution is

$$y = C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x + \cos 2x.$$

Hence, the correct option is

$$\boxed{\text{(A) } y = C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x + \cos 2x}$$

Mistakes to be avoided

- Check for resonance before computing the particular integral; here, $\pm 2i$ are not roots of the auxiliary equation, so no modification of the trial solution is required.

Q6.97.1 MCQ 1997 1M Ans: B ● ○ ○

Consider the differential equation

$$f(x,y)\frac{dy}{dx} + g(x,y) = 0$$

Rewrite the equation in differential form:

$$g(x,y)\,dx + f(x,y)\,dy = 0$$

Comparing with the standard differential equation $M(x,y)\,dx + N(x,y)\,dy = 0$, we identify

$$M(x,y) = g(x,y), \qquad N(x,y) = f(x,y)$$

For an exact differential equation, $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$.

Therefore,

$$\frac{\partial g}{\partial y} = \frac{\partial f}{\partial x}.$$

Therefore, the correct option is

$$\boxed{\text{(B) } \frac{\partial f}{\partial x} = \frac{\partial g}{\partial y}}$$

Key Points

- Correct identification of M and N is the most important step.

Q6.97.2 MCQ 1997 1M Ans: D ● ○ ○

Consider the differential equation

$$\frac{dy}{dx} + P\,y = Q$$

Method 1

A first-order linear differential equation (also called a Leibniz differential equation) has the standard form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P = P(x),\ Q = Q(x)$$

Thus, both P and Q are functions of the independent variable x (or constants).
Hence, the given equation is linear only if P and Q are functions of x or constants.

Method 2

A differential equation is said to be linear if

- the dependent variable y and its derivatives appear only to the first power,
- the coefficients of y and its derivatives depend only on the independent variable x.

Therefore, both P and Q must be functions of x only (or constants).
Therefore, the correct option is

$$\boxed{\text{(D) } P \text{ and } Q \text{ are functions of } x \text{ or constants}}$$

Key Points

- The coefficients may depend only on the independent variable.
- Constants are special cases of functions of the independent variable.

Q6.95.1 MCQ 1995 2M Ans: C ● ○ ○

Consider the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

Using the differential operator $D = \dfrac{d}{dx}$,

$$(D^2 + 3D + 2)y = 0$$

The auxiliary equation is

$$m^2 + 3m + 2 = 0$$

$$(m+1)(m+2) = 0$$

Hence, the roots are $m = -1,\ m = -2$.
Since the roots are real and distinct, the complementary function is

$$y = C_1 e^{-x} + C_2 e^{-2x}$$

As the differential equation is homogeneous, the particular integral is zero.
Therefore,

$$y = C_1 e^{-x} + C_2 e^{-2x}$$

Hence, the correct option is

$$\boxed{\text{(C) } y = C_1 e^{-x} + C_2 e^{-2x}}$$

Q6.95.2 MCQ 1995 1M Ans: C ● ○ ○

Consider the differential equation

$$y'' + \left(x^3 \sin x^5\right) y' + y = \cos x^3$$

Compare the given equation with the standard second-order linear differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = r(x)$$

Here, $a_2(x) = 1,\ a_1(x) = x^3 \sin x^5,\ a_0(x) = 1,\ r(x) = \cos x^3$.
Since

- the highest derivative is y'', the equation is of second order,
- y, y', and y'' appear only to the first power,
- the coefficients depend only on the independent variable x,
- the right-hand side is non-zero,

the equation is a second-order linear, non-homogeneous ordinary differential equation.
Hence, the correct option is

$$\boxed{\text{(C) second order linear}}$$

Mistakes to be avoided

- Do not classify the equation as having constant coefficients because the coefficient of y' is $x^3 \sin x^5$.
- The presence of trigonometric functions of x in the coefficients does not make the equation nonlinear.

Q6.94.1 MCQ 1994 2M Ans: A ● ○ ○

Consider the differential equation

$$\frac{d^4y}{dx^4} + P\frac{d^2y}{dx^2} + ky = 0,$$

where P and k are constants.
Compare the given equation with the standard linear differential equation of order n:

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = Q(x)$$

Here, $a_4 = 1,\ a_2 = P,\ a_0 = k,\ Q(x) = 0$.
Since

- the highest derivative is $\dfrac{d^4y}{dx^4}$,
- all derivatives appear only to the first power,
- the coefficients are constants,
- the right-hand side is zero,

the equation is a homogeneous linear differential equation of fourth order.
Hence, the correct option is

$$\boxed{\text{(A) Linear of fourth order}}$$

Q6.94.2 MCQ | 1994 | 1M | Ans: C | ● ○ ○

Consider the differential equation

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

A differential equation is said to be exact if there exists a scalar function $F(x,y)$ such that

$$dF = \frac{\partial F}{\partial x}\,dx + \frac{\partial F}{\partial y}\,dy = M\,dx + N\,dy$$

Hence,

$$M = \frac{\partial F}{\partial x}, \qquad N = \frac{\partial F}{\partial y}$$

Differentiating,

$$\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial y\,\partial x},$$

$$\frac{\partial N}{\partial x} = \frac{\partial^2 F}{\partial x\,\partial y}$$

Since mixed partial derivatives are equal (assuming continuity),

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Hence, the correct option is

$$\boxed{\text{(C) } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

Q6.26.1 NAT | 2026 | 2M | Ans: 2.25 | ● ● ○

The given differential equation is of Cauchy–Euler type:

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} - 3y = 0$$

Using the substitution $x = e^t$, $D = \frac{d}{dt}$, the derivatives transform as

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

Hence,

$$D(D-1)y + 3Dy - 3y = 0$$
$$(D^2 + 2D - 3)y = 0$$

The auxiliary equation is

$$m^2 + 2m - 3 = 0$$
$$m = 1, -3$$

Therefore, the general solution is given by

$$y = C_1e^t + C_2e^{-3t}$$

Since $e^t = x$,

$$y = C_1x + \frac{C_2}{x^3}$$

Differentiating,

$$\frac{dy}{dx} = C_1 - \frac{3C_2}{x^4}$$

Using the boundary conditions at $x = 1$:

$$C_1 + C_2 = 3,$$
$$C_1 - 3C_2 = -5$$

Solving,

$$C_2 = 2, \qquad C_1 = 1$$

Hence,

$$y = x + \frac{2}{x^3}$$

At $x = 2$,

$$y(2) = 2 + \frac{2}{8} = 2.25.$$

$$\boxed{2.25}$$

Key Points

- A Cauchy–Euler differential equation is transformed into a constant-coefficient equation using the substitution $x = e^t$.

Q6.25.1 MSQ 2025 2M Ans: B,D ● ● ○

Consider the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = 0$$

Separating the variables,

$$\frac{dy}{dx} = -\frac{y}{x}$$
$$\frac{dy}{y} = -\frac{dx}{x}$$

Integrating,

$$\ln|y| = -\ln|x| + C = \ln\left(\frac{C_1}{x}\right)$$

Hence, the non-trivial solution is

$$y = \frac{c}{x},$$

which matches option **B**.
Further,

$$y = 0 \quad \Rightarrow \quad \frac{dy}{dx} = 0,$$

and hence

$$\frac{dy}{dx} + \frac{y}{x} = 0 + \frac{0}{x} = 0$$

Therefore, $y = 0$ also satisfies the differential equation.
Thus, the correct options are

$$\boxed{\text{(B) } y = \frac{c}{x};\ c \text{ is a constant (D) } y = 0}$$

Key Points

- A first-order homogeneous linear differential equation has the general solution consisting of all non-trivial solutions and the trivial solution.
- The trivial solution $y = 0$ should always be checked explicitly, especially after dividing by y.

Q6.24.1 NAT 2024 2M Ans: 4.04 ● ● ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0$$

which is a Cauchy–Euler differential equation.
Using the substitution $x = e^t,\ D = \frac{d}{dt}$, the derivatives transform as

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

Hence,

$$D(D-1)y - Dy - 3y = 0$$
$$(D^2 - 2D - 3)y = 0$$

The auxiliary equation is

$$m^2 - 2m - 3 = 0$$
$$(m-3)(m+1) = 0$$
$$m = 3,\ -1$$

Therefore, the general solution becomes

$$y = C_1 e^{-t} + C_2 e^{3t}$$

Since $e^t = x$,

$$y = \frac{C_1}{x} + C_2 x^3$$

Using the boundary conditions,
At $x = 1, \ y = 2$,

$$C_1 + C_2 = 2$$

At $x = 2, \ y = \frac{17}{2}$,

$$\frac{C_1}{2} + 8C_2 = \frac{17}{2}$$

Solving,

$$C_2 = 1, \qquad C_1 = 1$$

Hence,

$$y = \frac{1}{x} + x^3$$

At $x = \frac{3}{2}$,

$$y\left(\frac{3}{2}\right) = \frac{2}{3} + \left(\frac{3}{2}\right)^3 = \frac{2}{3} + \frac{27}{8} = \frac{97}{24} \approx 4.0417$$

Rounded to two decimal places,

$$\boxed{4.04}$$

Q6.23.1 NAT | 2023 | 1M | Ans: -1 | ● ○ ○

The given differential equation is

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

The auxiliary equation is

$$m^2 + \omega^2 = 0$$

which gives the roots $m = \pm j\omega$.
Hence, the general solution is

$$x(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t)$$

Differentiating,

$$\frac{dx}{dt} = -C_1\omega \sin(\omega t) + C_2\omega \cos(\omega t)$$

Using the initial conditions,
At $t = 0, \ x(0) = 1 \ C_1 = 1$.
Also,

$$\left.\frac{dx}{dt}\right|_{t=0} = 0 \quad \Rightarrow \quad C_2\omega = 0 \quad \Rightarrow \quad C_2 = 0$$

Therefore,

$$x(t) = \cos(\omega t)$$

At $t = \frac{3\pi}{\omega}$,

$$x\left(\frac{3\pi}{\omega}\right) = \cos\left(\omega \cdot \frac{3\pi}{\omega}\right) = \cos(3\pi) = -1$$

$$\boxed{-1}$$

Key Points

- The equation $\frac{d^2x}{dt^2} + \omega^2 x = 0$ represents simple harmonic motion (SHM).

Mistakes to be avoided

- Do not forget the factor ω while differentiating $\sin(\omega t)$ and $\cos(\omega t)$.

Q6.22.1 MCQ 2022 2M Ans: A ● ● ●

Consider the partial differential equation

$$\frac{\partial u}{\partial t} = \frac{1}{\pi^2}\frac{\partial^2 u}{\partial x^2}$$

where, $t \geq 0$ and $x \in [0, 1]$.

Method 1

Assume a separable solution of the form $u(x,t) = X(x)T(t)$.
On Substituting gives

$$X(x)T'(t) = \frac{1}{\pi^2}X''(x)T(t)$$

Using the initial and boundary conditions,

$$u(x,0) = \sin(\pi x), \qquad u(0,t) = u(1,t) = 0,$$

the spatial function is

$$X(x) = \sin(\pi x)$$
$$X''(x) = -\pi^2 \sin(\pi x)$$

Hence,

$$\frac{T'}{T} = \frac{1}{\pi^2}\frac{X''}{X} = -1$$

Integrating,

$$T(t) = e^{-t}$$

Therefore,

$$u(x,t) = \sin(\pi x)e^{-t}$$

At $x = 0.5$,

$$u(0.5,t) = \sin\left(\frac{\pi}{2}\right)e^{-t} = e^{-t}$$

Thus,

$$\frac{u(0.5,t)}{u(0.5,0)} = \frac{e^{-t}}{1} = e^{-t}$$

Given $\frac{u(0.5,t)}{u(0.5,0)} = \frac{1}{e}$, we obtain

$$e^{-t} = \frac{1}{e} \quad \Rightarrow \quad t = 1.$$

Method 2

For the one-dimensional heat equation

$$\frac{\partial u}{\partial t} = \alpha^2\frac{\partial^2 u}{\partial x^2} \quad \text{with } u(x,0) = \sin(n\pi x),$$

the standard solution is

$$u(x,t) = \sin(n\pi x)\, e^{-n^2\pi^2\alpha^2 t}$$

Here, $\alpha^2 = \dfrac{1}{\pi^2}$, $n = 1$.
Hence,

$$u(x,t) = \sin(\pi x)e^{-t}$$

$$\Rightarrow \frac{u(0.5,t)}{u(0.5,0)} = e^{-t}$$

Since this ratio equals $1/e$, it follows immediately that

$$t = 1.$$

Therefore, the correct option is

$$\boxed{\text{(A) } 1}$$

Key Points

- The given PDE is the one-dimensional heat equation.
- For an initial profile $\sin(n\pi x)$, the spatial mode remains unchanged while its amplitude decays exponentially with time.
- Each eigenmode decays independently according to $e^{-n^2\pi^2\alpha^2 t}$.

Q6.21.1 NAT 2021 2M Ans: 0.167 ● ● ○

Consider the ordinary differential equation

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y = 1$$

with the initial conditions $y(0) = y'(0) = y''(0) = y'''(0) = 0$
We have to find $\lim_{t\to\infty} y(t)$.

Method 1

As $t \to \infty$, all transient components vanish if the characteristic roots have negative real parts.
The homogeneous equation is

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y = 0$$

with characteristic equation

$$m^3 + 6m^2 + 11m + 6 = 0$$

$$(m+1)(m+2)(m+3) = 0$$

Since all roots are negative, the complementary solution decays to zero as $t \to \infty$.
For the particular solution, assume $y_p = A$, where A is a constant.
Substituting into the differential equation,

$$6A = 1 \Rightarrow A = \frac{1}{6}$$

Hence,

$$\lim_{t\to\infty} y(t) = \frac{1}{6}.$$

Method 2

Using the Final Value Theorem,
Taking the Laplace transform with

$$(s^3 + 6s^2 + 11s + 6)Y(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(s+1)(s+2)(s+3)}$$

Since all poles of $sY(s)$ lie in the left-half plane,

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} \frac{1}{(s+1)(s+2)(s+3)} = \frac{1}{6}.$$

Rounded to three decimal places,

$$\boxed{0.167}$$

Key Points

- For a stable linear time-invariant system, the steady-state value equals the constant particular solution.
- A constant forcing function is balanced by a constant particular solution since all derivative terms vanish.
- The Final Value Theorem is applicable only when all poles of $sY(s)$ are in the left-half plane.

Mistakes to be avoided

- Do not solve for the complete time-domain solution when only the steady-state value is required.
- Verify that all characteristic roots have negative real parts before using the Final Value Theorem.

Q6.20.1 NAT 2020 2M Ans: 168 ● ○ ○

The given differential equation is

$$\frac{dy}{dx} = y - 20 \text{ with condition } y(0) = 40$$

Rearranging,

$$\frac{dy}{y-20} = dx$$

Integrating both sides,

$$\int \frac{dy}{y-20} = \int dx$$

$$\ln|y-20| = x + C$$

Using the initial condition $y(0) = 40$, $\ln 20 = C$.
Hence,

$$\ln|y-20| = x + \ln 20$$

$$y - 20 = 20e^x$$

$$y = 20 + 20e^x.$$

At $x = 2$,

$$y = 20 + 20e^2 \approx 20 + 20(7.389) \approx 167.78.$$

Rounding to nearest integer

$$\boxed{168}$$

Q6.19.1 MCQ 2019 2M Ans: D ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + 3y = 1 \text{ with the initial condition } y(0) = 1$$

Method 1

This is a first-order linear differential equation of the form

$$\frac{dy}{dx} + Py = Q, \quad \text{where } P = 3, \ Q = 1$$

The integrating factor is

$$\text{IF} = e^{\int 3\,dx} = e^{3x}$$

The complete solution is

$$y \times \text{IF} = \int Q \times \text{IF}\,dx + C$$

$$ye^{3x} = \int e^{3x} + C$$

$$ye^{3x} = \frac{e^{3x}}{3} + C$$

Hence,

$$y = \frac{1}{3} + Ce^{-3x}$$

Using the initial condition $y(0) = 1$,

$$1 = \frac{1}{3} + C \Rightarrow C = \frac{2}{3}$$

Therefore,

$$y = \frac{1}{3} + \frac{2}{3}e^{-3x} = \frac{1}{3}\left(1 + 2e^{-3x}\right).$$

Method 2

The steady-state solution is obtained by setting $\frac{dy}{dx} = 0$, which gives

$$y_s = \frac{1}{3}$$

Let $y = 1/3 + z$.
Substituting into the differential equation,

$$\frac{dz}{dx} + 3z = 0$$

$$\frac{dz}{z} = -3\,dx$$

Integrating,

$$z = Ce^{-3x}$$

Hence,

$$y = \frac{1}{3} + Ce^{-3x}$$

Using $y(0) = 1$, gives $C = \frac{2}{3}$.
Therefore,

$$y = \frac{1}{3}\left(1 + 2e^{-3x}\right).$$

Thus, the correct option is

$$\boxed{\text{(D) } \frac{1}{3}\left(1 + 2e^{-3x}\right)}$$

Key Points

- A first-order linear differential equation can be solved using the integrating factor method.
- The complete solution is the sum of the steady-state (particular) solution and the transient (homogeneous) solution.
- The transient term e^{-3x} decays to zero as $x \to \infty$.

Q6.18.1 MCQ 2018 1M Ans: C ● ○ ○

The given system of differential equations is

$$\frac{dx}{dt} + x + y = 0,$$

$$\frac{dy}{dt} - x = 0$$

From the second equation, $x = \frac{dy}{dt}$.
Substituting into the first equation,

$$\frac{d}{dt}\left(\frac{dy}{dt}\right) + \frac{dy}{dt} + y = 0$$

$$\frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0$$

Therefore, the correct option is

$$\boxed{\text{(C) } \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0}$$

Key Points

- A system of first-order differential equations can be reduced to a single higher-order differential equation by eliminating one variable.
- Different elimination paths (eliminating x or y) lead to equivalent second-order equations.
- The order of the resulting differential equation equals the number of coupled first-order equations.

Q6.17.1 NAT 2017 2M Ans: 0.5 ● ○ ○

The given initial value problem is

$$\frac{dx}{dt} = \sin t, \qquad x(0) = 0$$

Method 1

Separating the variables and Integrating on both sides,

$$dx = \sin t \, dt$$

$$x = -\cos t + C$$

Using the initial condition, $x(0) = 0$, gives

$$0 = -\cos 0 + C = -1 + C \Rightarrow C = 1$$

Hence,

$$x = 1 - \cos t.$$

At $t = \pi/3$,

$$x\left(\frac{\pi}{3}\right) = 1 - \cos\left(\frac{\pi}{3}\right) = 1 - \frac{1}{2} = \frac{1}{2}.$$

Method 2

By the Fundamental Theorem of Calculus,

$$x(t) = x(0) + \int_0^t \sin\tau \, d\tau.$$

Since $x(0) = 0$,

$$x(t) = \int_0^t \sin\tau \, d\tau = [-\cos\tau]_0^t = 1 - \cos t$$

Therefore,

$$x\left(\frac{\pi}{3}\right) = 1 - \frac{1}{2} = \frac{1}{2}.$$

$$\boxed{0.5}$$

Key Points

- The Fundamental Theorem of Calculus provides a direct way to solve first-order equations of the form $\dfrac{dx}{dt} = f(t)$.

Q6.16.1 MCQ 2016 2M Ans: D ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + y = 0 \text{ with the initial conditions } y|_{x=0} = 5, \left.\frac{dy}{dx}\right|_{x=0} = 10$$

Method 1

The auxiliary equation is

$$m^2 + 1 = 0$$
$$m = \pm i.$$

Hence, the complementary solution is

$$y = C_1 \cos x + C_2 \sin x$$

Using the initial conditions,
At $x = 0$, $y = 5$,

$$5 = C_1 \cos 0 + C_2 \sin 0 = C_1$$
$$C_1 = 5$$

Differentiating,

$$\frac{dy}{dx} = -C_1 \sin x + C_2 \cos x$$

Using $\left.\dfrac{dy}{dx}\right|_{x=0} = 10$, we get

$$10 = -5 \sin 0 + C_2 \cos 0 = C_2$$
$$C_2 = 10$$

Therefore,

$$y = 5\cos x + 10\sin x.$$

Method 2

Recognize that

$$\frac{d^2y}{dx^2} + y = 0$$

is the governing equation of simple harmonic motion.
The solution is of the form

$$y = A\cos x + B\sin x$$

The initial displacement gives $A = y(0) = 5$.
Since

$$\frac{dy}{dx} = -A\sin x + B\cos x,$$

the initial slope gives $B = \left.\frac{dy}{dx}\right|_{x=0} = 10$

Hence,

$$y = 5\cos x + 10\sin x.$$

Therefore, the correct option is

$$\boxed{\text{(D) } y = 5\cos x + 10\sin x}$$

Key Points

- The differential equation $\frac{d^2y}{dx^2} + y = 0$ has purely imaginary characteristic roots, leading to sinusoidal solutions.
- Initial displacement determines the coefficient of $\cos x$, while the initial slope determines the coefficient of $\sin x$.

Q6.15.1 MCQ 2015 1M Ans: D ● ○ ○

The given first-order linear differential equation is

$$\frac{dy}{dx} + p(x)y = r(x)$$

Its integrating factor is

$$\text{IF} = e^{\int p(x)\,dx},$$

which is non-zero since $p(x)$ is continuous.
Multiplying the equation by the integrating factor,

$$e^{\int p(x)\,dx}\frac{dy}{dx} + p(x)e^{\int p(x)\,dx}y = r(x)e^{\int p(x)\,dx}$$

Using the product rule,

$$\frac{d}{dx}\left(ye^{\int p(x)\,dx}\right) = r(x)e^{\int p(x)\,dx}.$$

This is an exact differential equation.
Therefore, the correct option is

$$\boxed{\text{(D) Exact differential equation}}$$

Key Points

- Multiplication by the integrating factor converts the left-hand side into the derivative of a product.
- Every first-order linear differential equation can be transformed into an exact differential equation using its integrating factor.

Mistakes to be avoided

- Remember that the integrating factor must be non-zero throughout the interval of interest.

Q6.14.1 MCQ 2014 2M Ans: A ● ○ ○

The given differential equation is

$$\frac{dy}{dx} - \frac{y}{1+x} = 1 + x$$

Comparing with the standard linear form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{we have } P(x) = -\frac{1}{1+x}$$

Hence, the integrating factor is

$$\text{IF} = e^{\int P(x)\,dx} = e^{\int -\frac{1}{1+x}\,dx} = e^{-\ln(1+x)} = \frac{1}{1+x}$$

Hence, the correct option is

$$\boxed{\text{(A) } \frac{1}{1+x}}$$

Mistakes to be avoided

- Remember that $e^{-\ln(1+x)} = \dfrac{1}{1+x}$, not $1+x$.
- Always rewrite the equation in the standard linear form before identifying $P(x)$.

Q6.14.2 MCQ 2014 2M Ans: B ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + x^2\frac{dy}{dx} + x^3y = e^x$$

The highest order derivative present is $\dfrac{d^2y}{dx^2}$.

The highest order derivative appears in power one.
Therefore, the degree of the differential equation is 1.
Each occurrence of y and its derivatives is of the first degree and appears linearly, with coefficients depending only on the independent variable x.
Hence, the given equation is a second-order linear differential equation.
Thus, the correct option is

$$\boxed{\text{(B) linear differential equation of first degree}}$$

Key Points

- The *degree* of a differential equation is determined by the power of the highest order derivative present.
- A differential equation is *linear* if the dependent variable and all its derivatives appear only to the first power and are not multiplied together.

Mistakes to be avoided

- The forcing term e^x affects homogeneity, not linearity or order.

Q6.13.1 MCQ 2013 2M Ans: B ● ○ ○

The given differential equation is

$$\frac{dy}{dx} = y^2 \text{ with the initial condition } y(0) = 1$$

By variable-separable method,

$$\frac{dy}{y^2} = dx$$

$$\int y^{-2}\, dy = \int dx$$

$$-\frac{1}{y} = x + C$$

Using the initial condition,

$$-\frac{1}{1} = 0 + C \Rightarrow C = -1$$

Hence,

$$-\frac{1}{y} = x - 1$$

$$\frac{1}{y} = 1 - x$$

Therefore,

$$y = \frac{1}{1-x}.$$

Thus, the correct option is

$$\boxed{\text{(B) } \frac{1}{1-x}}$$

Mistakes to be avoided

- Do not integrate $\frac{1}{y^2}$ as $\ln y$; its integral is $-\frac{1}{y}$.

Q6.13.2 MCQ 2013 2M Ans: C ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} + 0.25y = 0 \text{ with the initial conditions } y(0) = 0,\ \left.\frac{dy}{dx}\right|_{x=0} = 1$$

The auxiliary equation is

$$m^2 - m + 0.25 = 0$$

$$\left(m - \frac{1}{2}\right)^2 = 0$$

Thus, there is a repeated root $m = 1/2$.
Hence, the complementary solution is

$$y = (C_1 + C_2x)e^{x/2}$$

Differentiating,

$$\frac{dy}{dx} = e^{x/2}\left[\frac{1}{2}(C_1 + C_2x) + C_2\right]$$

Using the boundary conditions,

At $x = 0$, $y = 0$; $C_1 = 0$.
Also,

$$\left.\frac{dy}{dx}\right|_{x=0} = \frac{1}{2}C_1 + C_2 = 1,$$

$$C_2 = 1$$

Therefore,

$$y = xe^{x/2}.$$

Therefore, the correct option is

$$\boxed{\text{(C) } xe^{0.5x}}$$

Key Points

- A repeated root m of the auxiliary equation gives the complementary solution $(C_1 + C_2x)e^{mx}$.

Q6.12.1 MCQ 2012 1M Ans: D ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} - 4y = 0$$

The auxiliary equation is

$$m^2 - 4 = 0$$

$$m = \pm 2$$

Hence, the complementary solution is

$$y = C_1e^{2x} + C_2e^{-2x}$$

Using the identities

$$\cosh 2x = \frac{e^{2x} + e^{-2x}}{2}, \qquad \sinh 2x = \frac{e^{2x} - e^{-2x}}{2}$$

the solution can be rewritten as

$$y = a\cosh 2x + b\sinh 2x,$$

where a and b are arbitrary constants.
Therefore, the correct option is

$$\boxed{\text{(D) } y = a\cosh 2x + b\sinh 2x}$$

Key Points

- Hyperbolic functions are convenient alternative forms of exponential solutions:

$$\cosh z = \frac{e^z + e^{-z}}{2}, \qquad \sinh z = \frac{e^z - e^{-z}}{2}.$$

Mistakes to be avoided

- Do not confuse hyperbolic functions with trigonometric functions; real roots lead to cosh and sinh, while imaginary roots lead to cos and sin.

Q6.11.1 MCQ | 2011 | 2M | Ans: D | ● ● ○

The given differential equation is

$$\frac{dy}{dx} = \frac{y^2}{x} + \frac{y}{x} - \frac{2}{x}$$

$$\frac{dy}{dx} = \frac{y^2 + y - 2}{x} = \frac{(y-1)(y+2)}{x}$$

Method 1

Separating the variables,

$$\frac{dy}{(y-1)(y+2)} = \frac{dx}{x}$$

Using partial fractions,

$$\frac{1}{(y-1)(y+2)} = \frac{1}{3}\left(\frac{1}{y-1} - \frac{1}{y+2}\right)$$

Integrating,

$$\frac{1}{3}\ln\left|\frac{y-1}{y+2}\right| = \ln|x| + C$$

$$\ln\left|\frac{y-1}{y+2}\right| = \ln|Cx^3|$$

$$\frac{y-1}{y+2} = Cx^3$$

$$y(1 - Cx^3) = 1 + 2Cx^3$$

$$y = \frac{1 + 2Cx^3}{1 - Cx^3}$$

Redefining the arbitrary constant as $C = \frac{1}{c}$,

$$y = \frac{c + 2x^3}{c - x^3},$$

Method 2

Verify option **D** directly.
Let

$$y = \frac{c + 2x^3}{c - x^3}$$

Differentiating,

$$\frac{dy}{dx} = \frac{9cx^2}{(c - x^3)^2}$$

Also,

$$y - 1 = \frac{3x^3}{c - x^3}, \qquad y + 2 = \frac{3c}{c - x^3}$$

Therefore,

$$\frac{(y-1)(y+2)}{x} = \frac{9cx^6}{x(c - x^3)^2} = \frac{9cx^2}{(c - x^3)^2} = \frac{dy}{dx}.$$

Hence, option **D** satisfies the given differential equation.
Therefore, the correct option is

$$\boxed{\text{(D) } y = \frac{c + 2x^3}{c - x^3}}$$

Q6.10.1 | MCQ | 2010 | 2M | Ans: D | ● ○ ○

The given differential equation is

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 2y = 0 \text{ with the initial conditions } y(0) = 0, \left.\frac{dy}{dt}\right|_{t=0} = -1$$

The auxiliary equation is

$$m^2 + 2m + 2 = 0$$

$$m = -1 \pm i$$

Hence, the complementary solution is

$$y = e^{-t}\left(C_1 \cos t + C_2 \sin t\right)$$

Differentiating,

$$\frac{dy}{dt} = e^{-t}(-C_1 \sin t + C_2 \cos t) - e^{-t}(C_1 \cos t + C_2 \sin t)$$

Using the initial conditions,
At $t = 0,\ y = 0;\ C_1 = 0.$
Also,

$$\left.\frac{dy}{dt}\right|_{t=0} = C_2 - C_1 = -1 \Rightarrow C_2 = -1$$

Therefore,

$$y = -e^{-t} \sin t.$$

Therefore, the correct option is

$$\boxed{\text{(D)} \ -e^{-t} \sin t}$$

Key Points

- Complex conjugate roots $m = \alpha \pm i\beta$ produce the solution $y = e^{\alpha t}\left(C_1 \cos \beta t + C_2 \sin \beta t\right)$.
- The factor e^{-t} represents exponential decay, while the sine and cosine terms represent oscillatory motion.

Q6.09.1 | MCQ | 2009 | 2M | Ans: B | ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$$

Assume a trial solution $y = e^{mx}$.
Substituting into the differential equation,

$$m^2e^{mx} - me^{mx} - 6e^{mx} = 0.$$

Since $e^{mx} \neq 0$, we obtain the auxiliary equation

$$m^2 - m - 6 = 0$$

$$(m - 3)(m + 2) = 0$$

Hence, the roots are $m = 3,\ -2$.
Since the roots are real and distinct, the complementary solution is

$$y = C_1e^{3x} + C_2e^{-2x}$$

Therefore, the correct option is

$$\boxed{\text{(B) } C_1e^{3x} + C_2e^{-2x}}$$

Mistakes to be avoided

- Do not miss the negative root $m = -2$ while factoring the quadratic.

Q6.08.1 MCQ 2008 1M Ans: D ● ● ○

The given differential equation is

$$x\,dy - y\,dx = 0$$

It can be written in the differential form

$$-y\,dx + x\,dy = 0$$

Thus,

$$M(x, y) = -y, \qquad N(x, y) = x$$

Since

$$\frac{\partial M}{\partial y} = -1, \qquad \frac{\partial N}{\partial x} = 1$$

the equation is not exact.
We now test each proposed integrating factor.
From (A), for $\mu = \frac{1}{x^2}$,

$$M = -\frac{y}{x^2}, \qquad N = \frac{1}{x}$$

$$\frac{\partial M}{\partial y} = -\frac{1}{x^2}, \qquad \frac{\partial N}{\partial x} = -\frac{1}{x^2}$$

Hence, it is an integrating factor.
From (B), for $\mu = \frac{1}{y^2}$,

$$M = -\frac{1}{y}, \qquad N = \frac{x}{y^2}$$

$$\frac{\partial M}{\partial y} = \frac{1}{y^2}, \qquad \frac{\partial N}{\partial x} = \frac{1}{y^2}$$

Hence, it is also an integrating factor.
From (C). for $\mu = \frac{1}{xy}$,

$$M = -\frac{1}{x}, \qquad N = \frac{1}{y}$$

$$\frac{\partial M}{\partial y} = 0, \qquad \frac{\partial N}{\partial x} = 0$$

Thus, it is an integrating factor.
From (D), for $\mu = \frac{1}{x+y}$,

$$M = -\frac{y}{x+y}, \qquad N = \frac{x}{x+y}$$

$$\frac{\partial M}{\partial y} = -\frac{x}{(x+y)^2}, \qquad \frac{\partial N}{\partial x} = \frac{y}{(x+y)^2}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is *not* an integrating factor.
Therefore, the correct option is

$$\boxed{\text{(D) } \frac{1}{(x+y)}}$$

Key Points

- An integrating factor transforms a non-exact differential equation into an exact one.

Q6.08.2 MCQ 2008 1M Ans: D ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + y = 1$$

Method 1

The corresponding homogeneous equation is

$$\frac{d^2y}{dx^2} + y = 0$$

Its auxiliary equation is

$$m^2 + 1 = 0$$

$$m = \pm i$$

Hence, the complementary solution is

$$y_c = C_1 \cos x + C_2 \sin x$$

Since the forcing function is a constant, assume a constant particular solution $y_p = A$.
Substituting,

$$0 + A = 1 \Rightarrow \; A = 1$$

Therefore, the complete solution is

$$y = C_1 \cos x + C_2 \sin x + 1$$

Checking the options:

$$\begin{aligned}
&\textbf{A: } y = 1 && (C_1 = C_2 = 0),\\
&\textbf{B: } y = 1 + \cos x && (C_1 = 1,\ C_2 = 0),\\
&\textbf{C: } y = 1 + \sin x && (C_1 = 0,\ C_2 = 1).
\end{aligned}$$

All three satisfy the general solution.
However, (D) contains a constant term equal to 2, which cannot arise.
Hence, option **D** is *not* a solution.

Method 2

Verify option **D** directly.
Let

$$y = 2 + \sin x + \cos x$$

Then,

$$\frac{dy}{dx} = \cos x - \sin x$$

$$\frac{d^2y}{dx^2} = -\sin x - \cos x$$

Therefore,

$$\frac{d^2y}{dx^2} + y = (-\sin x - \cos x) + (2 + \sin x + \cos x) = 2 \neq 1.$$

Hence, option **D** does not satisfy the differential equation.

$$\boxed{\text{(D) } y = 2 + \sin x + \cos x}$$

Key Points

- For a constant forcing function, the particular integral is obtained by assuming a constant solution.

Q6.08.3 MCQ 2008 2M Ans: B ● ● ○

The given differential equation is

$$\frac{dy}{dx} + Ay^3 + By = 0$$

where A and B are constants.
Consider the transformation $u = y^{-2}$.
Differentiating,

$$\frac{du}{dx} = -2y^{-3}\frac{dy}{dx}$$

$$\frac{dy}{dx} = -\frac{y^3}{2}\frac{du}{dx}$$

Substituting into the given equation,

$$-\frac{y^3}{2}\frac{du}{dx} + Ay^3 + By = 0$$

$$-\frac{1}{2}\frac{du}{dx} + A + \frac{B}{y^2} = 0$$

Since $u = \frac{1}{y^2}$, the equation becomes

$$-\frac{1}{2}\frac{du}{dx} + Bu + A = 0$$

$$\frac{du}{dx} - 2Bu = 2A$$

This is a first-order linear differential equation in u.
Hence, the correct transformation is

$$u = y^{-2}.$$

Therefore, the correct option is

$$\boxed{\text{(B) } u = y^{-2}}$$

Key Points

- A suitable substitution should transform both the derivative term and the nonlinear terms into expressions involving only the new dependent variable.
- Such substitutions are commonly used to linearize Bernoulli- and Riccati-type differential equations.

Q6.07.1 MCQ 2007 1M Ans: D ● ● ○

The given differential equation is

$$(x^2 + 2x)\frac{dy}{dx} = 2(x+1)y \text{ with the initial condition } y(x_0) = y_0$$

Separating the variables,

$$\frac{dy}{y} = \frac{2(x+1)}{x(x+2)}\,dx$$

Observe that $\frac{d}{dx}(x^2 + 2x) = 2(x+1)$.

Hence, Integrating

$$\int \frac{dy}{y} = \int \frac{d(x^2+2x)}{x^2+2x}$$

$$\ln|y| = \ln|x^2+2x| + C$$

Therefore,

$$y = C(x^2+2x)$$

Using the initial condition,

$$y_0 = C(x_0^2+2x_0),$$

$$C = \frac{y_0}{x_0^2+2x_0},$$

provided $x_0^2 + 2x_0 \neq 0$.
Thus,

$$y(x) = \frac{(x^2+2x)y_0}{x_0^2+2x_0}$$

The above expression is valid only when $x_0 \neq 0, -2$.

For the initial condition $x_0 = -2,\ y_0 = 0$, the coefficient of $\frac{dy}{dx}$ and the right-hand side both vanish, giving $0 = 0$.

Hence, the differential equation imposes no restriction on the derivative at the initial point, leading to infinitely many solution curves passing through $(-2, 0)$.

Therefore, the correct option is

$$\boxed{\text{D } y(x=-2)=0}$$

Key Points

- A first-order differential equation may lose uniqueness at points where the coefficient of the highest derivative becomes zero.
- Infinite solutions occur when the differential equation reduces to the identity $0 = 0$ at the specified initial point.
- Always examine singular points separately instead of directly applying the general solution.

Q6.07.2 MCQ 2007 2M Ans: D ● ● ○

The given differential equation is

$$x\frac{dy}{dx} + y(x^2-1) = 2x^3$$

Dividing throughout by x,

$$\frac{dy}{dx} + \left(x - \frac{1}{x}\right)y = 2x^2,$$

which is a first-order linear differential equation.
Here,

$$P(x) = x - \frac{1}{x}, \qquad Q(x) = 2x^2$$

The integrating factor is

$$\text{IF} = e^{\int (x-\frac{1}{x})dx} = e^{x^2/2 - \ln x} = \frac{e^{x^2/2}}{x}$$

Hence,

$$y \cdot \frac{e^{x^2/2}}{x} = \int 2x^2 \cdot \frac{e^{x^2/2}}{x}\, dx + C$$

$$y \cdot \frac{e^{x^2/2}}{x} = \int 2xe^{x^2/2}\, dx + C$$

Let $t = \frac{x^2}{2}$, so that $dt = x\, dx$.
Then,

$$\int 2xe^{x^2/2}\, dx = 2\int e^t\, dt = 2e^t = 2e^{x^2/2}$$

Thus,

$$y \cdot \frac{e^{x^2/2}}{x} = 2e^{x^2/2} + C,$$

$$y = 2x + Cxe^{-x^2/2}.$$

Therefore, the correct option is

$$\boxed{\text{(D) } 2x + cxe^{-x^2/2}}$$

Mistakes to be avoided

- Simplify $e^{x^2/2-\ln x} = \frac{e^{x^2/2}}{x}$, not $xe^{x^2/2}$.

Q6.06.1 | MCQ | 2006 | 2M | Ans: C | ● ● ○

The given differential equation is

$$x^2\frac{d^3y}{dx^3} + 2x\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$$

Multiplying throughout by x,

$$x^3\frac{d^3y}{dx^3} + 2x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = 0$$

This is a Cauchy–Euler differential equation.
Using the substitution $x = e^t,\ D = \frac{d}{dt}$, the derivatives transform as

$$x\frac{dy}{dx} = Dy \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y, \qquad x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y.$$

Hence,

$$[D(D-1)(D-2) + 2D(D-1) - 2D]\, y = 0$$

$$(D^3 - 3D^2 + 2D + 2D^2 - 2D - 2D)y = 0$$

$$(D^3 - D^2 - 2D)y = 0$$

The auxiliary equation is

$$m^3 - m^2 - 2m = 0$$

$$m(m-2)(m+1) = 0$$

$$m = 0,\ 2,\ -1$$

Therefore,

$$y = C_1 + C_2x^2 + \frac{C_3}{x}$$

Rearranging the constants,

$$y = C_1x^2 + C_2x^{-1} + C_3$$

Therefore, the correct option is

$$\boxed{\text{(C) } y = C_1x^2 + C_2x^{-1} + C_3}$$

Mistakes to be avoided

- Do not forget to multiply the entire equation by x before applying the Cauchy–Euler transformations.

Q6.05.1 MCQ 2005 1M Ans: A ● ○ ○

The standard one-dimensional wave equation is

$$\frac{\partial^2 c}{\partial t^2} = a^2 \frac{\partial^2 c}{\partial x^2}$$

which corresponds to statement **4**.
The standard one-dimensional heat equation is

$$\frac{\partial c}{\partial t} = a^2 \frac{\partial^2 c}{\partial x^2}$$

which corresponds to statement **2**.
Hence, the correct matching is

$$P \to 4, \qquad Q \to 2$$

Thus, the correct option is

$$\boxed{\text{(A) P–4, Q–2}}$$

Key Points

- The wave equation is hyperbolic and is second-order in both time and space.
- The heat equation is parabolic and is first-order in time and second-order in space.
- The parameter a represents the wave speed in the wave equation and the thermal diffusivity coefficient (through a^2) in the heat equation.

Q6.05.2 MCQ 2005 2M Ans: C ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$$

The auxiliary equation is

$$m^2 + am + b = 0$$

$$m = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

The required solution is of the form

$$y = (C_1 + C_2 x)e^{mx}$$

which occurs when the auxiliary equation has a repeated root.
Therefore, the discriminant must be zero:

$$a^2 - 4b = 0$$

$$a^2 = 4b.$$

Therefore, the correct option is

$$\boxed{\text{(C) } a^2 = 4b}$$

Key Points

- A repeated root occurs only when the discriminant of the characteristic equation is zero.

Q6.05.3 MCQ | 2005 | 2M | Ans: A | ● ● ○

For the equation

$$\frac{dy}{dx} + 2xy + 2x = 0$$

rewrite it as

$$\frac{dy}{dx} + 2x(y + 1) = 0$$

Let $u = y + 1$. Then,

$$\frac{du}{dx} = \frac{dy}{dx},$$

and the equation becomes

$$\frac{du}{dx} + 2xu = 0$$

Separating the variables and Integrating,

$$\frac{du}{u} = -2x\,dx$$

$$\ln|u| = -x^2 + C$$

$$u = Ce^{-x^2}$$

Hence,

$$f(x) = Ce^{-x^2} - 1.$$

Similarly, for

$$\frac{dy}{dx} + 2xy - 2x = 0$$

rewrite it as

$$\frac{dy}{dx} + 2x(y - 1) = 0$$

Let $v = y - 1$
Then,

$$\frac{dv}{dx} + 2xv = 0,$$

whose solution is $v = Ce^{-x^2}$.
Therefore,

$$g(x) = Ce^{-x^2} + 1.$$

Since the constant of integration is the same in both solutions,

$$g(x) - f(x) = 2$$

$$g(x) = f(x) + 2.$$

Hence, the correct option is

$$\boxed{\text{(A) } g(x) = f(x) + 2}$$

Q6.04.1 MCQ | 2004 | 1M | Ans: C | ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + \sin x\,\frac{dy}{dx} + ye^x = \sinh x$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, therefore the order of the differential equation is 2.

Further, dependent variable y and its derivatives appear to the first power and are not multiplied together.
Hence, the differential equation is linear.
Therefore, it is a second-order linear differential equation.
Therefore, the correct option is

$$\boxed{\text{(C) second order and linear}}$$

Q6.04.2 MCQ 2004 2M Ans: B ● ○ ○

The given differential equation is

$$\frac{dX}{dt} + \frac{X}{20} = 10$$

where X denotes the amount of salt (kg) in the tank.

Method 1

This is a first-order linear differential equation of the form

$$\frac{dX}{dt} + P(t)X = Q(t), \quad \text{where } P(t) = \frac{1}{20}, \; Q(t) = 10$$

The integrating factor is

$$\text{IF} = e^{\int 1/20\, dt} = e^{t/20}$$

Multiplying the equation by the integrating factor,

$$\frac{d}{dt}\left(Xe^{t/20}\right) = 10e^{t/20}$$

Integrating,

$$Xe^{t/20} = 10\int e^{t/20}\, dt + C = 200e^{t/20} + C$$

Hence,

$$X = 200 + Ce^{-t/20}$$

Using the initial condition, $X(0) = 0$, gives

$$0 = 200 + C \Rightarrow \; C = -200$$

Therefore,

$$X = 200\left(1 - e^{-t/20}\right)$$

When $X = 100$,

$$100 = 200\left(1 - e^{-t/20}\right)$$

Thus,

$$e^{-t/20} = \frac{1}{2}$$

$$e^{t/20} = 2$$

Hence,

$$t = 20 \ln 2.$$

Method 2

The equation can be rewritten as

$$\frac{dX}{dt} = \frac{200 - X}{20}$$

The equilibrium (steady-state) amount of salt is $X_s = 200$ kg
For a first-order system,

$$X(t) = X_s + [X(0) - X_s]\, e^{-t/20}$$

Using $X(0) = 0$, we obtain

$$X(t) = 200 - 200e^{-t/20}$$

Setting $X(t) = 100$, gives

$$e^{-t/20} = \frac{1}{2}$$

$$t = 20 \ln 2.$$

Hence, the correct option is

$$\boxed{\text{(B) } 20 \ln 2}$$

Key Points

- The steady-state value is obtained by setting $\frac{dX}{dt} = 0$, giving $X_s = 200$ kg.
- The transient term decays exponentially with time constant 20 minutes.

Q6.04.3 MCQ 2004 2M Ans: D ● ● ○

The given differential equation is

$$\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2} = 0$$

Method 1

Let $P = dy/dx$.
Using the chain rule,

$$\frac{d^2y}{dx^2} = \frac{dP}{dx} = \frac{dP}{dy}\frac{dy}{dx} = P\frac{dP}{dy}$$

Substituting,

$$P^2 + yP\frac{dP}{dy} = 0$$

Since $P \neq 0$ for the non-trivial solution,

$$P + y\frac{dP}{dy} = 0$$

Hence,

$$\frac{dP}{P} = -\frac{dy}{y}$$

Integrating,

$$\ln|P| = -\ln|y| + \ln|\alpha|,$$

where α is the constant of integration.
Therefore,

$$P = \frac{\alpha}{y}$$
$$\frac{dy}{dx} = \frac{\alpha}{y}.$$

Method 2

Observe that

$$\frac{d}{dx}\left(y\frac{dy}{dx}\right) = \left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}$$

Hence, the given differential equation becomes

$$\frac{d}{dx}\left(y\frac{dy}{dx}\right) = 0$$

Integrating directly,

$$y\frac{dy}{dx} = \alpha,$$

where α is a constant.
Therefore,

$$\frac{dy}{dx} = \frac{\alpha}{y}.$$

Hence, the correct option is

$$\boxed{\text{(D)}\ \left(\frac{dy}{dx}\right) = \frac{\alpha}{y}}$$

Key Points

- Introducing $P = \frac{dy}{dx}$ converts equations involving y'' into first-order equations in P using $\frac{dP}{dx} = P\frac{dP}{dy}$.
- The given equation is also recognizable as the derivative of the product $y\frac{dy}{dx}$, providing a much shorter solution.

Q6.03.1 MCQ 2003 2M Ans: D ● ● ○

The given differential equation is

$$(4t^2+1)\frac{dy}{dt} + 8ty - t = 0 \text{ with the initial condition } y(1) = 0$$

Dividing throughout by $4t^2 + 1$,

$$\frac{dy}{dt} + \frac{8t}{4t^2+1}y = \frac{t}{4t^2+1}$$

This is a first-order linear differential equation with

$$P(t) = \frac{8t}{4t^2+1}, \qquad Q(t) = \frac{t}{4t^2+1}$$

The integrating factor is

$$\text{IF} = e^{\int 8t/(4t^2+1)\,dt}$$

Let $u = 4t^2 + 1$, so that $du = 8t\,dt$.
Hence,

$$\text{IF} = e^{\int du/u} = 4t^2 + 1$$

The complete solution is

$$y \times (4t^2+1) = \int \frac{t}{4t^2+1} \times 4t^2 + 1 \cdot dt + C$$

$$(4t^2+1)y = \frac{t^2}{2} + C$$

Using $y(1) = 0$, gives

$$0 = \frac{\frac{1}{2} + C}{5} \implies C = -\frac{1}{2}$$

Therefore,

$$y = \frac{t^2-1}{2(4t^2+1)}.$$

Now,

$$\lim_{t\to\infty} y = \lim_{t\to\infty} \frac{t^2-1}{2(4t^2+1)} = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8}.$$

Therefore, the correct option is

$$\boxed{\text{(D) } 1/8}$$

Q6.03.2 MCQ 2003 1M Ans: A ● ○ ○

The given differential equation is

$$\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0$$

The auxiliary equation is

$$m^2 + 10m + 25 = 0$$

$$(m + 5)^2 = 0$$

Thus, the characteristic equation has a repeated root $m = -5$.
Hence, the complementary solution is

$$x = (C_1 + C_2t)e^{-5t}.$$

Since the equation is homogeneous, no particular integral is required.
Therefore, the solution is

$$x = (C_1 + C_2t)e^{-5t}.$$

Hence, the correct option is

$$\boxed{\text{(A) } (C_1 + C_2t)e^{-5t}}$$

Q6.00.1 MCQ 2000 1M Ans: A ● ○ ○

The given differential equation is

$$\cos^2 x \frac{dy}{dx} + y = \tan x$$

Dividing throughout by $\cos^2 x$,

$$\frac{dy}{dx} + \sec^2 x\, y = \frac{\tan x}{\cos^2 x}$$

This is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P(x) = \sec^2 x$$

Hence, the integrating factor is

$$\text{IF} = e^{\int \sec^2 x\, dx} = e^{\tan x}.$$

Therefore, the correct option is

$$\boxed{\text{(A) } e^{\tan x}}$$

Key Points

- The standard integral $\int \sec^2 x\, dx = \tan x$is frequently used in differential equations.

Q6.00.2 MCQ 2000 2M Ans: D ● ● ○

The given differential equation is

$$\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$$

The auxiliary equation is

$$m^4 + 2m^2 + 1 = 0$$

$$(m^2 + 1)^2 = 0$$

$$m = \pm i,$$

each with multiplicity 2.

For repeated complex roots $\alpha \pm i\beta$ of multiplicity 2, the complementary solution is

$$y = e^{\alpha x}\left[(C_1 + C_2x)\cos\beta x + (C_3 + C_4x)\sin\beta x\right]$$

Here, $\alpha = 0,\ \beta = 1$.
Therefore,

$$y = (C_1 + C_2x)\cos x + (C_3 + C_4x)\sin x.$$

Therefore, the correct option is

$$\boxed{\text{(D) } (c_1 + c_2x)\cos x + (c_3 + c_4x)\sin x}$$

Key Points

- A repeated pair of complex conjugate roots $\alpha \pm i\beta$ produces the solution

$$e^{\alpha x}\left[(C_1 + C_2x)\cos\beta x + (C_3 + C_4x)\sin\beta x\right].$$

Mistakes to be avoided

- Remember that the factor x is introduced for repeated roots, whether real or complex.

Q6.26.1 NAT | 2026 | 2M | Ans: 14.78 | ● ● ○

The given differential equation is

$$\frac{dy}{dt} = 2y \text{ with the initial condition } y(0) = 2$$

Separating the variables and Integrating,

$$\frac{dy}{y} = 2\,dt$$

$$\int \frac{dy}{y} = \int 2\,dt$$

$$\ln|y| = 2t + C$$

$$y = Ce^{2t}$$

Using the initial condition, we get

$$2 = Ce^0 \Rightarrow \ C = 2$$

Hence,

$$y = 2e^{2t}.$$

At $t = 1$,

$$y(1) = 2e^2 \approx 2(7.3891) = 14.7782.$$

Rounded to two decimal places,

$$\boxed{14.78}$$

Key Points

- The differential equation $\frac{dy}{dt} = ky$ models exponential growth (for $k > 0$) or exponential decay (for $k < 0$).

Q6.25.1 MCQ | 2025 | 1M | Ans: B | ● ○ ○

A differential equation is said to be *linear* if the dependent variable and its derivatives

- appear only to the first power,
- are not multiplied together, and
- have coefficients that are constants or functions of the independent variable only.

Examine each option:

A:

$$\frac{dy}{dx} + 2x = y^2$$

Since the dependent variable appears as y^2, the equation is nonlinear.

B.

$$x^3\frac{dy}{dx} + xy = x^2$$

This equation satisfies the above properties.
Therefore, is linear.

C.

$$x\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} = 0$$

The product $y\frac{dy}{dx}$ makes the equation nonlinear.

D.

$$\left(\frac{dy}{dx}\right)^2 + 2x = y$$

Since $\left(\frac{dy}{dx}\right)^2$ appears, the equation is nonlinear.
Therefore, the correct option is

$$\boxed{\text{(B) } x^3\frac{dy}{dx} + xy = x^2}$$

Key Points

- A linear differential equation is linear in the dependent variable and all its derivatives.
- Coefficients of y and its derivatives may be functions of the independent variable.
- Powers of the dependent variable or its derivatives, and products between them, make the equation nonlinear.

Q6.25.2 MCQ 2025 2M Ans: A ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + y = e^x \text{ with the initial condition } y(0) = 0$$

Method 1

This is a first-order linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P(x) = 1,\ Q(x) = e^x$$

The integrating factor is

$$\text{IF} = e^{\int 1\,dx} = e^x$$

The complete solution is

$$y \times \text{IF} = \int Q \times \text{IF} \cdot dx + C$$

$$ye^x = \int e^x \times e^x \cdot dx + C$$

$$ye^x = \frac{e^{2x}}{2} + C$$

$$y = \frac{e^x}{2} + Ce^{-x}$$

Using the initial condition,

$$0 = \frac{1}{2} + C \Rightarrow \ C = -\frac{1}{2}$$

Therefore,

$$y = \frac{1}{2}e^x - \frac{1}{2}e^{-x}.$$

Method 2

The complementary solution of is

$$y_c = Ce^{-x}$$

For the particular solution, since the forcing function is e^x, assume $y_p = Ae^x$.
Substituting into differential equation,

$$Ae^x + Ae^x = e^x$$

$$2A = 1 \Rightarrow A = \frac{1}{2}$$

Hence,

$$y = \frac{1}{2}e^{x} + Ce^{-x}$$

Using $y(0) = 0$, we obtain

$$C = -\frac{1}{2}$$

Thus,

$$y = \frac{1}{2}e^{x} - \frac{1}{2}e^{-x}.$$

Therefore, the correct option is

$$\boxed{\text{(A) } y = \frac{1}{2}e^{x} - \frac{1}{2}e^{-x}}$$

Key Points

- A first-order linear differential equation can be solved using either the integrating factor method or the complementary function–particular integral approach.
- When the forcing function is an exponential, assuming a particular solution of the same exponential form is often the quickest approach.

Q6.24.1 NAT 2024 2M Ans: 0.064 ● ● ○

The given differential equation is

$$\frac{d^2x}{dt^2} + 4x = e^{-t} \text{ with the initial conditions } x(0) = 0, \left.\frac{dx}{dt}\right|_{t=0} = 0$$

The auxiliary equation is

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

Hence, the complementary solution is

$$x_c = C_1 \cos 2t + C_2 \sin 2t$$

For the particular solution, assume $x_p = Ae^{-t}$.
Substituting into the differential equation,

$$Ae^{-t} + 4Ae^{-t} = e^{-t}$$

$$5A = 1 \Rightarrow A = \frac{1}{5}$$

Thus,

$$x = C_1 \cos 2t + C_2 \sin 2t + \frac{1}{5}e^{-t}$$

Using the initial conditions,
At $t = 0$,

$$0 = C_1 + \frac{1}{5} \Rightarrow C_1 = -\frac{1}{5}$$

Differentiating,

$$\frac{dx}{dt} = -2C_1 \sin 2t + 2C_2 \cos 2t - \frac{1}{5}e^{-t}$$

Using $\left.\dfrac{dx}{dt}\right|_{t=0} = 0$, gives

$$2C_2 - \frac{1}{5} = 0 \Rightarrow C_2 = \frac{1}{10}$$

Hence,

$$x(t) = -\frac{1}{5}\cos 2t + \frac{1}{10}\sin 2t + \frac{1}{5}e^{-t}.$$

Therefore,

$$x\left(\frac{\pi}{8}\right) = -\frac{1}{5\sqrt{2}} + \frac{1}{10\sqrt{2}} + \frac{e^{-\pi/8}}{5} \approx -0.14142 + 0.07071 + 0.13505 = 0.06434.$$

Rounded to three decimal places,

$$\boxed{0.064}$$

Key Points

- For complex roots $\alpha \pm i\beta$, the complementary solution is $e^{\alpha t}\left(C_1 \cos \beta t + C_2 \sin \beta t\right)$.
- For an exponential forcing function e^{at}, the particular integral can be obtained quickly using

$$\frac{1}{f(D)}e^{at} = \frac{e^{at}}{f(a)},$$

provided $f(a) \neq 0$.

Q6.23.1 MCQ 2023 1M Ans: B ● ○ ○

A quick test for linearity is:

- No powers such as y^2, $\left(\dfrac{dy}{dx}\right)^2$.
- No products such as $y\dfrac{dy}{dx}$.
- Coefficients may be constants or functions of the independent variable.

Among the given equations, only

$$\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 10y = 0$$

violates these conditions because the first derivative appears squared.
Therefore, the correct option is

$$\boxed{\text{(B) } \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 10y = 0}$$

Mistakes to be avoided

- Do not mistake variable coefficients such as $10x$ or $10xy$ (where $10x$ is the coefficient of y) for nonlinearity.

Q6.22.1 | NAT | 2022 | 1M | Ans: 1 | ● ○ ○

The given differential equation is

$$\left(\frac{dy}{dx}\right)^2 + 5\frac{dy}{dx} + 4y = 5x^3$$

The *order* of a differential equation is the order of the highest derivative present in the equation.
Here, the highest derivative present is $\frac{dy}{dx}$, which is the first derivative.
Therefore, the order of the differential equation is 1.
Hence, the order is

$$\boxed{1}$$

Key Points

- The **order** of a differential equation is determined solely by the highest order derivative present.
- The **degree** depends on the power of the highest order derivative (provided the equation is polynomial in derivatives).
- The power of a derivative does not affect the order of the differential equation.

Mistakes to be avoided

- Do not confuse *order* with *degree*; the squared first derivative does not make the order equal to 2.
- Ignore powers of derivatives while determining the order.

Q6.22.2 | MCQ | 2022 | 2M | Ans: D | ● ○ ○

The given differential equation is

$$4\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 0$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} + \frac{1}{4}y = 0$$

The auxiliary equation is

$$m^2 - m + \frac{1}{4} = 0$$

$$\left(m - \frac{1}{2}\right)^2 = 0$$

Thus, the characteristic equation has a repeated root $m = \frac{1}{2}$.
Hence, the complementary solution is

$$y = (C_1 + C_2x)\, e^{x/2}.$$

Hence, the correct option is

$$\boxed{\text{(D) } y = (c_1 + c_2x)e^{x/2}}$$

Key Points

- A repeated root of the auxiliary equation produces the solution $(C_1 + C_2x)e^{mx}$.

Q6.20.1 MCQ 2020 1M Ans: D ● ○ ○

The given differential equation is

$$\frac{dy}{dx} + my = e^{-mx}$$

Comparing with the standard linear differential equation

$$\frac{dy}{dx} + P(x)y = Q(x),$$

we identify $P(x) = m$.
Hence, the integrating factor is

$$\text{IF} = e^{\int P(x)\,dx} = e^{\int m\,dx} = e^{mx}.$$

Therefore, the correct option is

$$\boxed{\text{(D) } e^{mx}}.$$

Key Points

- The integrating factor depends only on the coefficient of the dependent variable y.

Q6.20.2 MCQ 2020 2M Ans: C ● ● ○

The given differential equation is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$$

This is a Cauchy–Euler differential equation.

Method 1

Using the substitution $x = e^t,\ D = \frac{d}{dt}$, the derivatives transform as

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

Substituting,

$$D(D-1)y + Dy - y = 0$$
$$(D^2 - 1)y = 0$$

The auxiliary equation is

$$m^2 - 1 = 0$$
$$m = \pm 1$$

Hence,

$$y = C_1e^t + C_2e^{-t}$$

Since $e^t = x$, the solution becomes

$$y = C_1x + \frac{C_2}{x}.$$

Method 2

Assume a trial solution $y = x^m$.
Then,

$$\frac{dy}{dx} = mx^{m-1}, \qquad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Substituting into the differential equation gives

$$m(m-1)x^m + mx^m - x^m = 0$$

$$(m^2 - 1)x^m = 0$$

Since $x^m \neq 0$, the indicial equation is

$$m^2 - 1 = 0$$

$$m = \pm 1$$

Therefore,

$$y = C_1 x + \frac{C_2}{x}.$$

Hence, the correct option is

$$\boxed{\text{(C) } y = C_1 x + \frac{C_2}{x}}$$

Key Points

- A Cauchy–Euler differential equation can be solved either by the substitution $x = e^t$ or by assuming a trial solution $y = x^m$.

Q6.19.1 MCQ 2019 1M Ans: B ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$$

Method 1

The auxiliary equation is

$$m^2 + am + b = 0$$

Since the roots are stated to be *real and equal*, let the repeated root be m.
From the sum of roots,

$$m + m = -a \Rightarrow m = -\frac{a}{2}$$

Hence, the repeated root is $m = -\frac{a}{2}$.
For a repeated real root, the complementary solution is

$$y = (C_1 + C_2 x)e^{mx}$$

$$y = (C_1 + C_2 x)e^{-ax/2}.$$

Method 2

For the auxiliary equation

$$m^2 + am + b = 0,$$

equal roots occur when the discriminant is zero:

$$a^2 - 4b = 0$$

The repeated root is $m = \frac{-a}{2}$.

$$y = (C_1 + C_2 x)e^{-ax/2}.$$

Hence, the correct option is

$$\boxed{\text{(B) } y = (c_1 + c_2 x)e^{-ax/2}}$$

Q6.19.2 | MCQ | 2019 | 2M | Ans: C | ● ● ○

The given Cauchy–Euler differential equation is

$$x^2\frac{d^2y}{dx^2} - 7x\frac{dy}{dx} + 16y = 0$$

Using the transformation $x = e^t,\ D = \frac{d}{dt}$, the derivatives transform as

$$x\frac{dy}{dx} = Dy, \qquad x^2\frac{d^2y}{dx^2} = D(D-1)y.$$

Substituting,

$$D(D-1)y - 7Dy + 16y = 0$$

$$(D^2 - 8D + 16)y = 0$$

The auxiliary equation is

$$m^2 - 8m + 16 = 0$$

$$(m-4)^2 = 0$$

Hence, the repeated root is $m = 4$.
The complementary solution in the transformed variable is

$$y = (C_1 + C_2t)e^{4t}$$

Since $t = \ln x,\ e^{4t} = x^4$, we obtain

$$y = (C_1 + C_2 \ln x)x^4.$$

Therefore, the correct option is

$$\boxed{\text{(C) } y = (C_1 + C_2 \ln x)x^4}$$

Mistakes to be avoided

- For repeated roots, remember that the second solution is $x^m \ln x$, not $x\,x^m$.

Q6.18.1 | NAT | 2018 | 2M | Ans: 4.55 | ● ● ○

Method 1

The given differential equation is

$$2\frac{d^2y}{dt^2} + 8y = 0 \text{ with the initial conditions } y(0) = 0,\ \left.\frac{dy}{dt}\right|_{t=0} = 10$$

Dividing throughout by 2,

$$\frac{d^2y}{dt^2} + 4y = 0$$

The auxiliary equation is

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

Hence, the complementary solution is

$$y = C_1 \cos 2t + C_2 \sin 2t$$

Differentiating,

$$\frac{dy}{dt} = -2C_1 \sin 2t + 2C_2 \cos 2t$$

Using the initial conditions,
At $t = 0$,

$$0 = C_1 \Rightarrow C_1 = 0$$

Also,

$$10 = 2C_2 \Rightarrow C_2 = 5$$

Therefore,

$$y = 5 \sin 2t$$

At $t = 1$,

$$y(1) = 5 \sin 2 \approx 5(0.909297) = 4.54649.$$

Method 2

The equation

$$\frac{d^2y}{dt^2} + 4y = 0$$

represents simple harmonic motion with angular frequency $\omega = 2$.
The standard solution is

$$y = A \cos 2t + B \sin 2t$$

Using $y(0) = 0$, gives $A = 0$.
Also,

$$\frac{dy}{dt} = 2B \cos 2t$$

Applying $\left.\frac{dy}{dt}\right|_{t=0} = 10$, yields

$$2B = 10 \Rightarrow B = 5$$

Hence,

$$y = 5 \sin 2t$$

$$y(1) = 5 \sin 2 \approx 4.54649.$$

Rounded to two decimal places,

$$\boxed{4.55}$$

Key Points

- The initial displacement determines the cosine coefficient, while the initial velocity determines the sine coefficient.
- The equation represents simple harmonic motion with angular frequency $\omega = 2$.

Q6.16.1 MCQ 2016 1M Ans: C ● ○ ○

The given functions are

$$f(x, y) = x^3 - 3xy^2$$

$$g(x, y) = 3x^2y - y^3$$

Compute the partial derivatives of f:

$$\frac{\partial f}{\partial x} = 3x^2 - 3y^2, \qquad \frac{\partial f}{\partial y} = -6xy$$

Now compute the partial derivatives of g:

$$\frac{\partial g}{\partial x} = 6xy, \qquad \frac{\partial g}{\partial y} = 3x^2 - 3y^2$$

Comparing,

$$\frac{\partial f}{\partial x} = \frac{\partial g}{\partial y}, \qquad \frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}$$

Hence, the correct relation is

$$\frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}.$$

Therefore, the correct option is

$$\boxed{\text{(C) } \frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}}$$

Q6.15.1 MCQ | 2015 | 2M | Ans: D | ● ○ ○

The given differential equation is

$$6y\frac{dy}{dx} - 25x = 0$$

Separating the variables,

$$6y\frac{dy}{dx} = 25x$$

$$6y\,dy = 25x\,dx$$

Integrating both sides,

$$6\int y\,dy = 25\int x\,dx,$$

$$6\left(\frac{y^2}{2}\right) = 25\left(\frac{x^2}{2}\right) + C$$

Hence,

$$3y^2 = \frac{25}{2}x^2 + C$$

$$6y^2 - 25x^2 = C_1,$$

where C_1 is an arbitrary constant.
Since the equation is of the form $Ay^2 - Bx^2 = C$, it represents a family of hyperbolas.
Therefore, the correct option is

$$\boxed{\text{(D) family of hyperbolas}}$$

Key Points

- The standard forms of conic sections are:

$$Ax^2 + By^2 = C \quad \Rightarrow \quad \text{ellipse (or circle)},$$

$$Ax^2 - By^2 = C \quad \Rightarrow \quad \text{hyperbola},$$

$$y = ax^2 + bx + c \quad \Rightarrow \quad \text{parabola}.$$

Q6.13.1 MCQ | 2013 | 1M | Ans: D | ● ○ ○

The given partial differential equation is

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2}$$

The highest order derivative present is $\frac{\partial^2 u}{\partial x^2}$,
which is a second-order derivative.
Hence, the order of the partial differential equation is 2.

Further, the dependent variable u is multiplied by its first derivative: $u\frac{\partial u}{\partial x}$.
Since the dependent variable appears in a product with its derivative, the equation is nonlinear.
Therefore, the given equation is a second-order nonlinear partial differential equation.
Therefore, the correct option is

$$\boxed{\text{(D) non-linear equation of order 2}}$$

Key Points

- This equation is the well-known *viscous Burgers' equation*, a fundamental nonlinear PDE in fluid mechanics and applied mathematics.

Q6.11.1 MCQ 2011 1M Ans: C ● ● ○

The given differential equation is

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 9x + 6$$

The corresponding homogeneous equation is

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0$$

The auxiliary equation is

$$m^2 + 6m + 9 = 0$$

$$(m + 3)^2 = 0$$

Hence, the repeated root is $m = -3$.
Therefore, the complementary solution is

$$y_c = (C_1 + C_2x)e^{-3x}$$

For the particular solution, assume $y_p = Ax + B$.
Then,

$$\frac{dy_p}{dx} = A, \qquad \frac{d^2y_p}{dx^2} = 0$$

Substituting into the differential equation,

$$0 + 6A + 9(Ax + B) = 9x + 6$$

Comparing coefficients,
For x:

$$9A = 9 \quad \Rightarrow \quad A = 1$$

For constants:

$$6A + 9B = 6 \quad \Rightarrow \quad B = 0$$

Hence,

$$y_p = x$$

Therefore, the complete solution is

$$y = (C_1 + C_2x)e^{-3x} + x.$$

Therefore, the correct option is

$$\boxed{\text{(C) } y = (C_1 + C_2x)e^{-3x} + x}$$

Key Points

- For a polynomial forcing function, assume a polynomial particular solution of the same degree.
- If the assumed particular solution overlaps with the complementary function, multiply it by an appropriate power of x. (Not required in this problem.)

Q6.10.1 MCQ 2010 1M Ans: C ● ○ ○

The given solution is

$$y = 5\sin\left(3x + \frac{\pi}{3}\right)$$

Differentiating with respect to x,

$$\frac{dy}{dx} = 15\cos\left(3x + \frac{\pi}{3}\right)$$

Differentiating once again,

$$\frac{d^2y}{dx^2} = -45\sin\left(3x + \frac{\pi}{3}\right)$$

Since

$$y = 5\sin\left(3x + \frac{\pi}{3}\right),$$

we have

$$\frac{d^2y}{dx^2} = -9y$$

Therefore,

$$\frac{d^2y}{dx^2} + 9y = 0.$$

Therefore, the correct option is

$$\boxed{\text{(C) } \frac{d^2y}{dx^2} + 9y = 0}$$

Key Points

- The function $A\sin(kx + \phi)$ or $A\cos(kx + \phi)$ always satisfies $\dfrac{d^2y}{dx^2} + k^2y = 0$.

Q6.10.2 MCQ 2010 2M Ans: D ● ● ○

The given differential equation is

$$\frac{dy}{dx} - y^2 = 1 \text{ with the initial condition } y(0) = 1$$

Method 1

Separating the variables,

$$\frac{dy}{dx} = 1 + y^2$$

$$\frac{dy}{1 + y^2} = dx$$

Integrating both sides,

$$\int \frac{dy}{1 + y^2} = \int dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

Using the initial condition,

$$1 = \tan C \Rightarrow C = \frac{\pi}{4}$$

Therefore,

$$y = \tan\left(x + \frac{\pi}{4}\right).$$

Method 2

Verify option **D**.
Let $y = \tan\left(x + \frac{\pi}{4}\right)$.
Differentiating,

$$\frac{dy}{dx} = \sec^2\left(x + \frac{\pi}{4}\right)$$

$$\frac{dy}{dx} = 1 + \tan^2\left(x + \frac{\pi}{4}\right) = 1 + y^2$$

$$\frac{dy}{dx} - y^2 = 1$$

Also,

$$y(0) = \tan\frac{\pi}{4} = 1$$

Hence, option **D** satisfies both the differential equation and the initial condition.
Therefore, the correct option is

$$\boxed{\text{(D) } y = \tan\left(x + \frac{\pi}{4}\right)}$$

Key Points

- The standard integral

$$\int \frac{dy}{1 + y^2} = \tan^{-1} y + C$$

is frequently used in separable differential equations.

Q6.09.1 MCQ 2009 1M Ans: A ● ○ ○

The given differential equation is

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = r,$$

where p, q, and r are constants.
The corresponding homogeneous equation is

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = 0$$

For any quadratic equation $am^2 + bm + c = 0$,the roots are

$$\begin{cases} \text{Real and distinct,} & b^2 - 4ac > 0, \\ \text{Real and equal,} & b^2 - 4ac = 0, \\ \text{Complex conjugate,} & b^2 - 4ac < 0. \end{cases}$$

The auxiliary equation is

$$m^2 + pm + q = 0,$$

where $a = 1,\ b = p,\ c = q$.

Therefore, $\Delta = p^2 - 4q$.
For real and distinct roots,

$$p^2 - 4q > 0.$$

Therefore, the correct option is

$$\boxed{\text{(A) } p^2 - 4q > 0}$$

Mistakes to be avoided

- Remember:

$$\Delta > 0 \Rightarrow \text{real and distinct roots,}$$

not $\Delta \geq 0$.

Q6.09.2 MCQ 2009 2M Ans: C ● ● ○

The given differential equation is

$$\frac{d^2y}{dx^2} = 0 \text{ with the boundary conditions } \left.\frac{dy}{dx}\right|_{x=0} = 1, \ \left.\frac{dy}{dx}\right|_{x=1} = 1$$

Integrating once,

$$\frac{dy}{dx} = C_1$$

Applying the first boundary condition, $1 = C_1$.
The second boundary condition is also satisfied since $\dfrac{dy}{dx} = 1$ for every value of x.
Integrating again,

$$y = \int 1\,dx = x + C_2$$

Since no condition is given on y, the constant C_2 remains arbitrary.
Therefore,

$$y = x + C, \ C \text{ is arbitrary}.$$

Therefore, the correct option is

$$\boxed{\text{(C) } y = x + C, \text{ where } C \text{ is an arbitrary constant}}$$

Q6.08.1 MCQ 2008 2M Ans: A ● ● ○

The given partial differential equation is

$$\frac{\partial^2 u}{\partial x^2} = \pi^2 \frac{\partial u}{\partial t},$$

with the boundary conditions $u(0,t) = 0, \ u(1,t) = 0$,
and the initial condition $u(x,0) = \sin(\pi x)$.
Using separation of variables, let $u(x,t) = X(x)T(t)$.
Substituting into the differential equation gives

$$X''T = \pi^2 XT'$$

Dividing by XT,

$$\frac{X''}{X} = \pi^2 \frac{T'}{T} = -\lambda$$

The boundary conditions $X(0) = X(1) = 0$ give the eigenfunction

$$X(x) = \sin(\pi x), \ \text{with } \lambda = \pi^2$$

Hence,

$$\frac{T'}{T} = -\frac{\lambda}{\pi^2} = -1$$

so $T = e^{-t}$.
Therefore,

$$u(x,t) = e^{-t}\sin(\pi x).$$

Therefore, the correct option is

$$\boxed{\text{(A) } e^{-t}\sin(\pi x)}$$

Key Points

- For heat-type equations, solutions are commonly obtained using the method of separation of variables.
- The boundary conditions $u(0,t) = u(1,t) = 0$ lead to sine eigen functions.
- The initial condition determines the amplitude of the separated solution.
- Direct substitution is often the quickest verification method.

Q6.08.2 MCQ 2008 2M Ans: A ● ● ○

The given differential equation is

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$$

Assume a trial solution $y = e^{mx}$.
Substituting into the differential equation,

$$m^2e^{mx} + 2me^{mx} + 2e^{mx} = 0$$

Since $e^{mx} \neq 0$, the auxiliary equation is

$$m^2 + 2m + 2 = 0$$

$$m = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$$

For complex roots $\alpha \pm i\beta$, the complementary solution is

$$y = e^{\alpha x}\left(C_1\cos\beta x + C_2\sin\beta x\right).$$

Here, $\alpha = -1,\ \beta = 1$.
Thus,

$$y = e^{-x}(C_1\cos x + C_2\sin x)$$

Using Euler's identities,

$$\cos x = \frac{e^{ix} + e^{-ix}}{2},$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

the solution can be written as a linear combination of $e^{(-1+i)x}$ and $e^{(-1-i)x}$.
Hence, the independent solutions are

$$e^{(-1+i)x},\ e^{(-1-i)x}.$$

Therefore, the correct option is

$$\boxed{\text{(A) } e^{-(1+i)x},\ e^{-(1-i)x}}$$

Key Points

- Using Euler's formula, $e^{(\alpha \pm i\beta)x} = e^{\alpha x}(\cos \beta x \pm i \sin \beta x)$, the same solution can be expressed in exponential form.
- The two exponential solutions corresponding to distinct roots are linearly independent.

Q6.05.1 | MCQ | 2005 | 1M | Ans: C | ● ○ ○

The given differential equation is

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = C^2\left(\frac{d^2y}{dx^2}\right)^2$$

The highest order derivative present is $\frac{d^2y}{dx^2}$.

Hence, the **order** of the differential equation is 2.

Since the equation is polynomial in derivatives, the degree is the power of the highest order derivative after removing radicals and fractions.

Here, the highest order derivative appears as power of 2.

Therefore, the **degree** of the differential equation is 2.

Therefore, the correct option is

$$\boxed{\text{(C) 2nd order and 2nd degree}}$$

Mistakes to be avoided

- Do not confuse the exponent 3 on $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3$ with the degree; the degree depends only on the highest order derivative.

Q6.94.1 | MCQ | 1994 | 1M | Ans: A | ● ○ ○

The given differential equation is

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0 \text{ with the initial conditions } y(0) = 1,\ y'(0) = -2$$

The auxiliary equation is

$$m^2 + 2m + 1 = 0$$
$$(m+1)^2 = 0$$

Hence, the repeated root is $m = -1$.

Therefore, the complementary solution is

$$y = (C_1 + C_2 t)e^{-t}.$$

Differentiating,

$$y' = C_2 e^{-t} - (C_1 + C_2 t)e^{-t} = (C_2 - C_1 - C_2 t)\, e^{-t}$$

Using the initial conditions,

At $t = 0,\ 1 = C_1$.

Also,

$$-2 = C_2 - C_1 \Rightarrow C_2 = -1$$

Hence,

$$y = (1-t)e^{-t}.$$

Therefore, the correct option is

$$\boxed{\text{(A) } (1-t)e^{-t}}$$

DEBUG INFO

Differential Equations

Engineering Mathematics

Total Questions : 252

MCQ : 205

MSQ : 8

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